

Dr Mehmet Mercangöz | 2026

AI and Autonomous Industrial Systems Summer School

Towards Autonomous Industrial Systems: The Evolving Role of AI in Operations

What do humans do in a process plant



Perception

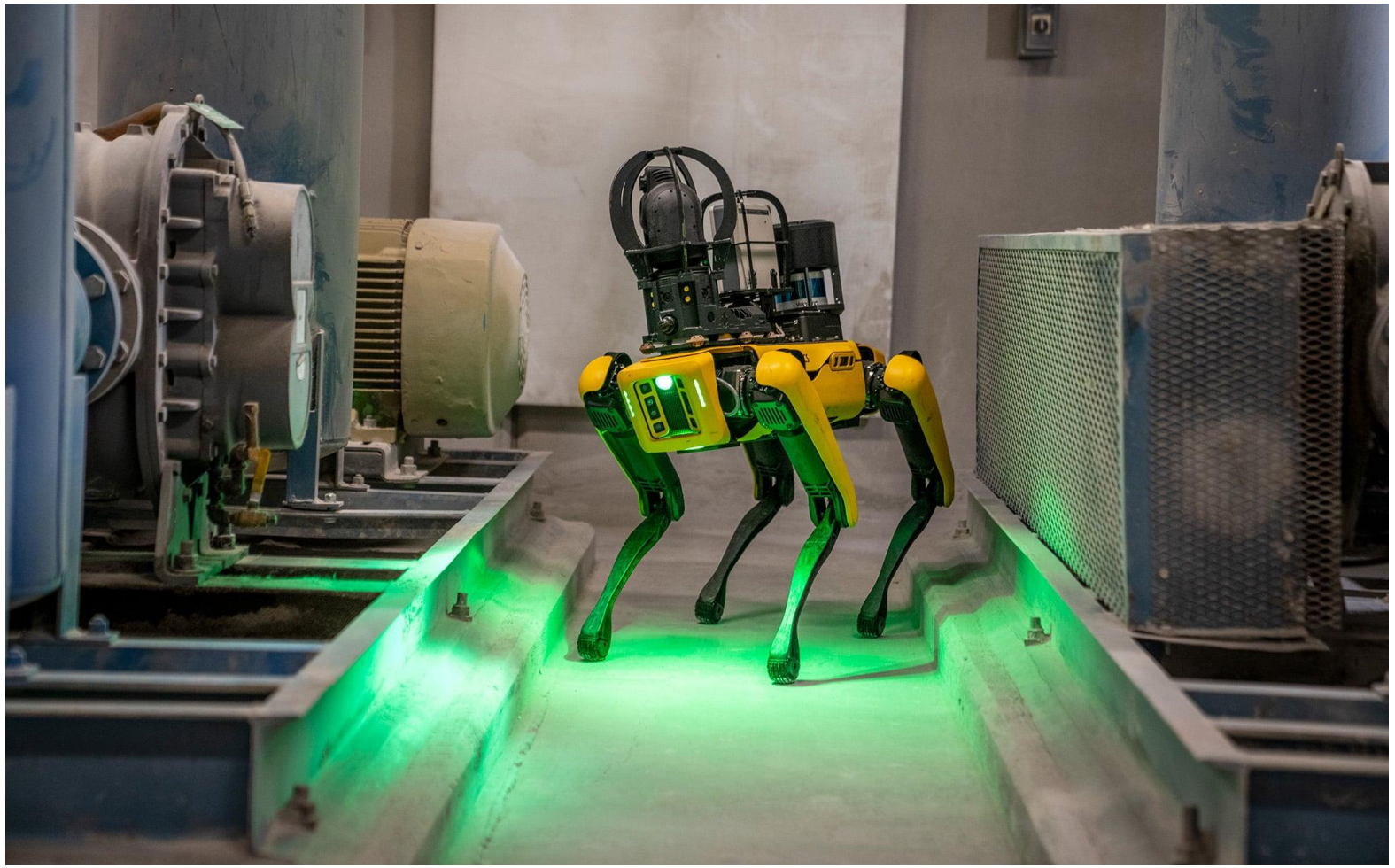


Understanding, analysis, decision making



Manipulation

Perception



What about manual work?

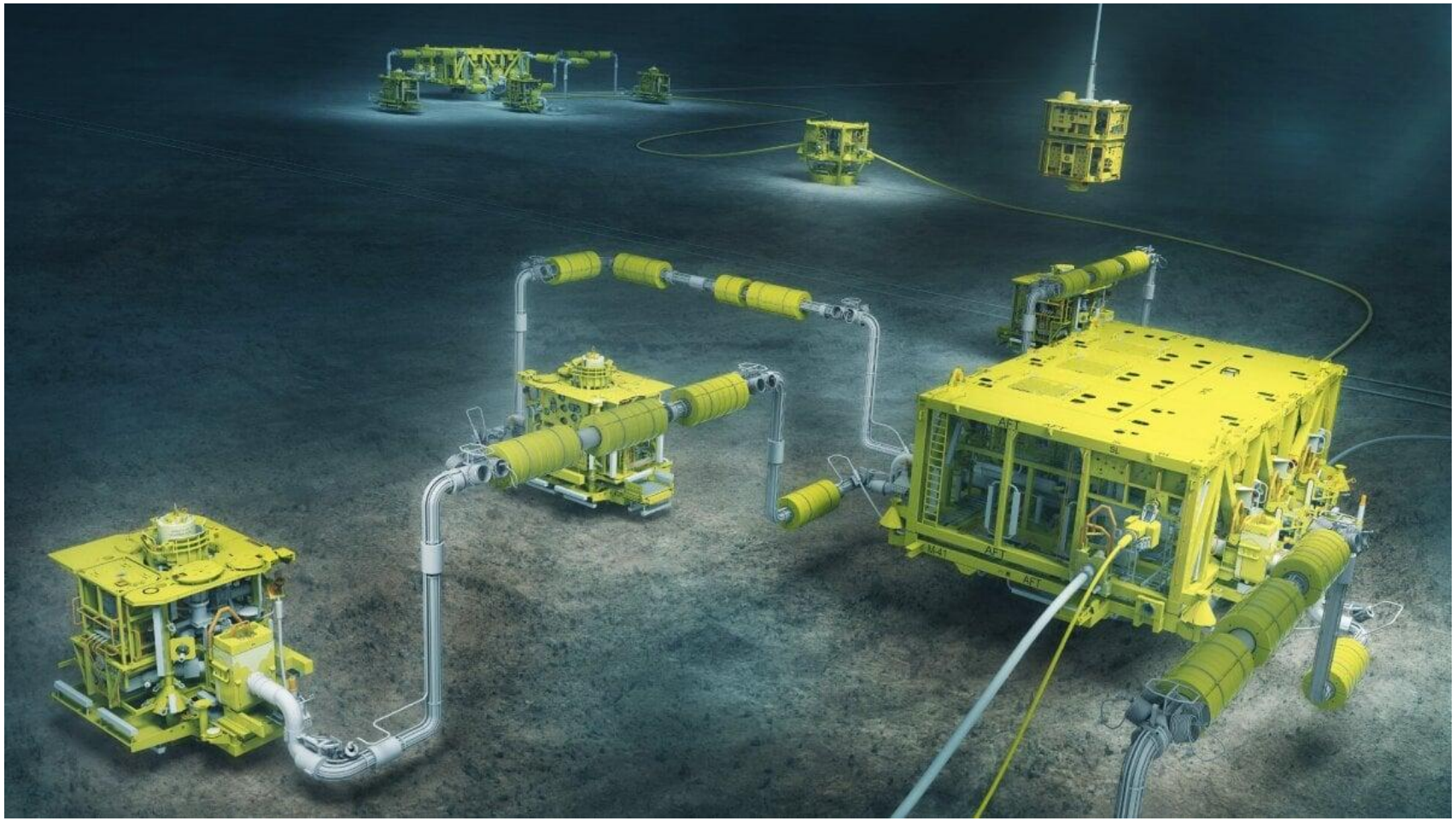


Understanding, analysis, decision making



Manipulation

What about manual work?



Subsea gas processing plant – no maintenance for 20 years

What about cognitive work?



Understanding, analysis, decision making

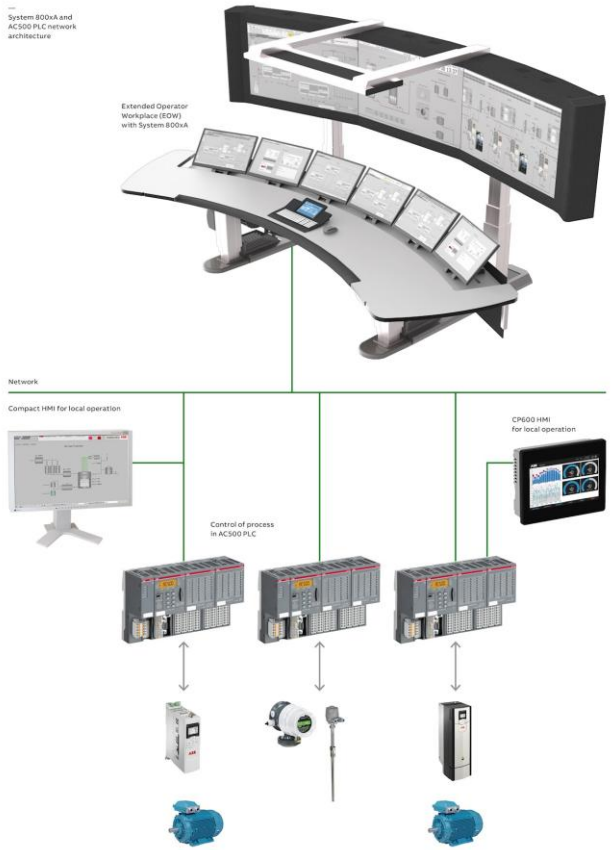
How do we operate a process plant?



Panel operator



Safety system



Control system

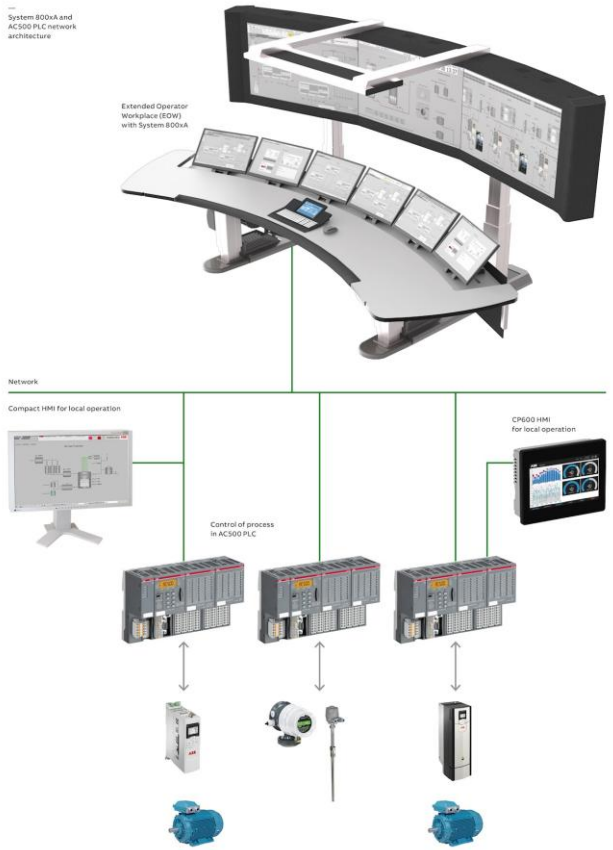
How do we operate a process plant?



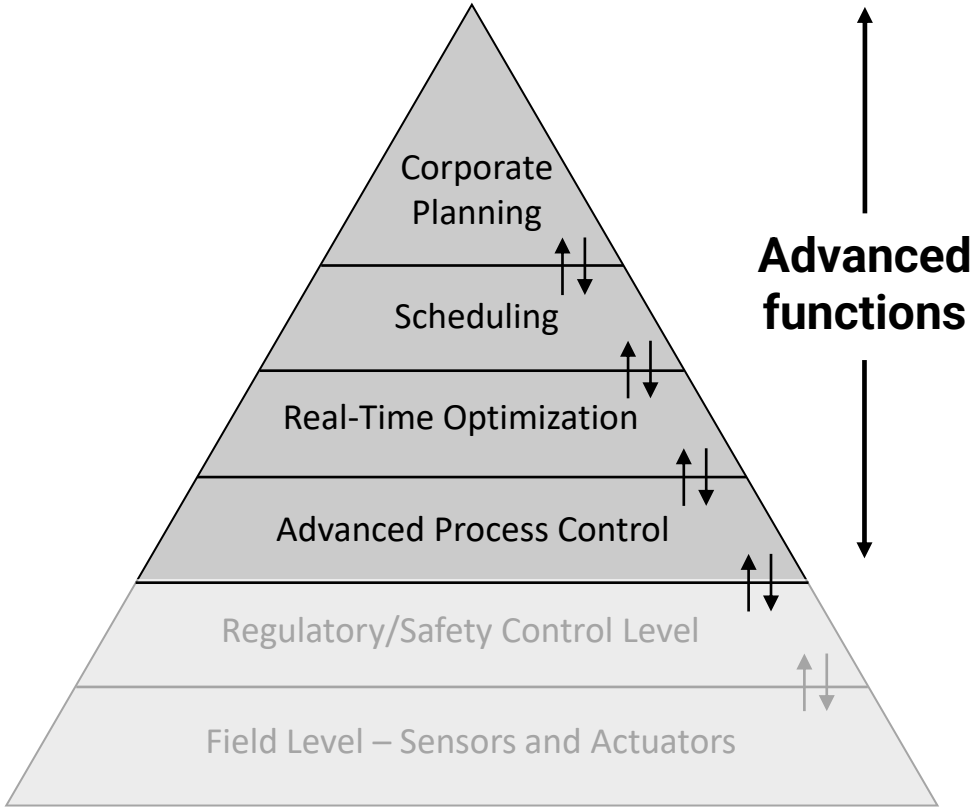
Panel operator



Safety system



Control system



Automation pyramid

Cognitive workload



Building models
Tuning controllers
Setting up optimization problems
…



Monitoring the process
Characterizing the process state
Making decisions for routine operations
Making decisions for handling anomalies
…

Need to replicate the ability to do the things that humans are doing today in their cognitive capacity

Cognitive workload



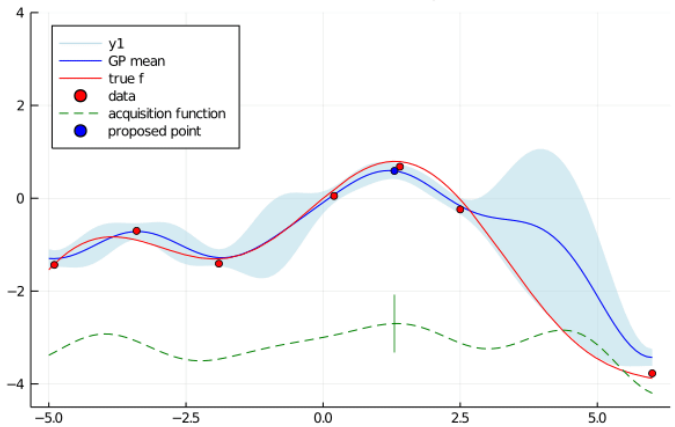
Building models
Tuning controllers
Setting up optimization problems
…



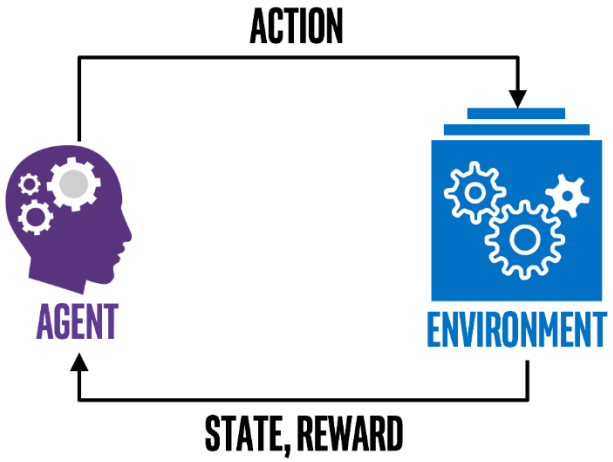
Monitoring the process
Characterizing the process state
Making decisions for routine operations
Making decisions for handling anomalies
…

Need to replicate the ability to do the things that humans are doing today in their cognitive capacity

AI for autonomy



Learning a model
(+ model-based algorithms)

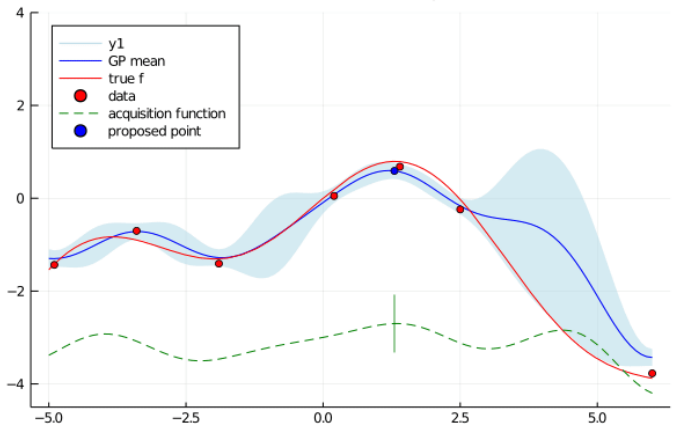


Learning a policy
(reinforcement learning)



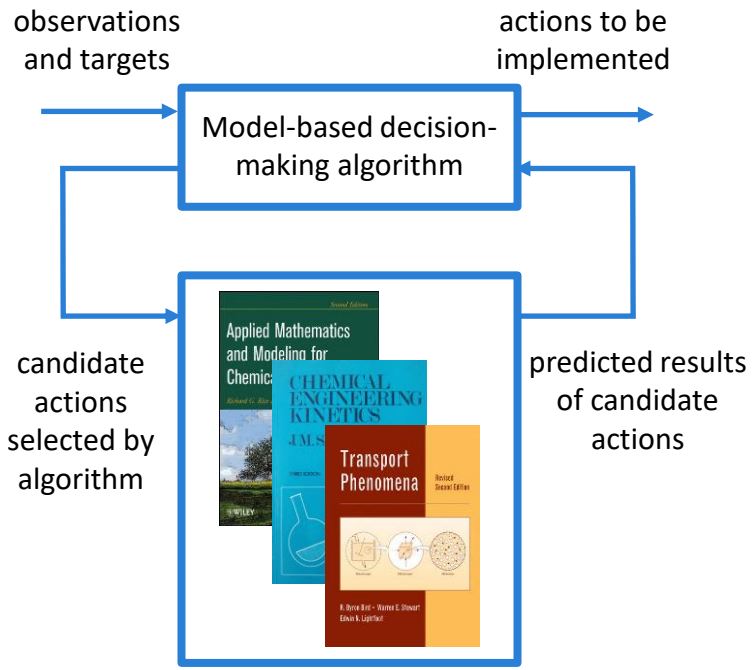
Learning how to mimic
human reasoning?

AI for autonomy

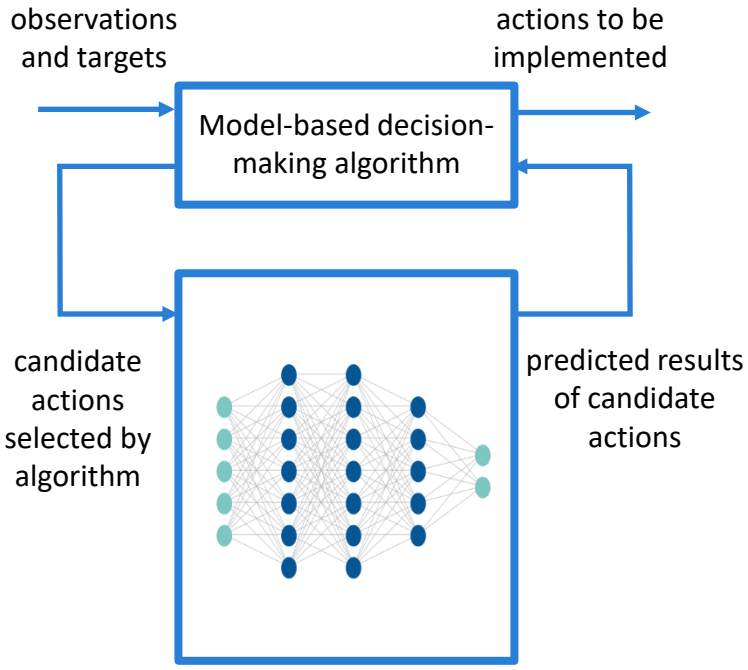


Learning a model
(+ model-based algorithms)

AI for autonomy: AI building models for model-based algorithms

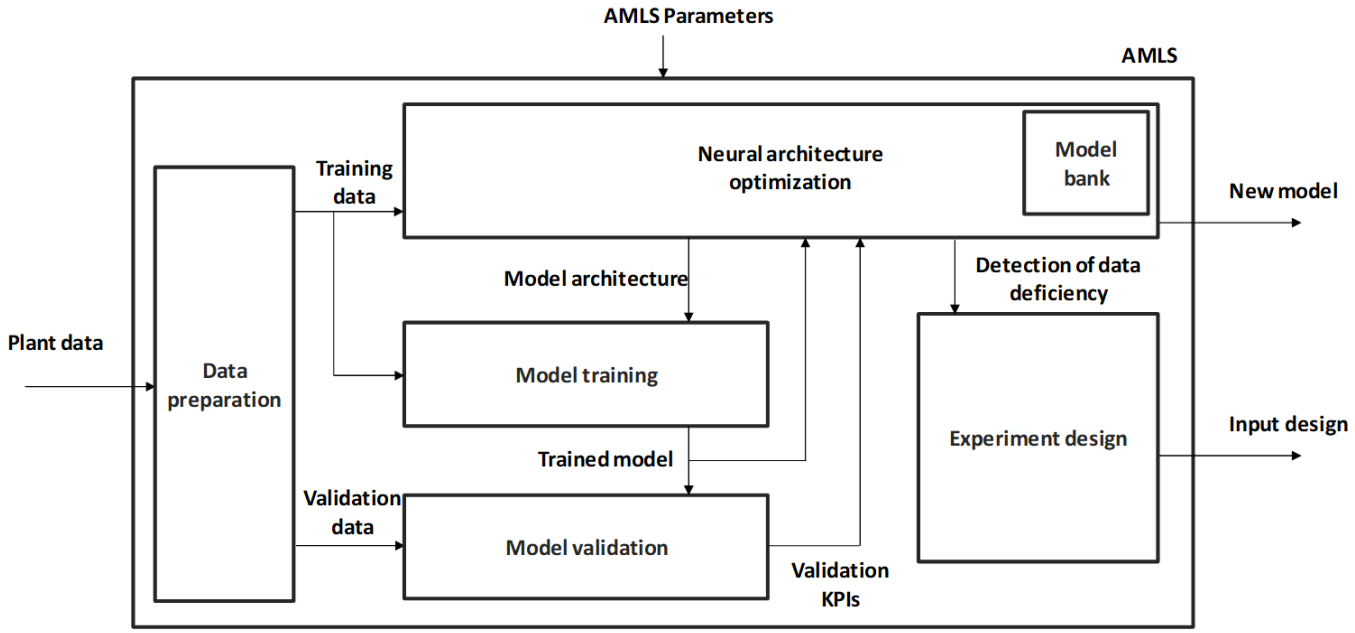


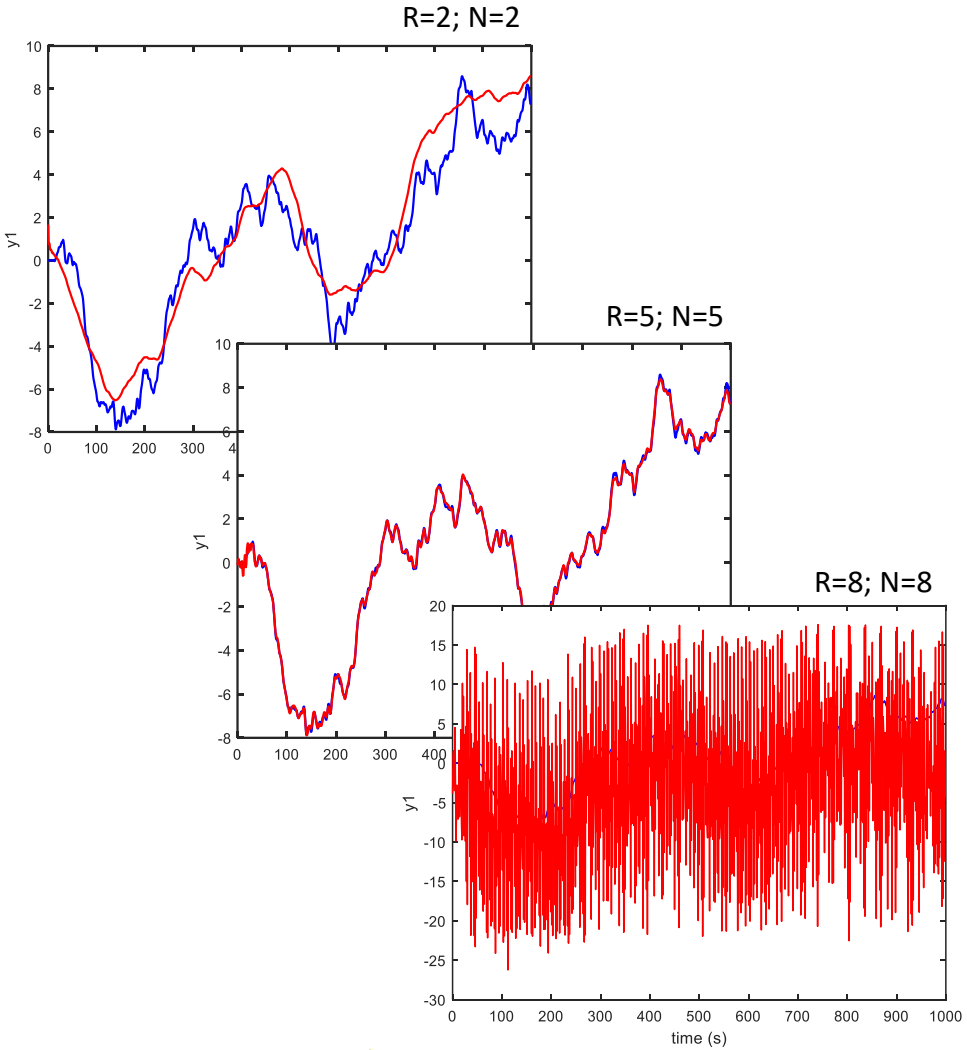
First principles / gray box



Data-driven

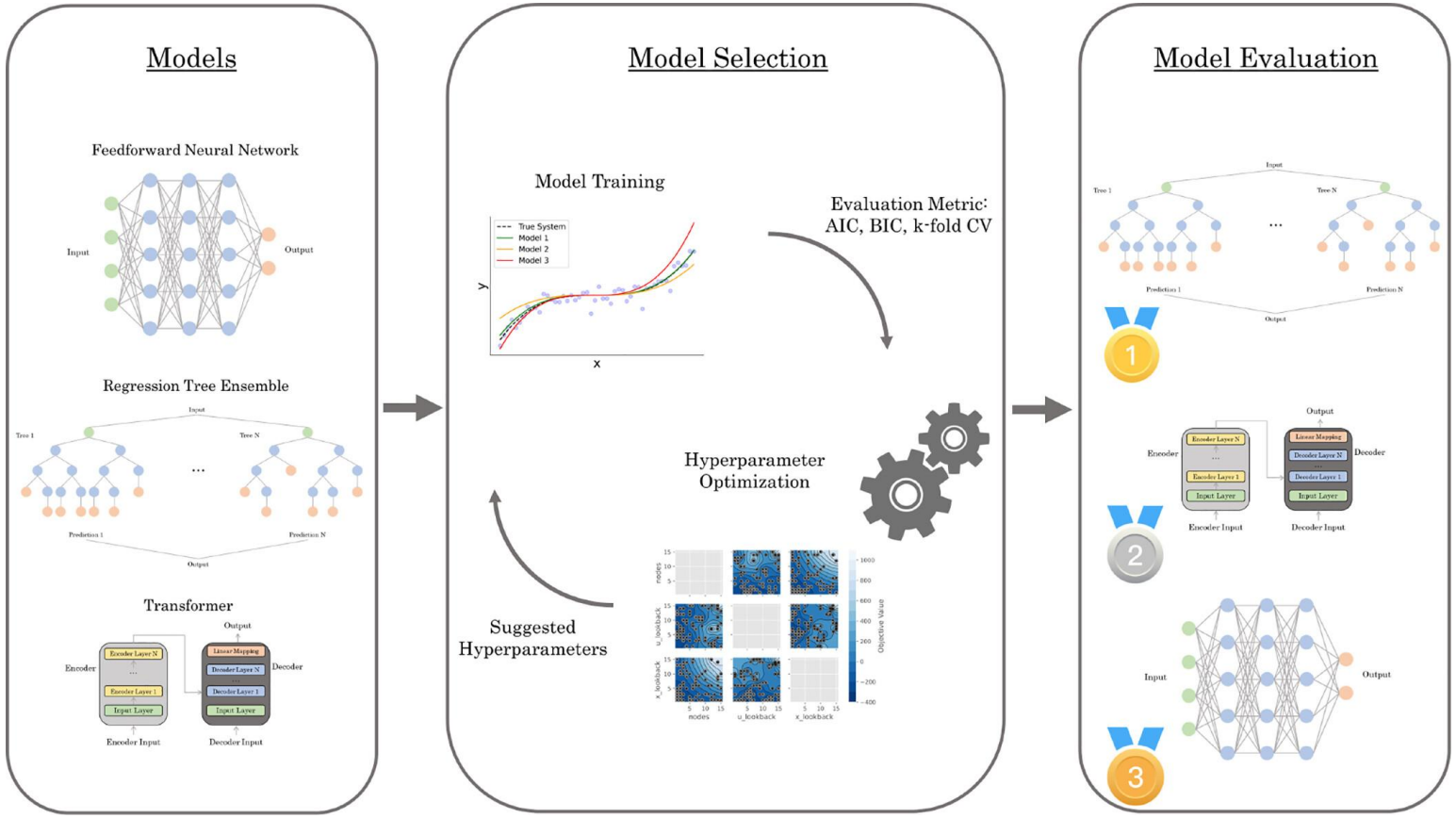
AI for autonomy: AI building models for model-based algorithms





Autonomous Process Model Identification using RNNs and Hyperparameter Optimization M Mercangöz, A Cortinovis, S Schönborn
2020 21st IFAC World Congress, Berlin

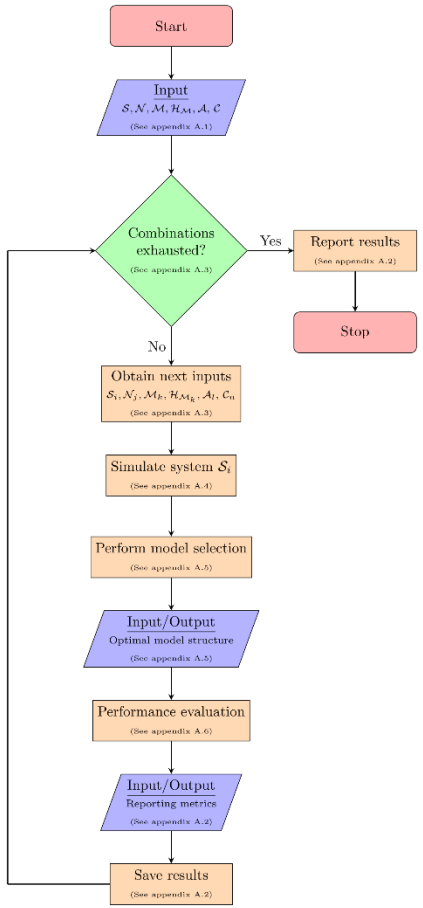
AI for autonomy: AI building models for model-based algorithms

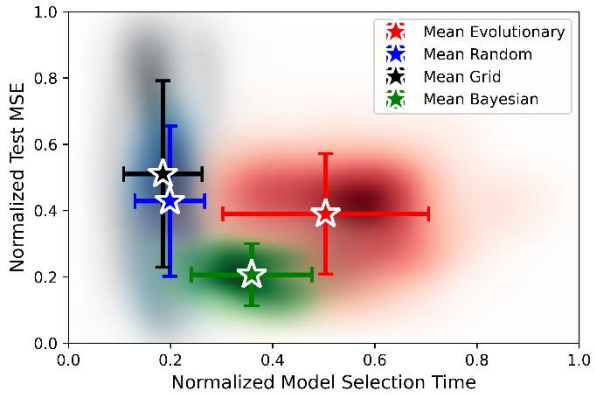


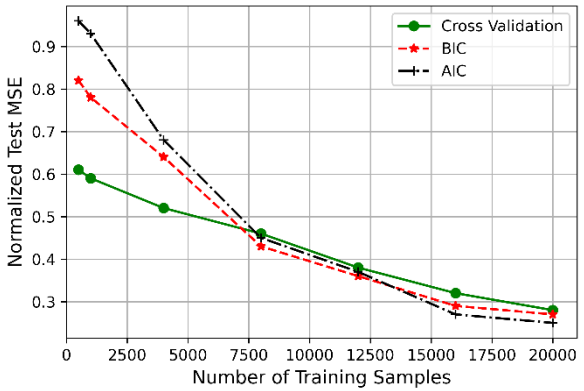
Ahmed, A., del Rio-Chanona, E. A., & Mercangöz, M. (2025). Comparative Study of Machine Learning and System Identification for Process Systems Engineering Dynamics. Industrial & Engineering Chemistry Research.

AI for autonomy: AI building models for model-based algorithms

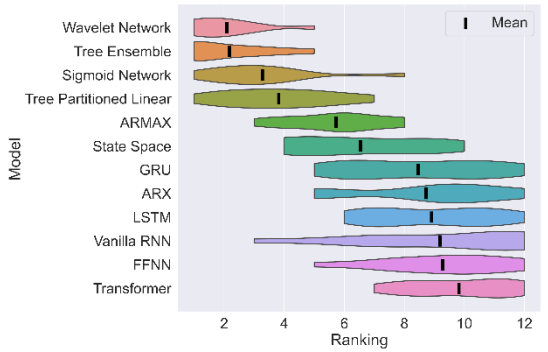
We can make the pipeline autonomous!

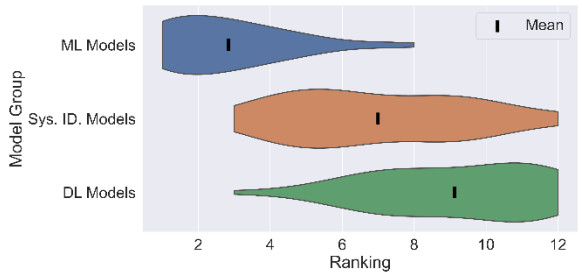




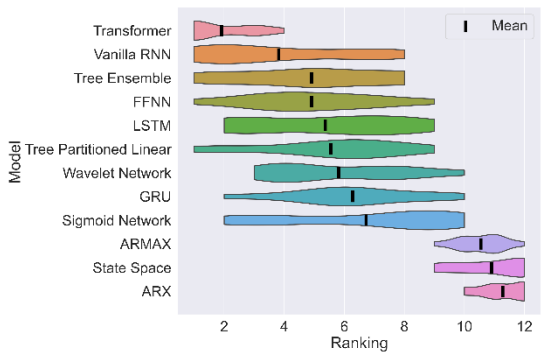


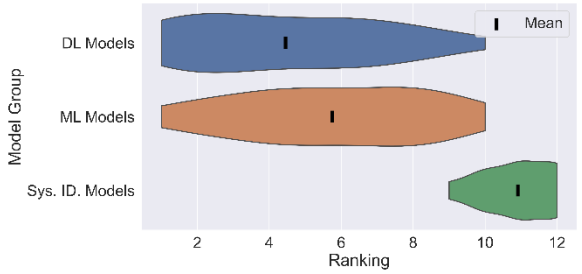
20,000 training samples





500 training samples

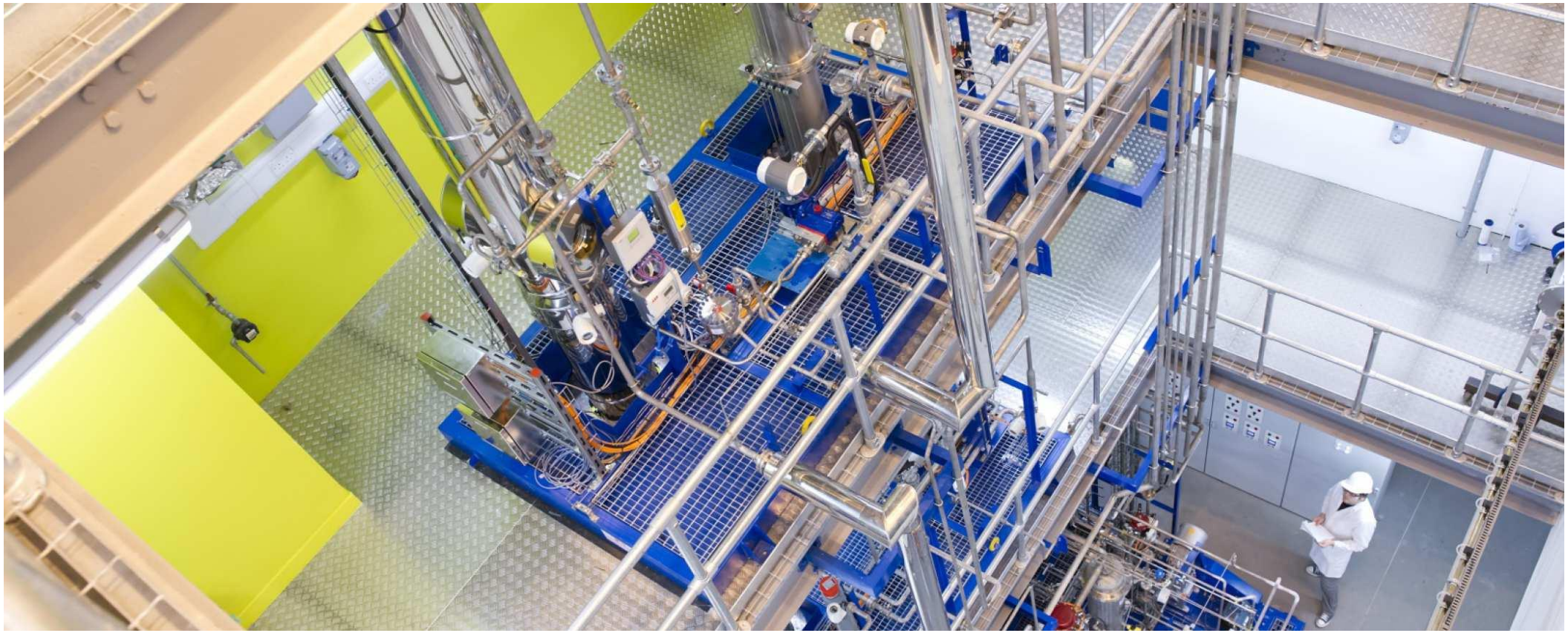




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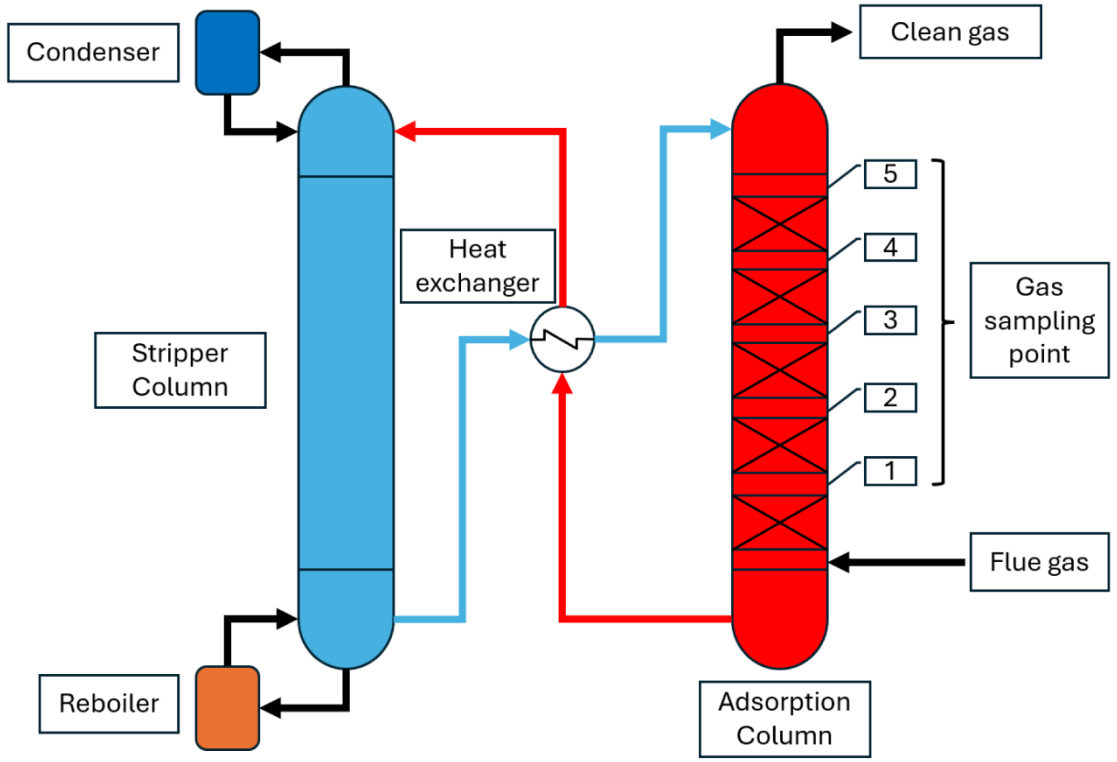
Imperial Chem Eng Carbon Capture Pilot Plant

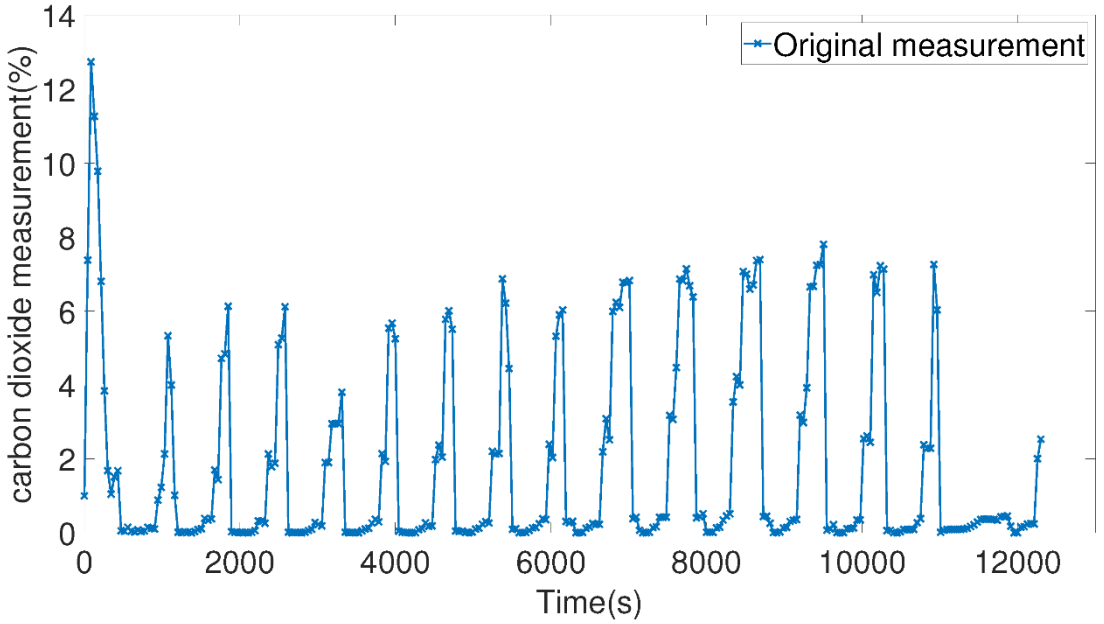


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Imperial Chem Eng Carbon Capture Pilot Plant

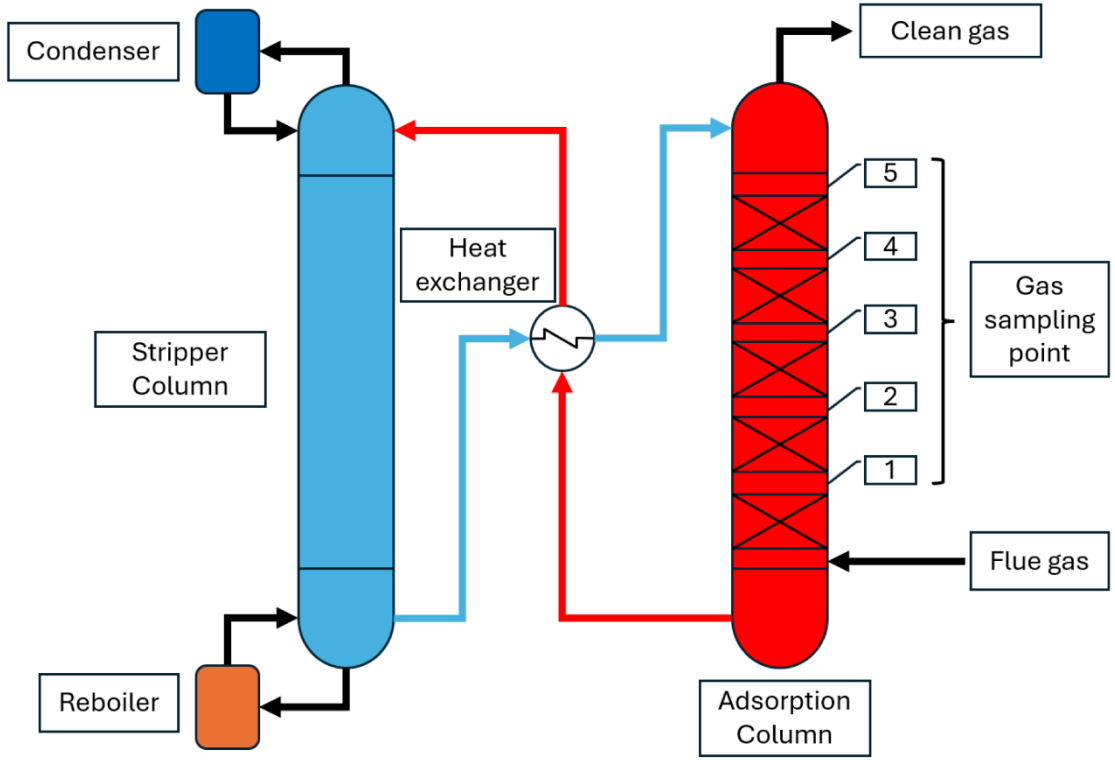


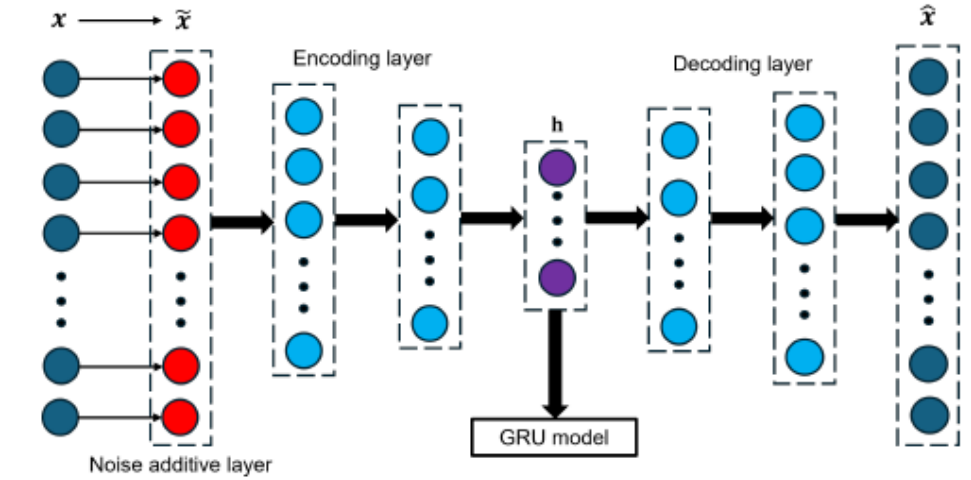


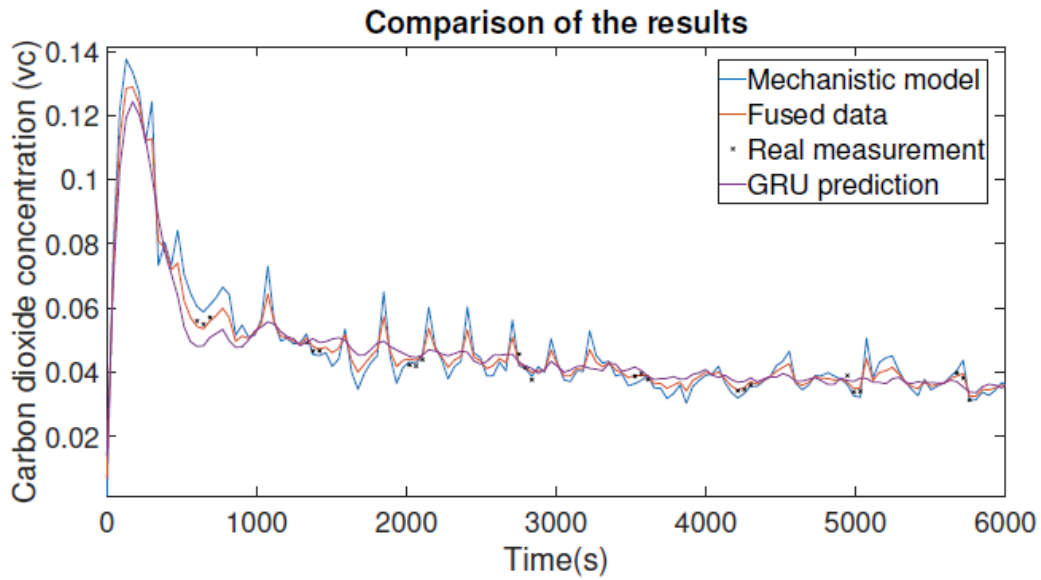
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Imperial Chem Eng Carbon Capture Pilot Plant







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Chai, S., Guo, S., & Mercangöz, M. (2026). Hybrid Data-Driven and Mechanistic CO2 Soft Sensor with MHE-Imputed Labels and Covariance-Weighted Fusion in a Pilot-Scale Absorber. Processes, 14(6), 916.

External Parameters

Prices, production targets, ..



PI Loops Set-points

Regulatory Control Loops

Inputs (PI Loops)

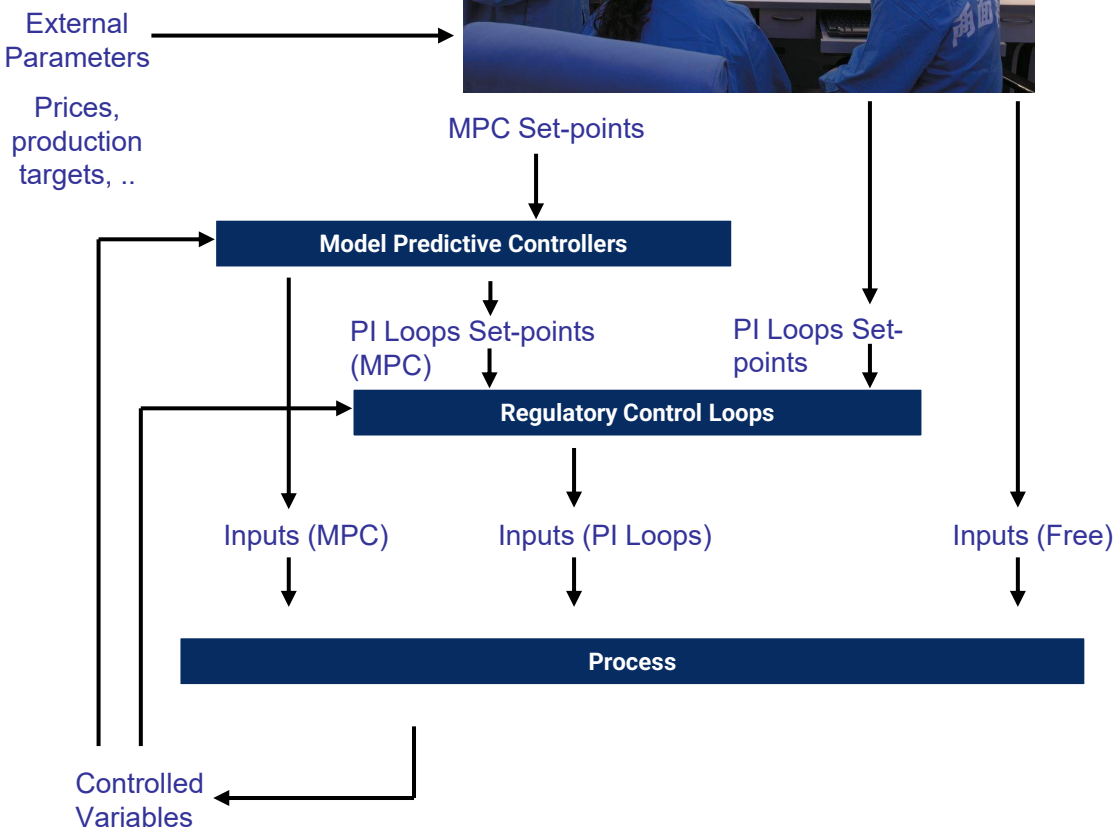
Inputs (Free)

Process

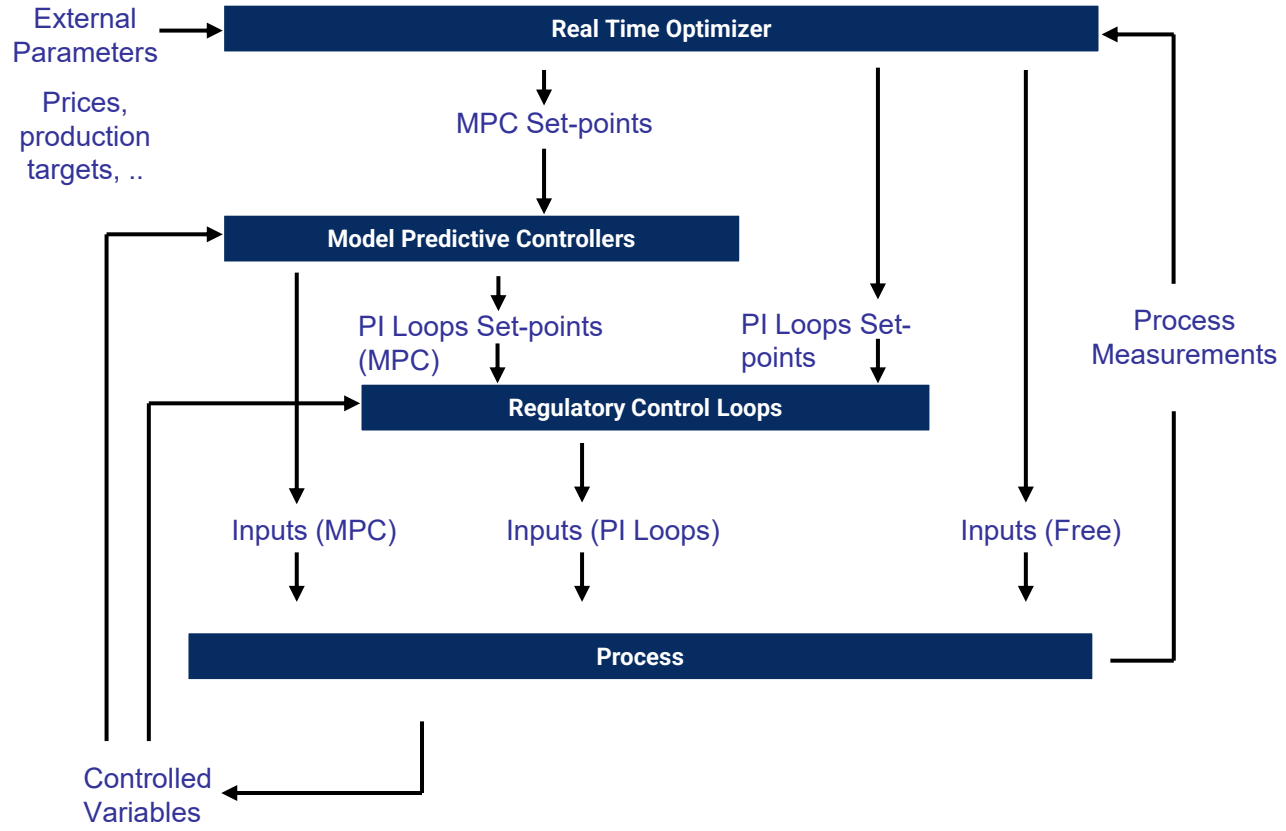
Controlled Variables



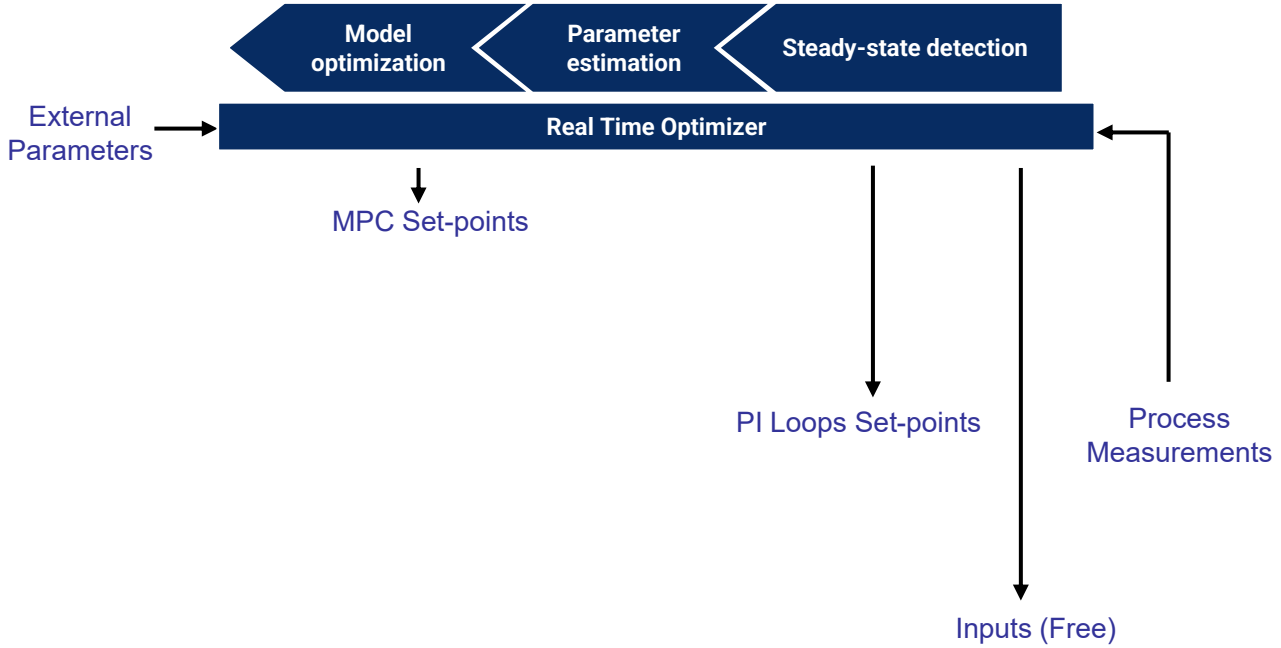



External Parameters
Prices, production targets, ..
MPC Set-points
Model Predictive Controllers
PI Loops Set-points (MPC)
PI Loops Set-points
Regulatory Control Loops
Inputs (MPC)
Inputs (PI Loops)
Inputs (Free)
Process
Controlled Variables




External Parameters
Prices, production targets, ..
Real Time Optimizer
MPC Set-points
Model Predictive Controllers
PI Loops Set-points (MPC)
PI Loops Set-points
Process Measurements
Regulatory Control Loops
Inputs (MPC)
Inputs (PI Loops)
Inputs (Free)
Process
Controlled Variables

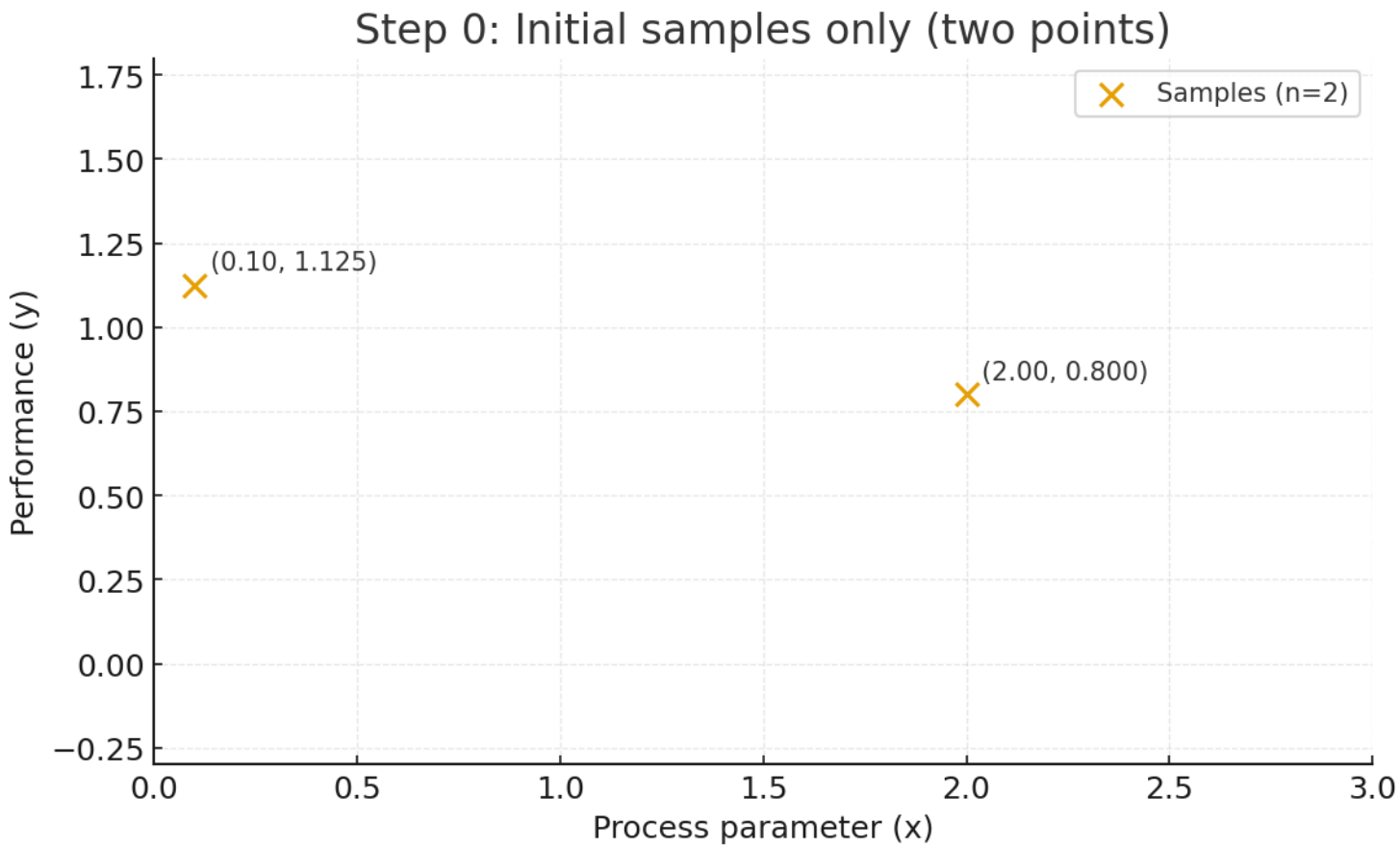




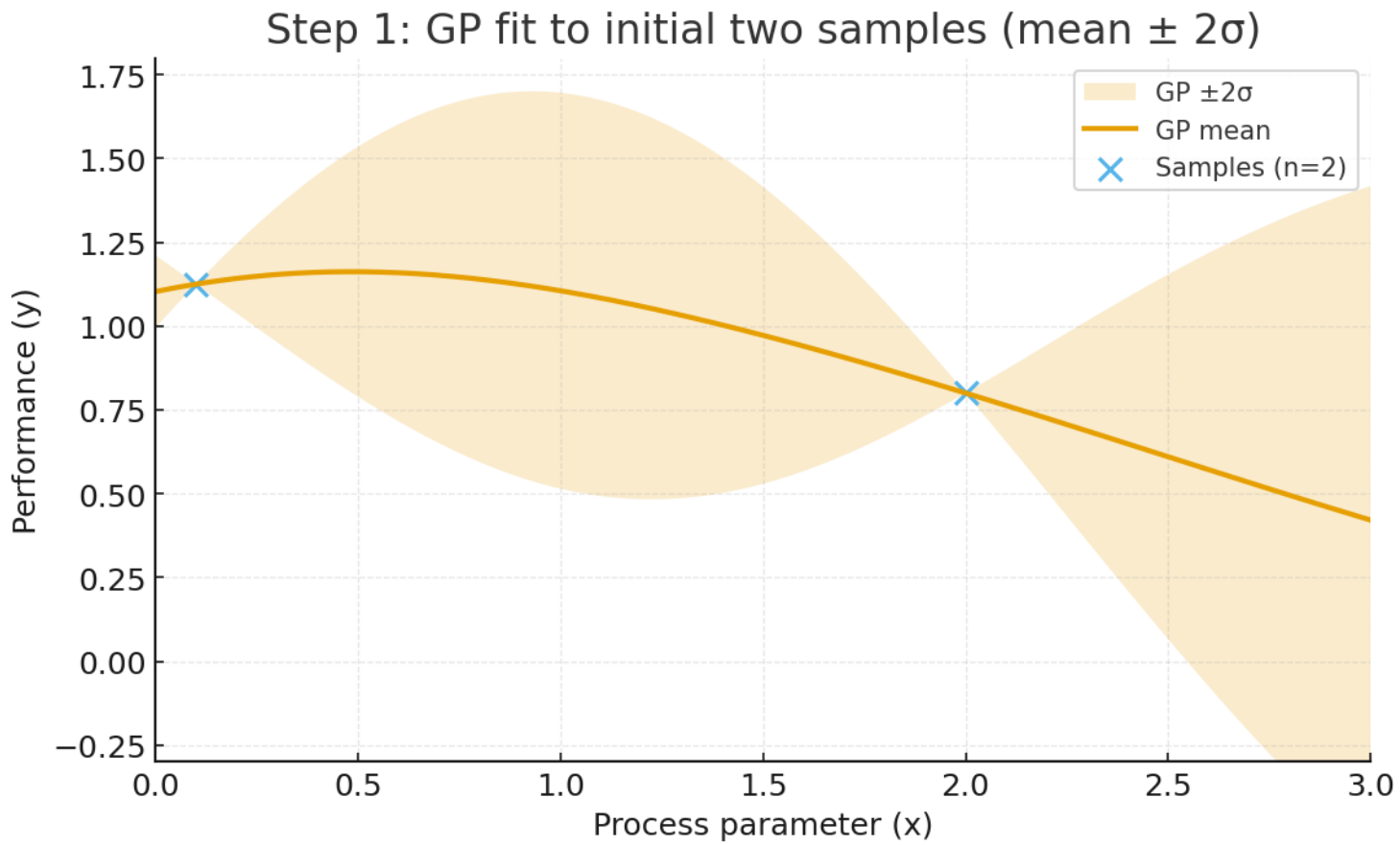
RTO layer

- Model is usually of nonlinear first-principles type but can also be data-driven
- Parameters of nonlinear model are estimated regularly to minimize mismatch between model predictions and process measurements
- Often a further compensation term (bias) is used to force model predictions from previous iteration to match with most recent measurements
- Constrained optimization is used to determine optimal operating points (control references or free MV values)

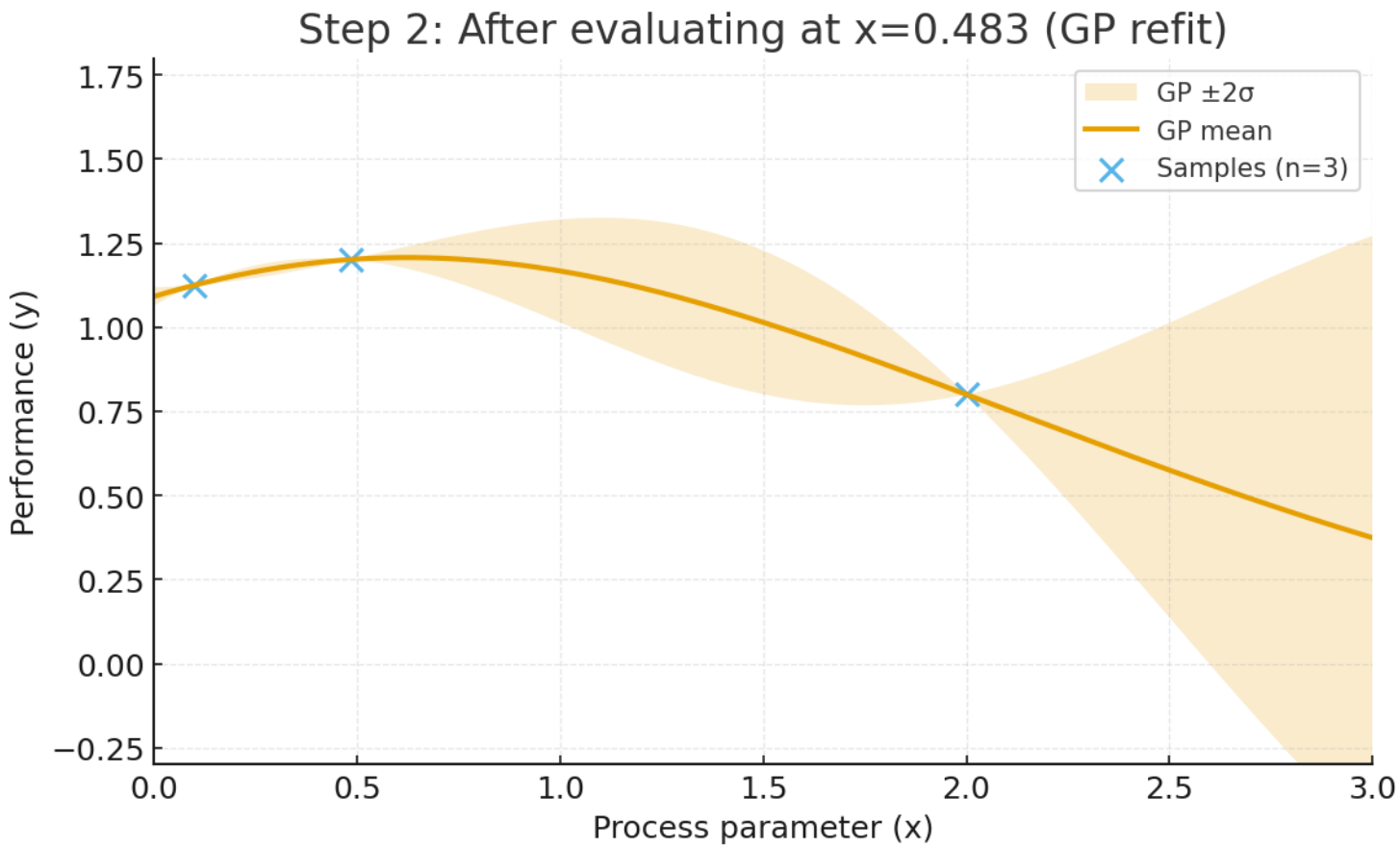
AI for autonomy: AI for “safe” active learning



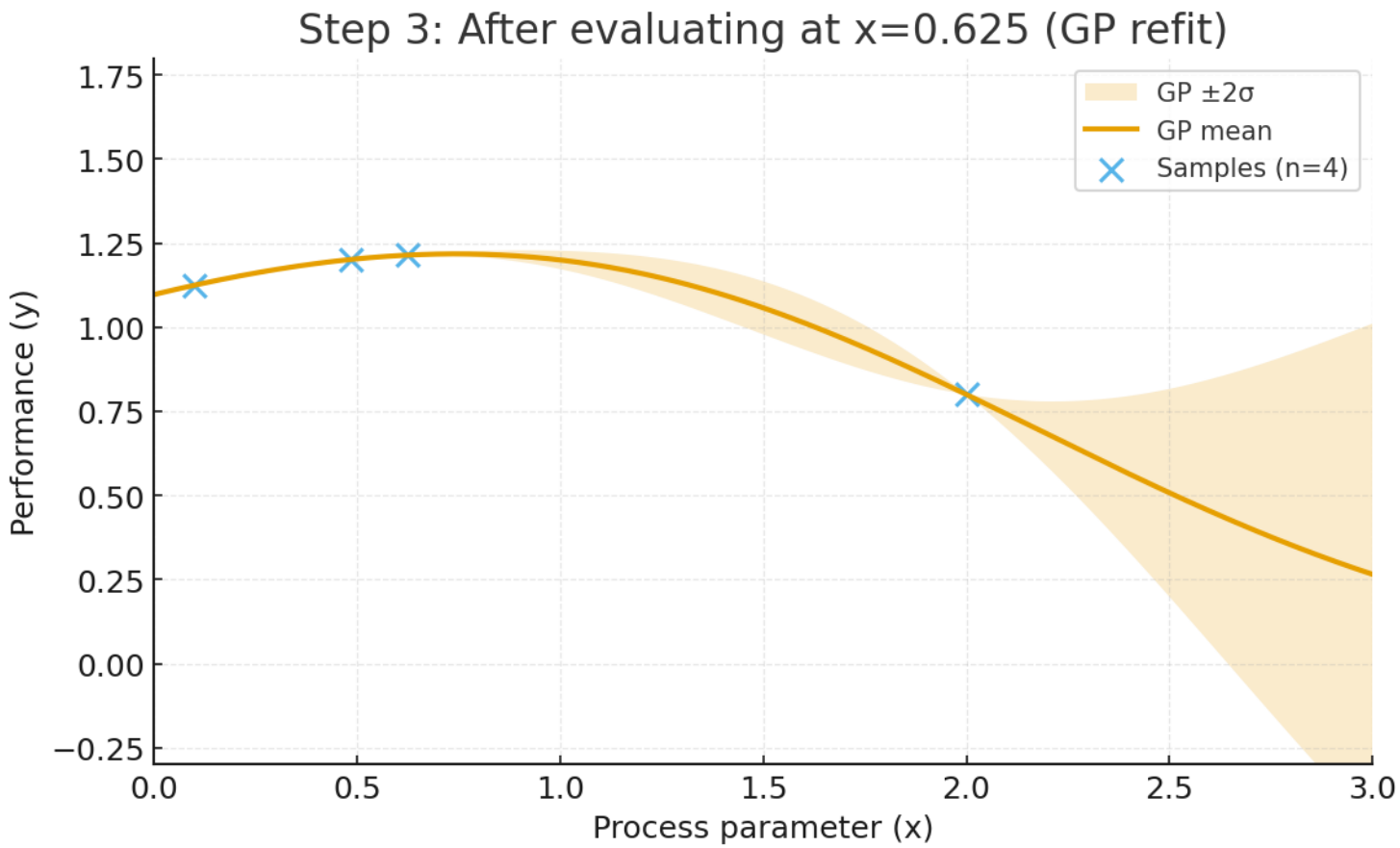
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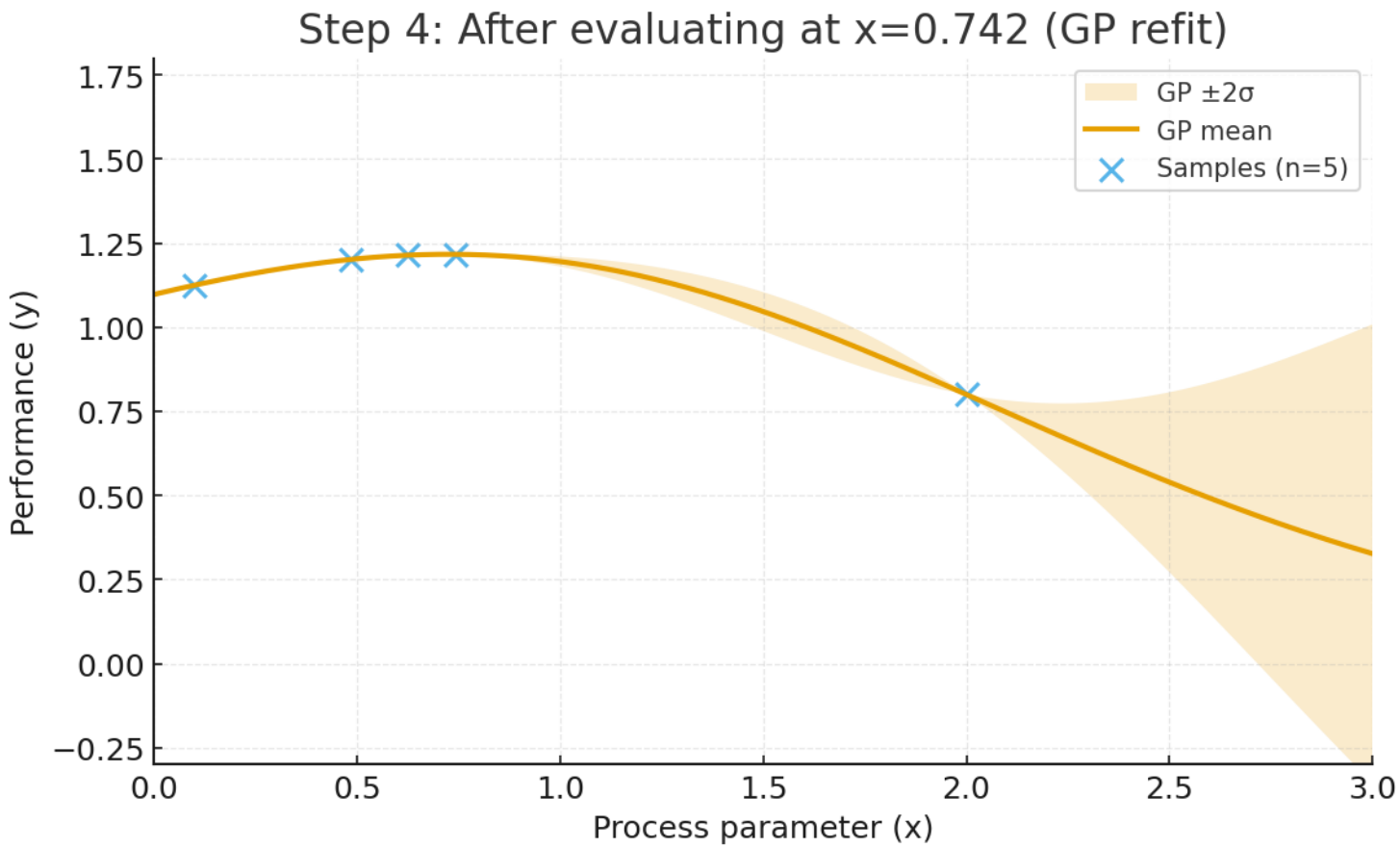
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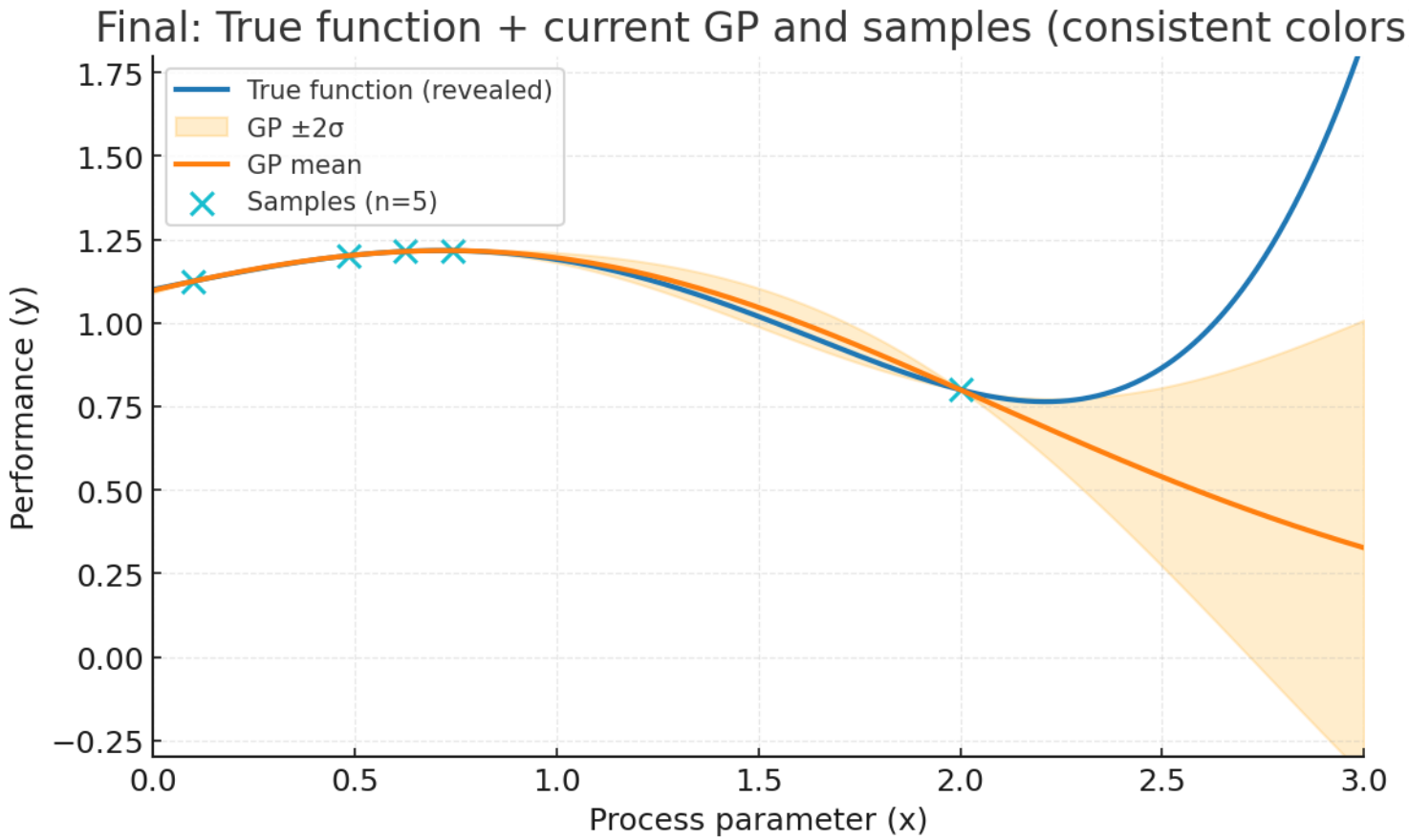
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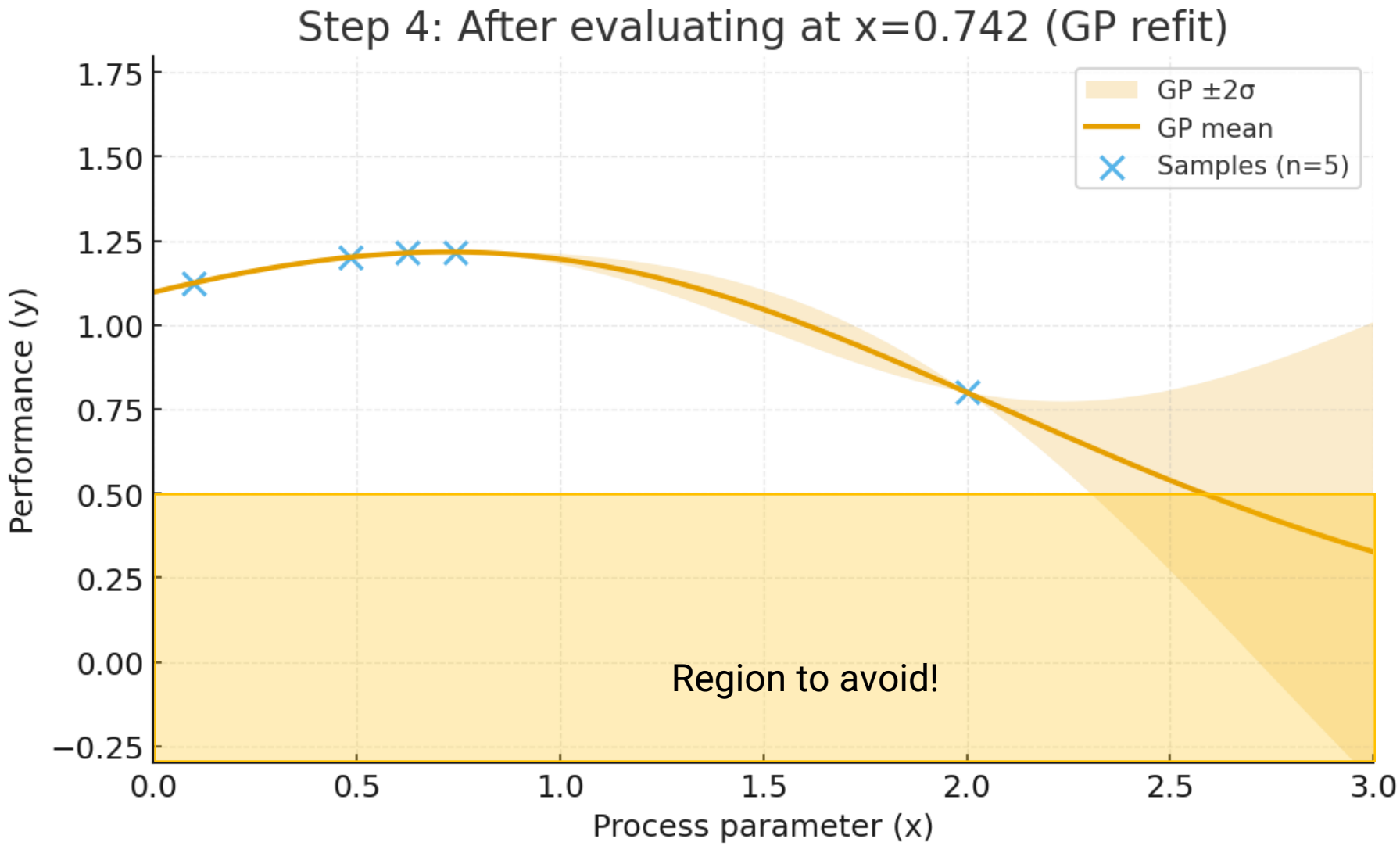
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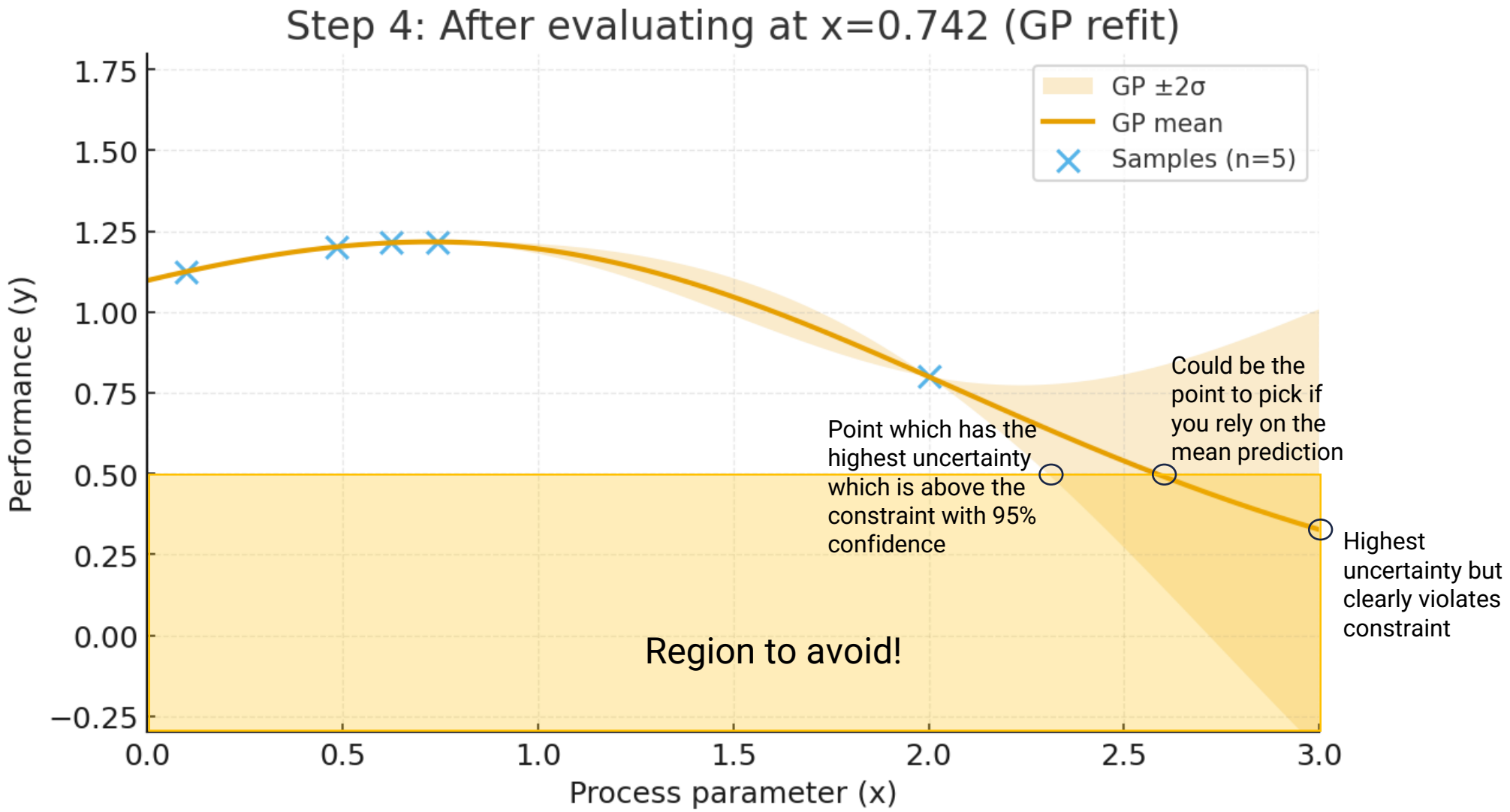
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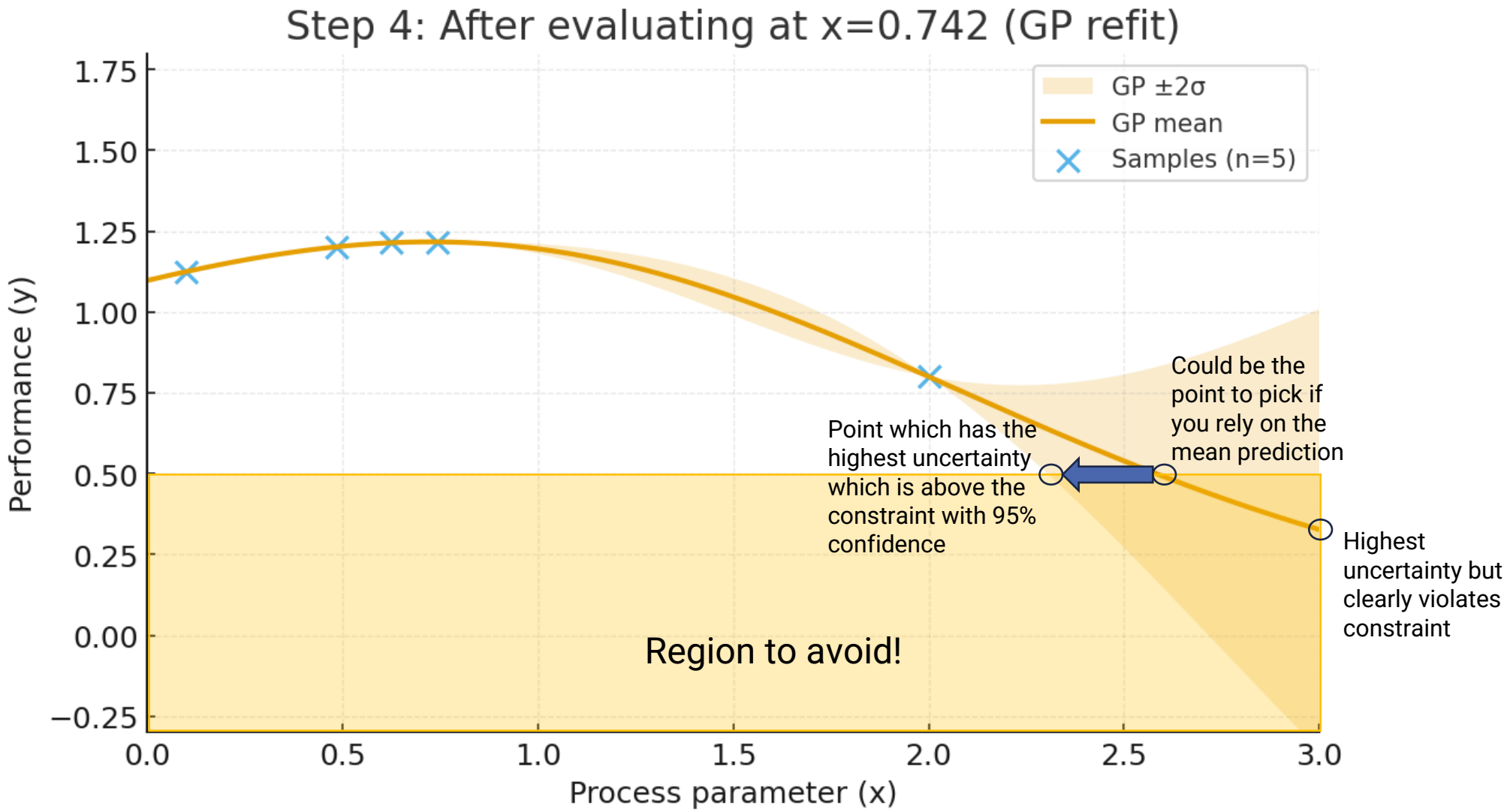
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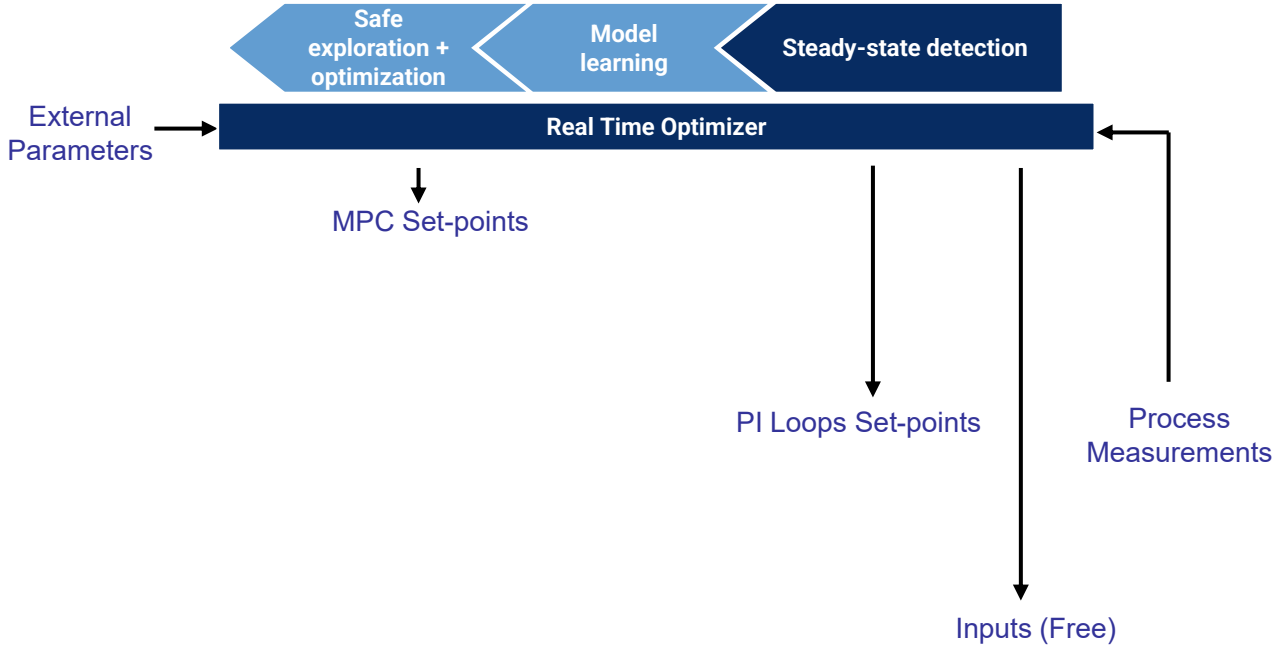


AI for autonomy: AI for “safe” active learning



AI for autonomy: AI for "safe" active learning





RTO to ARTEO

- Use GPs to model unknown process characteristics (outputs) as a function of RTO decision variables
- Use confidence intervals of GPs to enforce constraints with safety margins
- Use the confidence intervals to guide exploration as an (adjustable) part of the optimisation objective

Adaptive Real-Time Exploration and Optimization (ARTEO)

Korkmaz, B. S., Zagórowska, M., & Mercangöz, M. (2023). Safe optimization of an industrial refrigeration process using an adaptive and explorative framework. *IFAC-PapersOnLine*, *56*(2), 1400-1405.

Adaptive Real-Time Exploration and Optimization (ARTEO)

1. at t Given initial points + any collected data → learn $\mathcal{P}_{l,t}$

2. Receive any external inputs defining f and solve:

$$X_t^* = \underset{X_t}{\operatorname{argmin}}\, f\left(X_t, \mathcal{P}_{l,t}\right)$$

$$s.t.\, g\left(X_t, \mathcal{P}_{l,t}{}^{UCB}\right) \leq h$$

where

$$f\left(X_t, \mathcal{P}_{l,t}\right) = C\left(X_t, \mathcal{P}_{l,t}\right) - zU\left(X_t, \mathcal{P}_{l,t}\right)$$

and U is an uncertainty function

$$U\left(X_t, \mathcal{P}_{l,t}\right) = \sum_{l=1}^{n} \sigma_{\mathcal{P}_{l,t}}(X_t)$$

- n Unknown system characteristics

$$\mathcal{P}_{l,t}$$

- Cost function

$$C\left(X_t, \mathcal{P}_{l,t}\right)$$

Constraint function

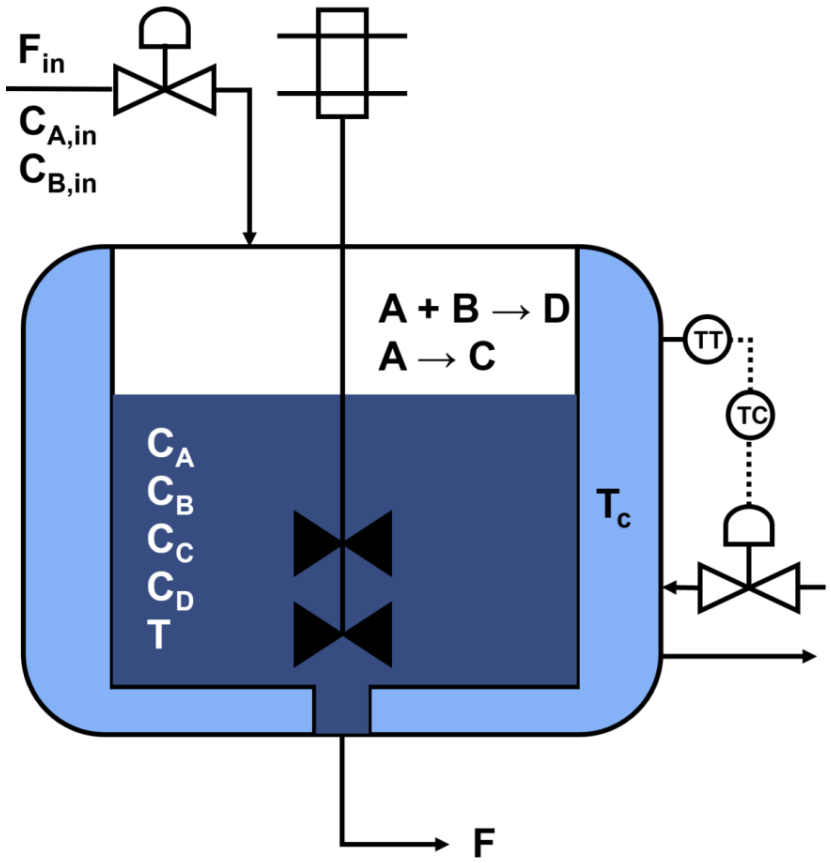
$$g\left(X_t, \mathcal{P}_{l,t}{}^{UCB}\right)$$

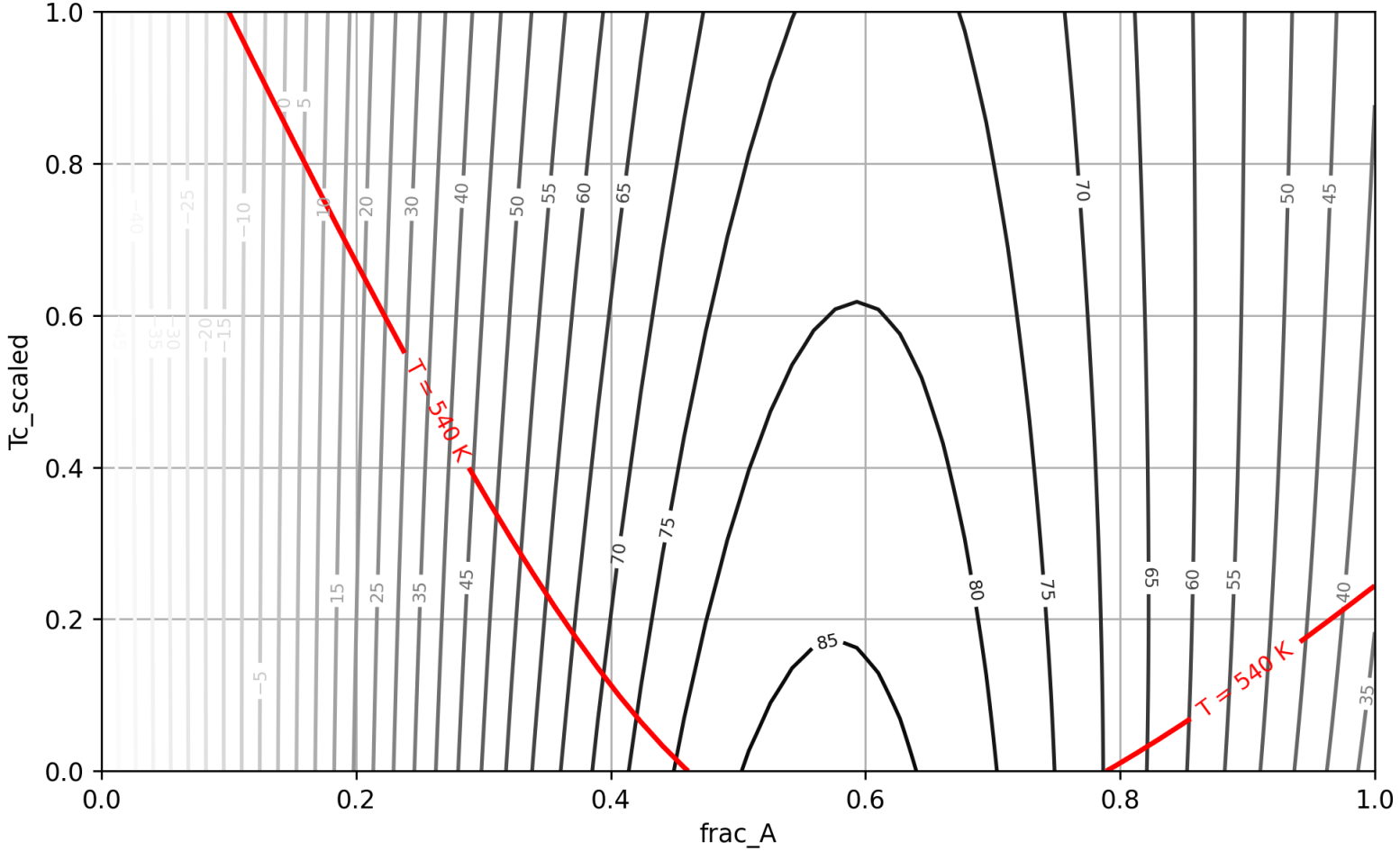
3. Implement X_t^* → collect resulting data

4. Repeat

Korkmaz, B. S., Zagórowska, M., & Mercangöz, M. (2023). Safe optimization of an industrial refrigeration process using an adaptive and explorative framework. *IFAC-PapersOnLine*, *56*(2), 1400-1405.

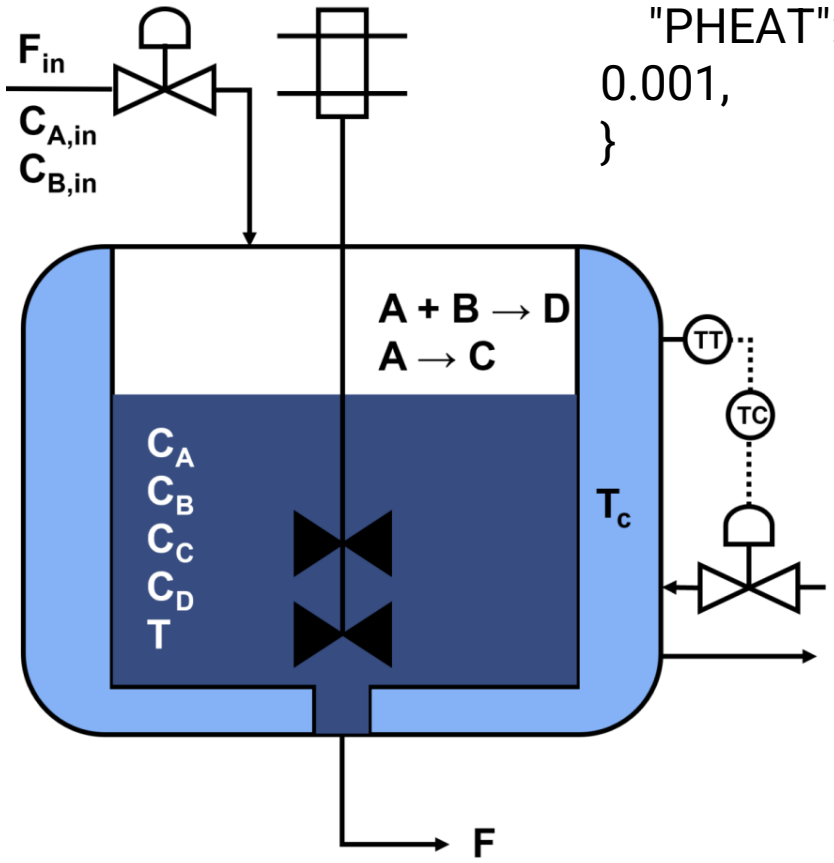
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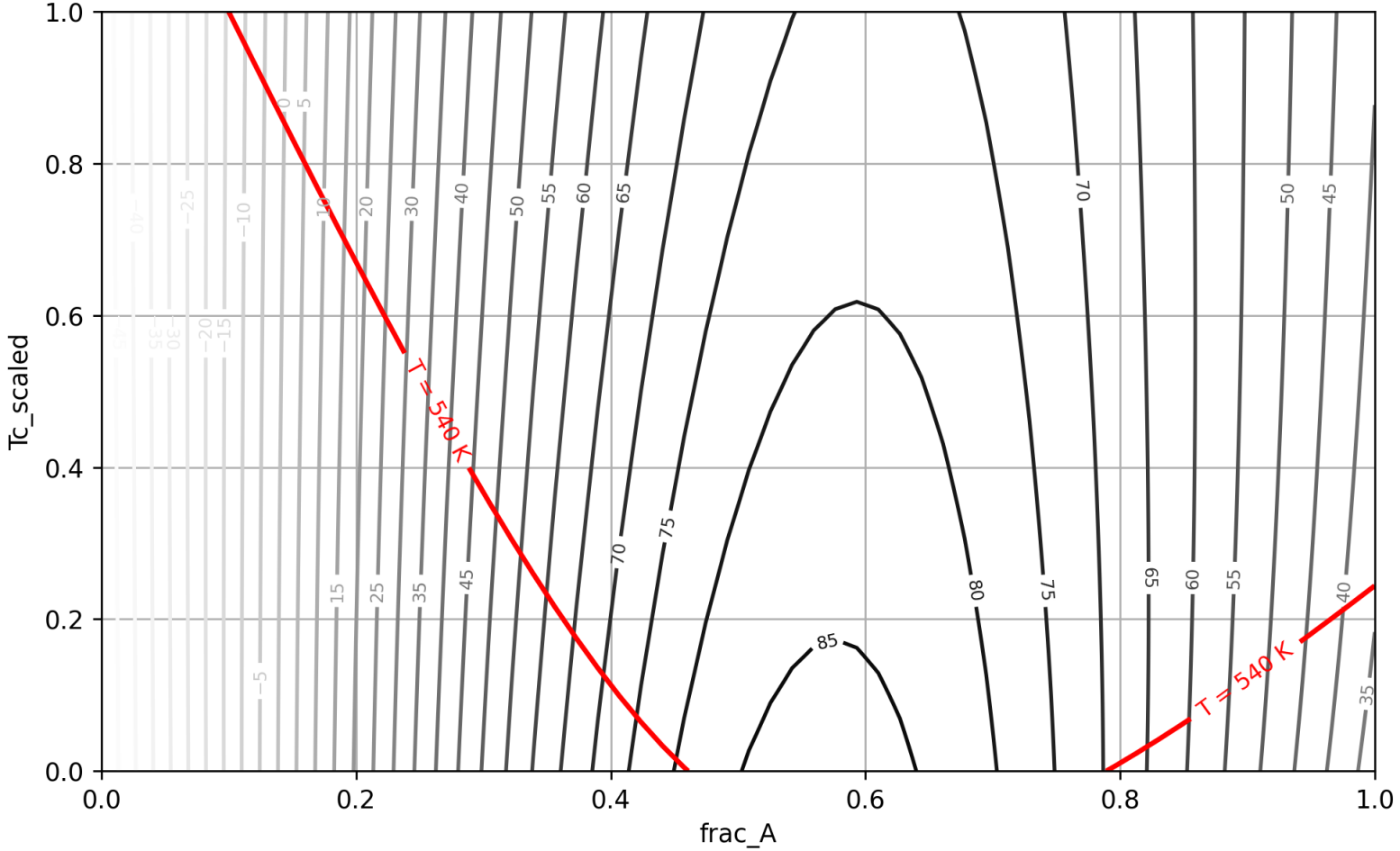


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}
```



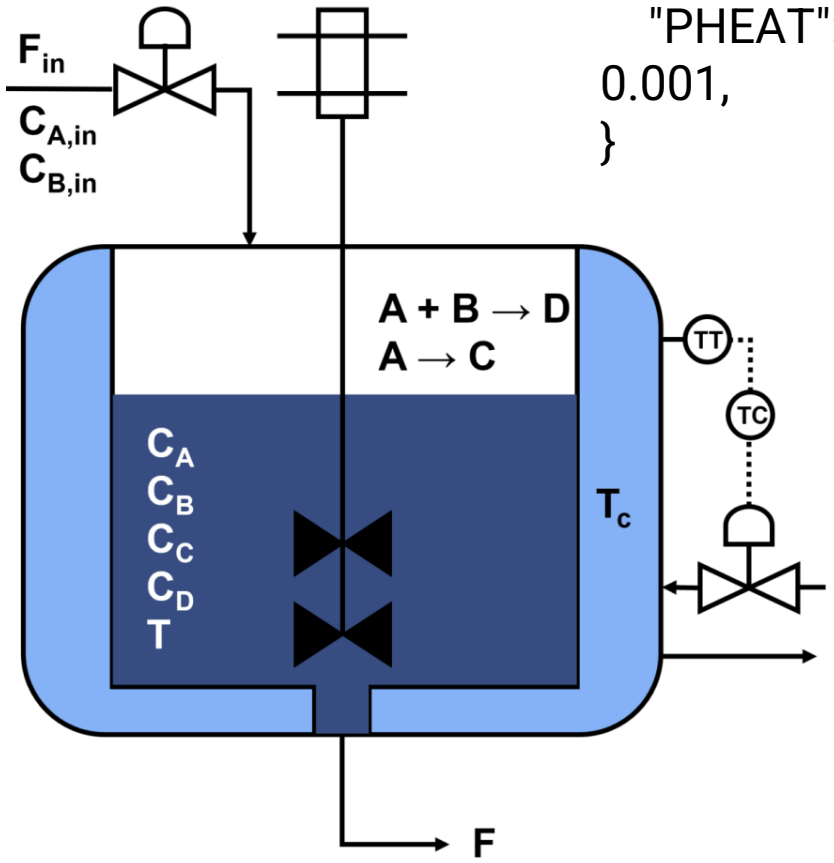
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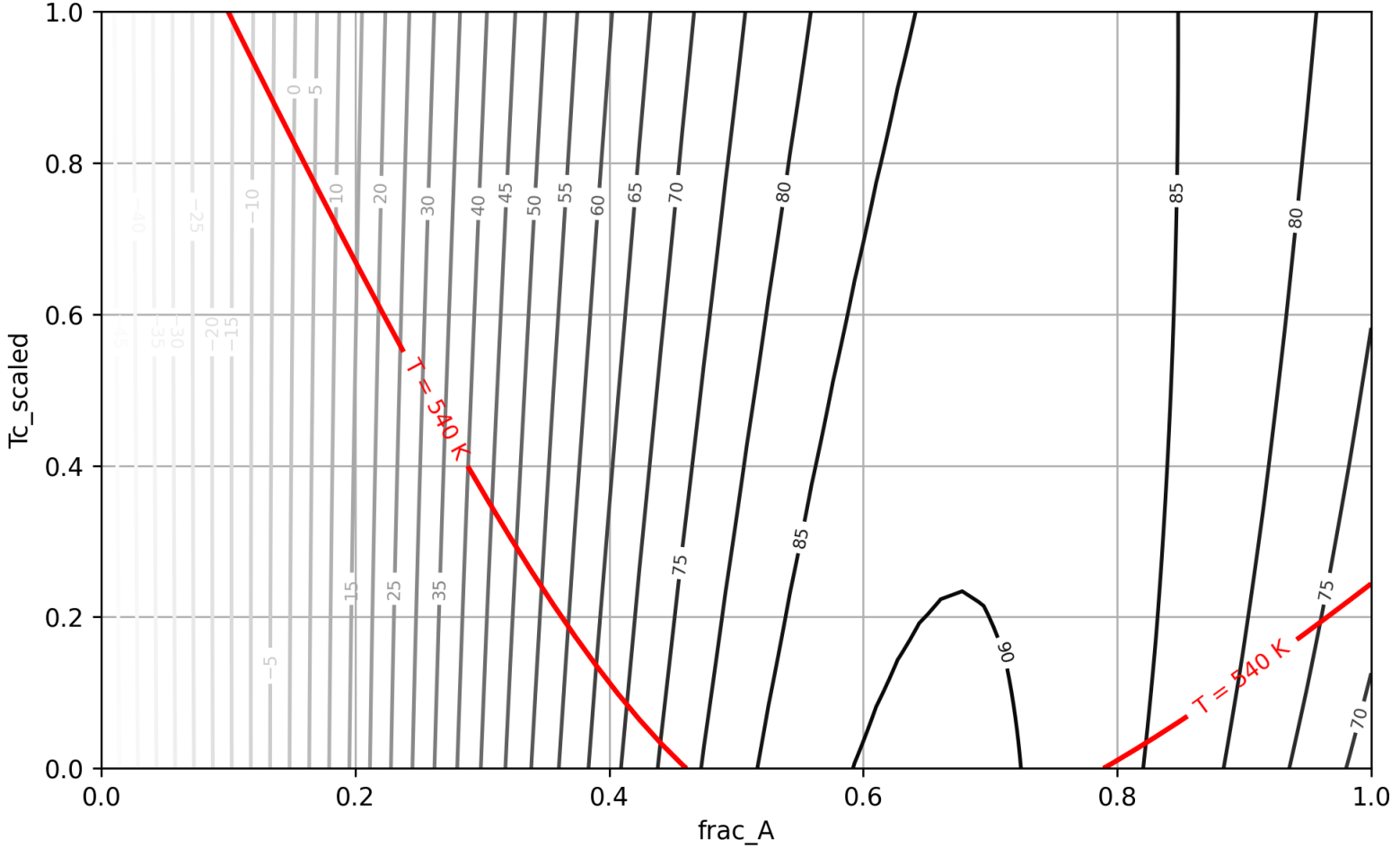
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```
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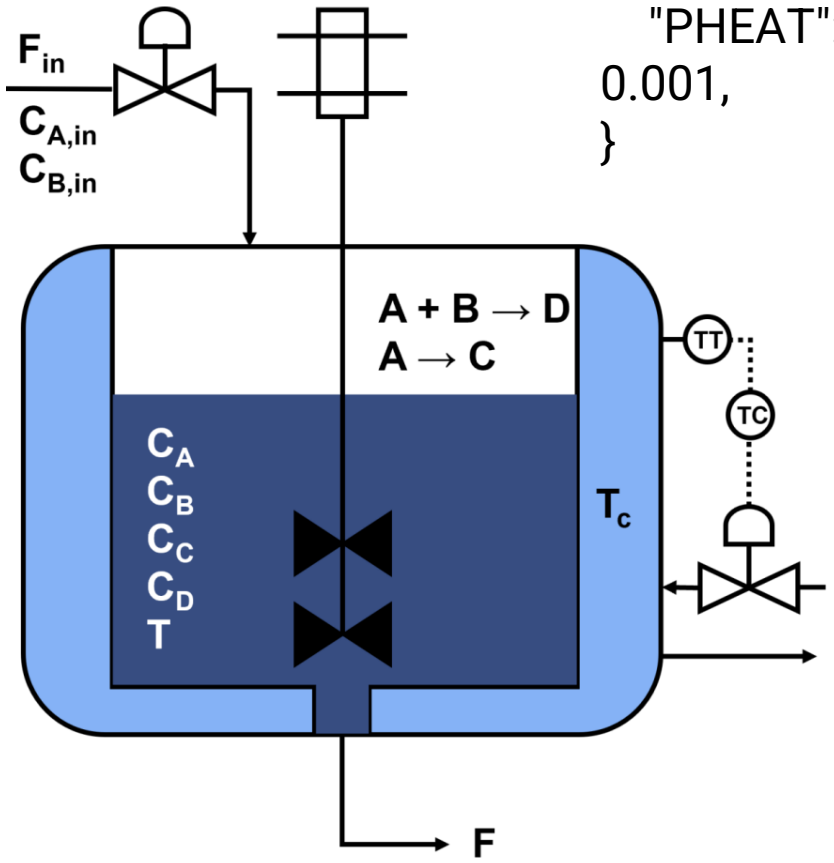


AI for autonomy: AI for “safe” active learning

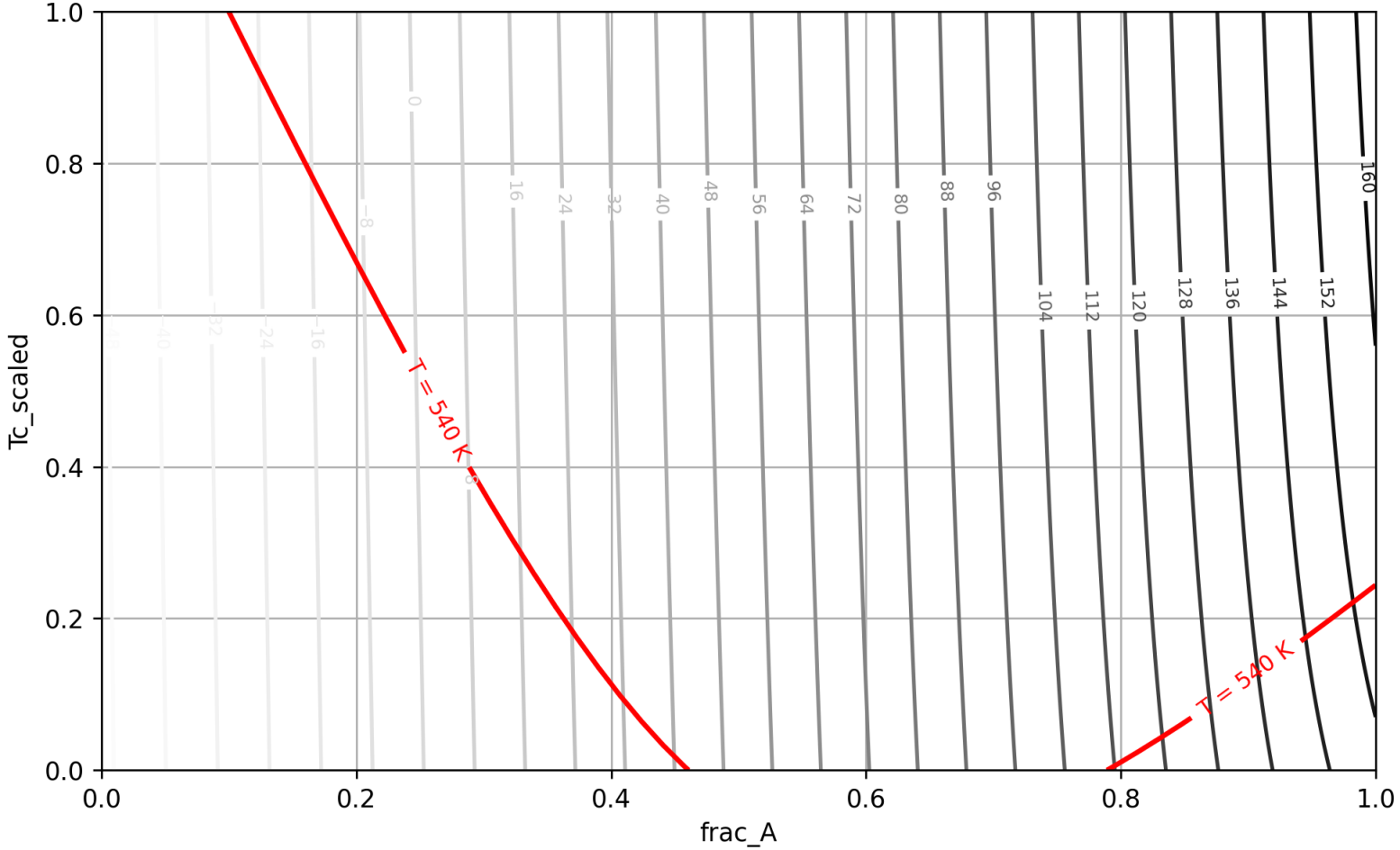


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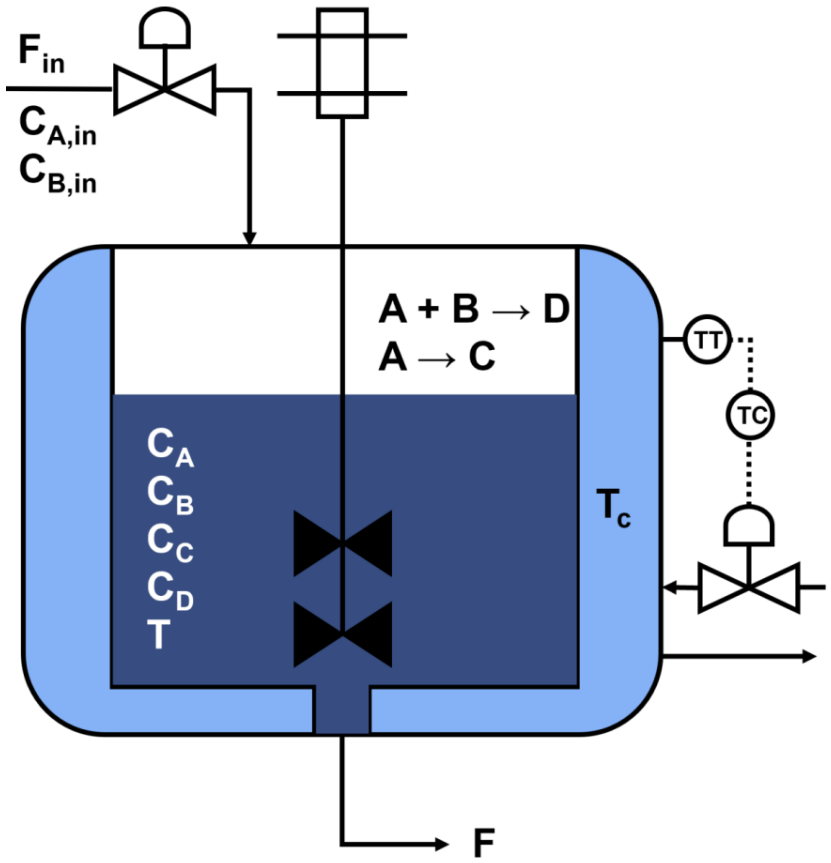
```
PRICES = {
    "PA": 5.0,
    "PB": 12.0,
    "PC": 12.0,
    "PD": 2.0,
    "PHEAT":
0.001,
}
```



AI for autonomy: AI for “safe” active learning



AI for autonomy: AI for “safe” active learning




```python
# — physical constants & geometry ————————————————————————————
R, rho, Cp, UA, V = 8.314, 1_000.0, 2_000.0, 550.0, 1.0
W_A, W_s, T_feed = 100.0, 80.0, 350.0              # feeds
MW_A, MW_I = 0.100, 0.165
A_d, E_d, A_p, E_p, A_t, E_t = 1e9, 125e3, 4e4, 25e3, 1e6, 15e3
ΔH_rxn = 100e3                                      # J mol⁻¹

# — scaling constants (inputs & states) ——————————————————————————
TC_MIN, TC_MAX = 280.0, 400.0                       # K
I_MIN,  I_MAX  = 0.0,   2.5                         # kg h⁻¹
S_CONC, S_TEMP = 100.0, 350.0                       # state divisors

# — Arrhenius helper ——————————————————————————————————————————
def _k(A, E, T): return A * np.exp(-E / (R * T))

# — scaled ODE system ————————————————————————————————————————
def reactor_odes_scaled(t, y_s, u_I_s, u_Tc_s):
    A, I, R_, P, T = (*y_s[:4] * S_CONC, y_s[4] * S_TEMP)
    f_I = I_MIN + u_I_s * (I_MAX - I_MIN)           # kg h⁻¹
    T_c = TC_MIN + u_Tc_s * (TC_MAX - TC_MIN)       # K

    F = (f_I + W_A + W_s) / (rho*3600)      # m³ s⁻¹, placeholder
    D = F / V

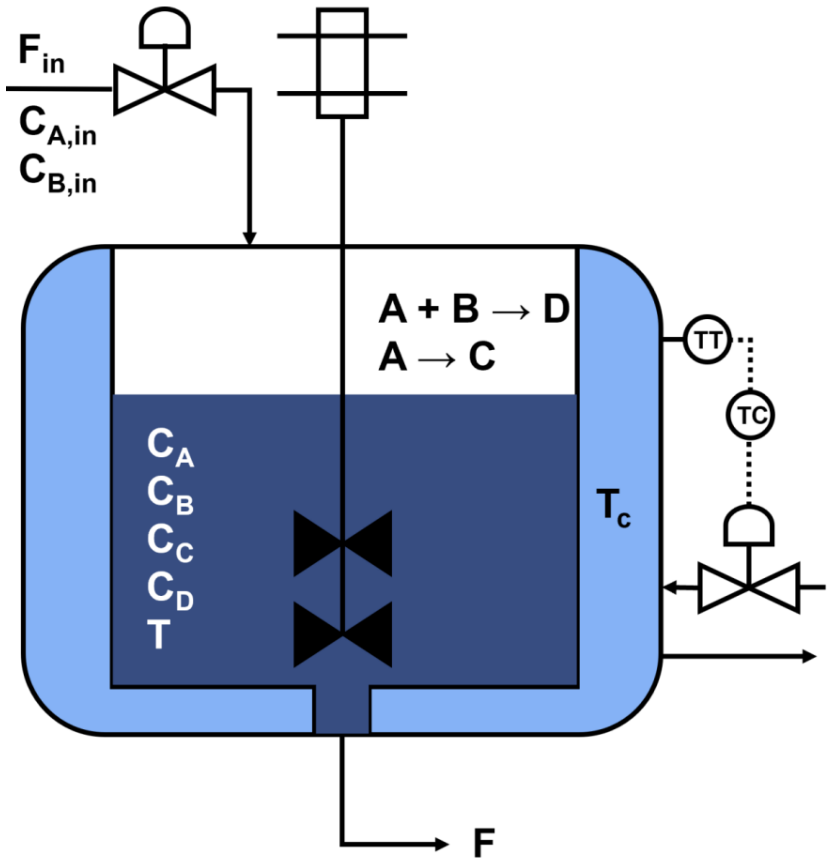
    k_d, k_p, k_t = _k(A_d, E_d, T), _k(A_p, E_p, T), _k(A_t, E_t, T)
    r_d = k_d * I / MW_I
    r_p = k_p * R_ * A / MW_A
    r_t = k_t * R_**2

    dA = (W_A/3600)/V - D*A - r_p*MW_A
    dI = (f_I/3600)/V - D*I - r_d*MW_I
    dR =  2*r_d - 2*r_t
    dP =  r_p*MW_A - D*P

    Q_rxn  = ΔH_rxn * r_p * V
    Q_cool = UA * (T - T_c)
    Q_flow = rho*Cp*F * (T_feed - T)
    dT     = (Q_rxn - Q_cool + Q_flow) / (rho*Cp*V)

    return [dA/S_CONC, dI/S_CONC, dR/S_CONC, dP/S_CONC, dT/S_TEMP]
```

AI for autonomy: AI for “safe” active learning




```python
# — physical constants & geometry ————————————————————————————
R, rho, Cp, UA, V = 8.314, 1_000.0, 2_000.0, 550.0, 1.0
W_A, W_s, T_feed = 100.0, 80.0, 350.0              # feeds
MW_A, MW_I = 0.100, 0.165
A_d, E_d, A_p, E_p, A_t, E_t = 1e9, 125e3, 4e4, 25e3, 1e6, 15e3
ΔH_rxn = 100e3                                      # J mol⁻¹

# — scaling constants (inputs & states) ——————————————————————————
TC_MIN, TC_MAX = 280.0, 400.0                       # K
I_MIN,  I_MAX  =   0.0,   2.5                       # kg h⁻¹
S_CONC, S_TEMP = 100.0, 350.0                       # state divisors

# — Arrhenius helper ——————————————————————————————————————————
def _k(A, E, T): return A * np.exp(-E / (R * T))

# — scaled ODE system ——————————————————————————————————————————
def reactor_odes_scaled(t, y_s, u_I_s, u_Tc_s):
    A, I, R_, P, T = (*y_s[:4] * S_CONC, y_s[4] * S_TEMP)
    f_I = I_MIN + u_I_s * (I_MAX - I_MIN)           # kg h⁻¹
    T_c = TC_MIN + u_Tc_s * (TC_MAX - TC_MIN)       # K

    F = (f_I + W_A + W_s) / (rho*3600)      # m³ s⁻¹, placeholder
    D = F / V

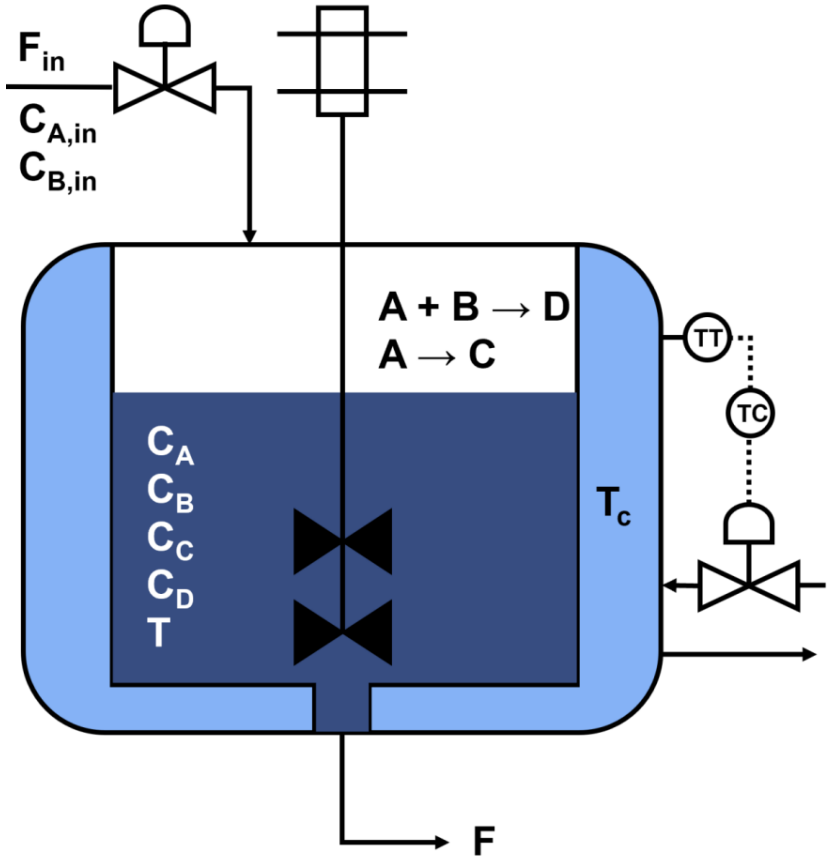
    k_d, k_p, k_t = _k(A_d, E_d, T), _k(A_p, E_p, T), _k(A_t, E_t, T)
    r_d = k_d * I / MW_I
    r_p = k_p * R_ * A / MW_A
    r_t = k_t * R_**2

    dA = (W_A/3600)/V - D*A - r_p*MW_A
    dI = (f_I/3600)/V - D*I - r_d*MW_I
    dR =  2*r_d - 2*r_t
    dP =  r_p*MW_A - D*P

    Q_rxn  = ΔH_rxn * r_p * V
    Q_cool = UA * (T - T_c)
    Q_flow = rho*Cp*F * (T_feed - T)
    dT     = (Q_rxn - Q_cool + Q_flow) / (rho*Cp*V)

    return [dA/S_CONC, dI/S_CONC, dR/S_CONC, dP/S_CONC, dT/S_TEMP]
```

AI for autonomy: AI for “safe” active learning

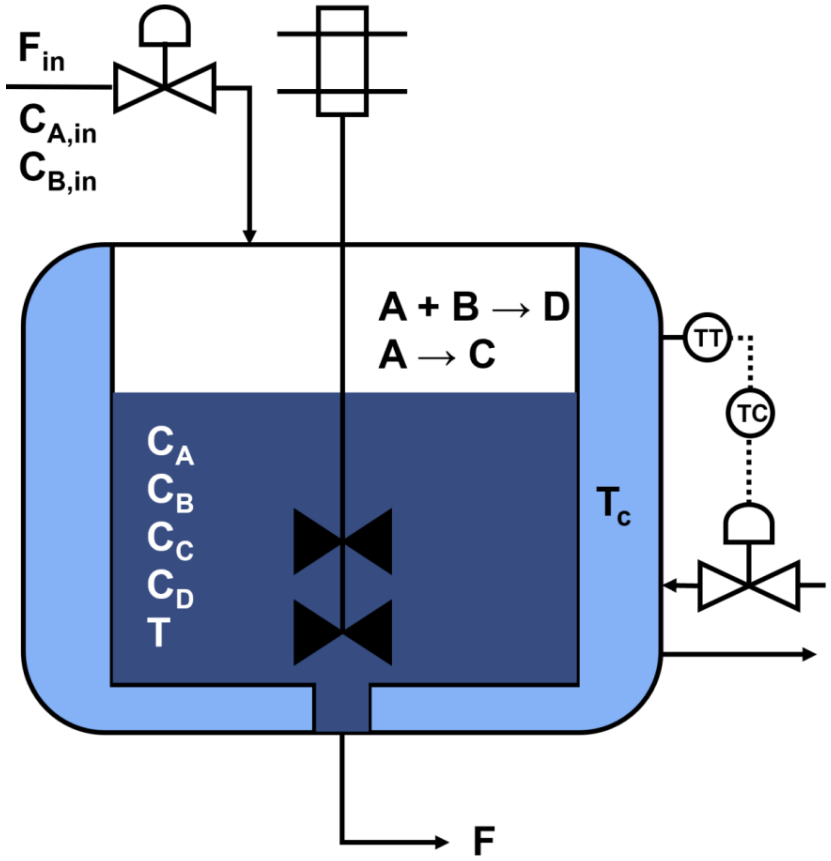


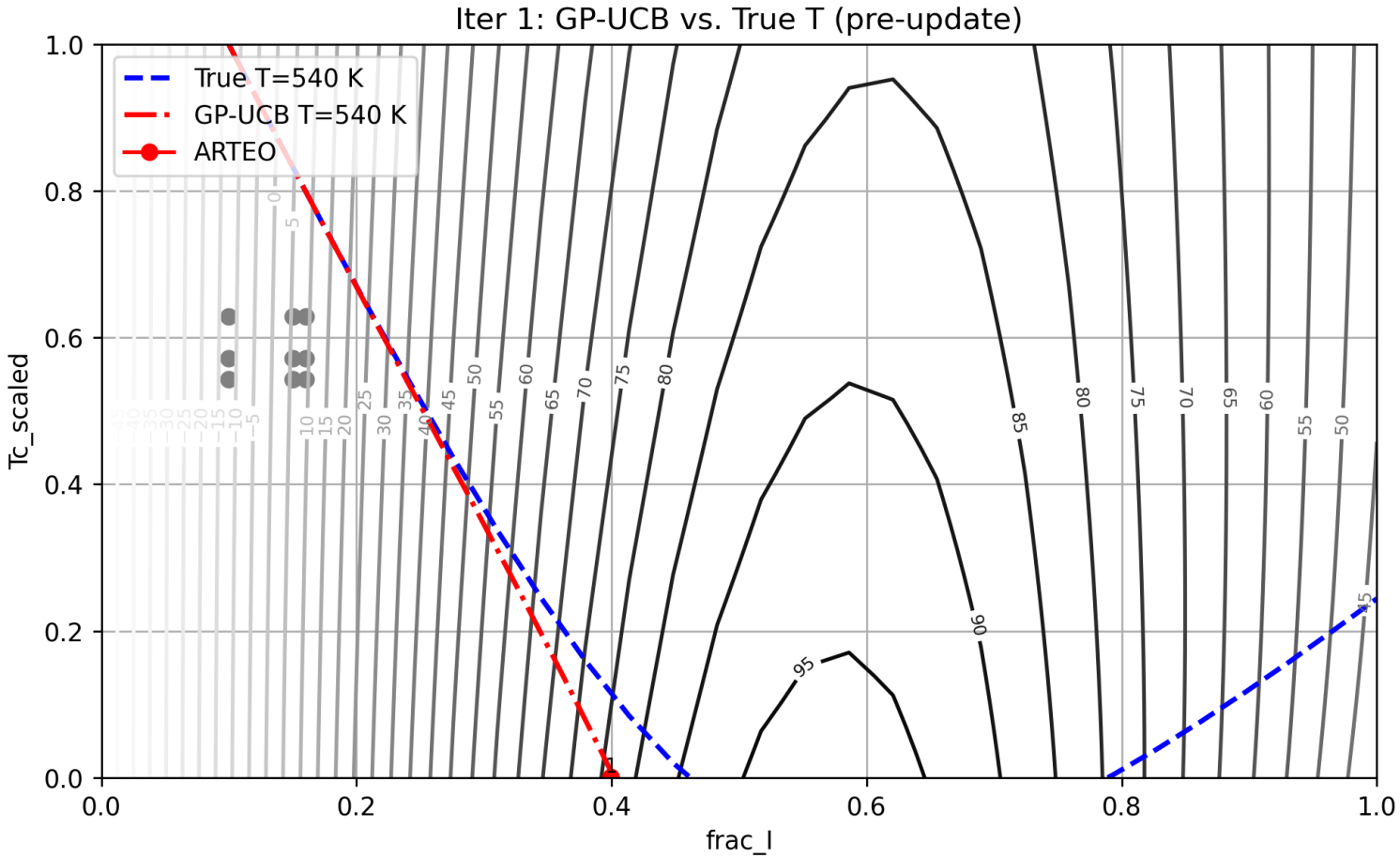
Process model is unknown!

We don’t see where those contours or the T limits are

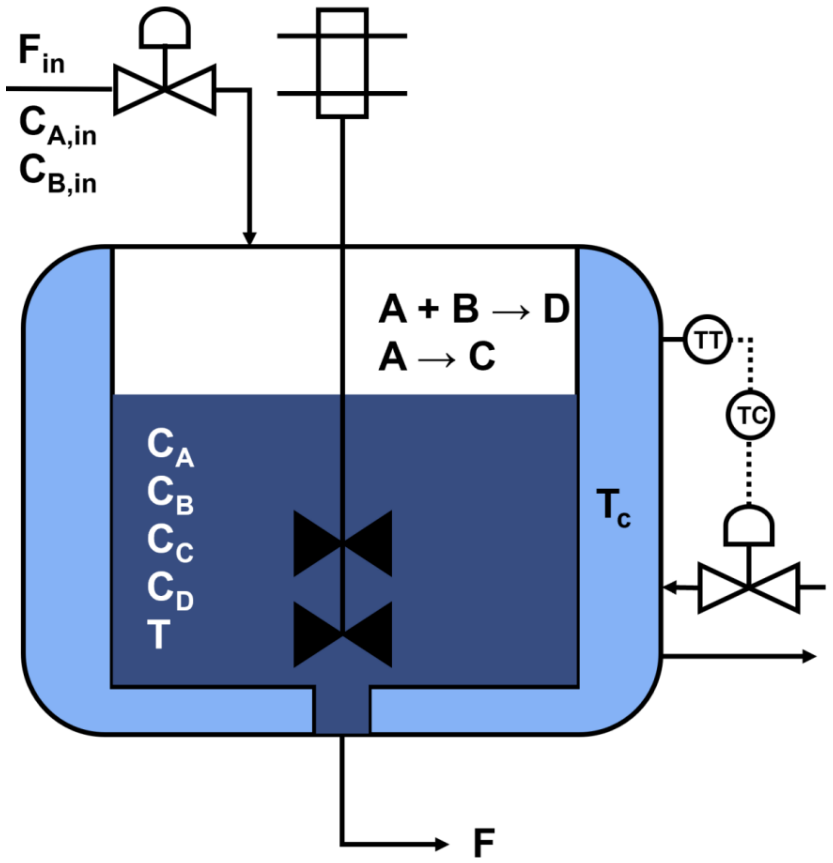
We will learn it as we go (safely!)

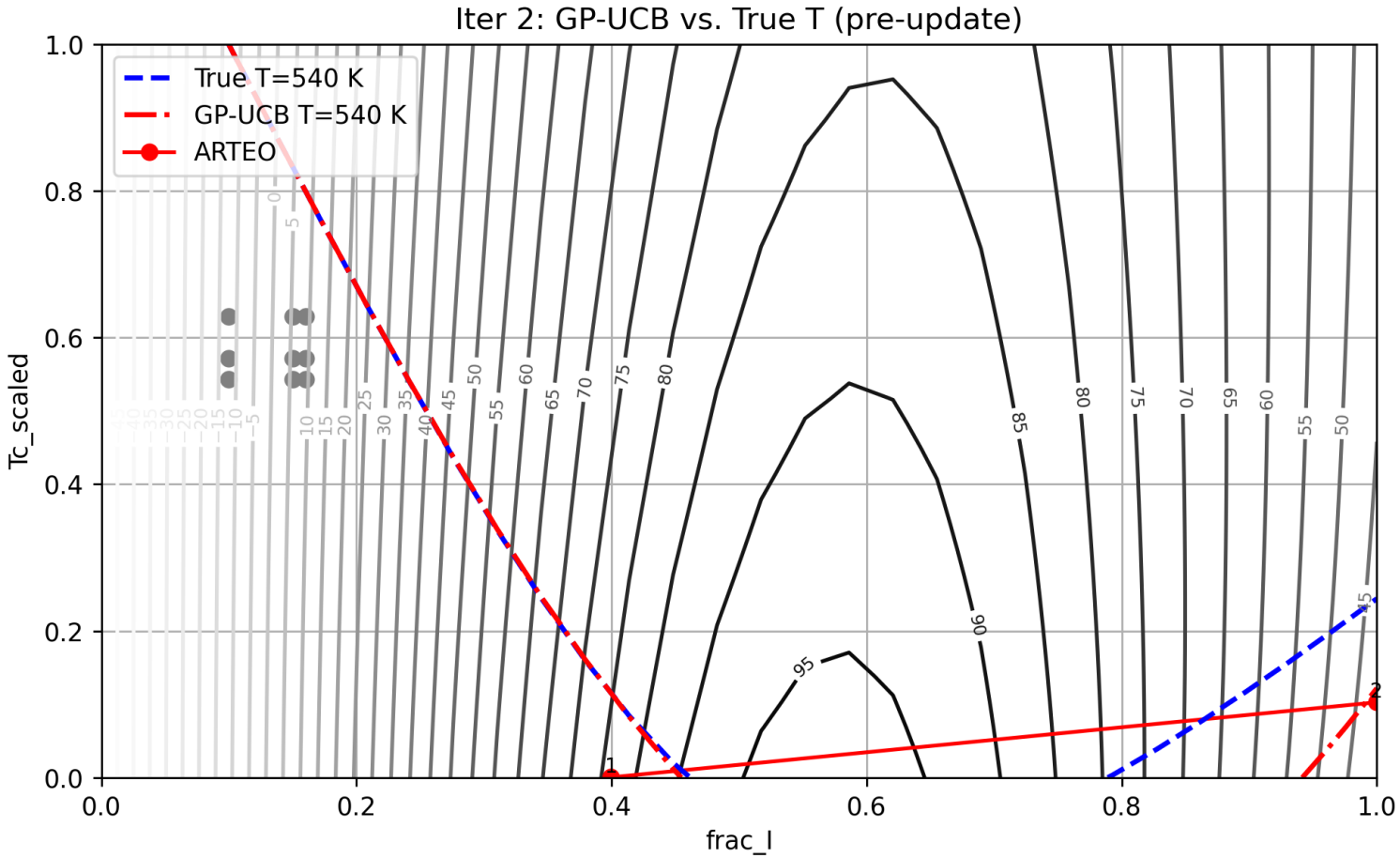
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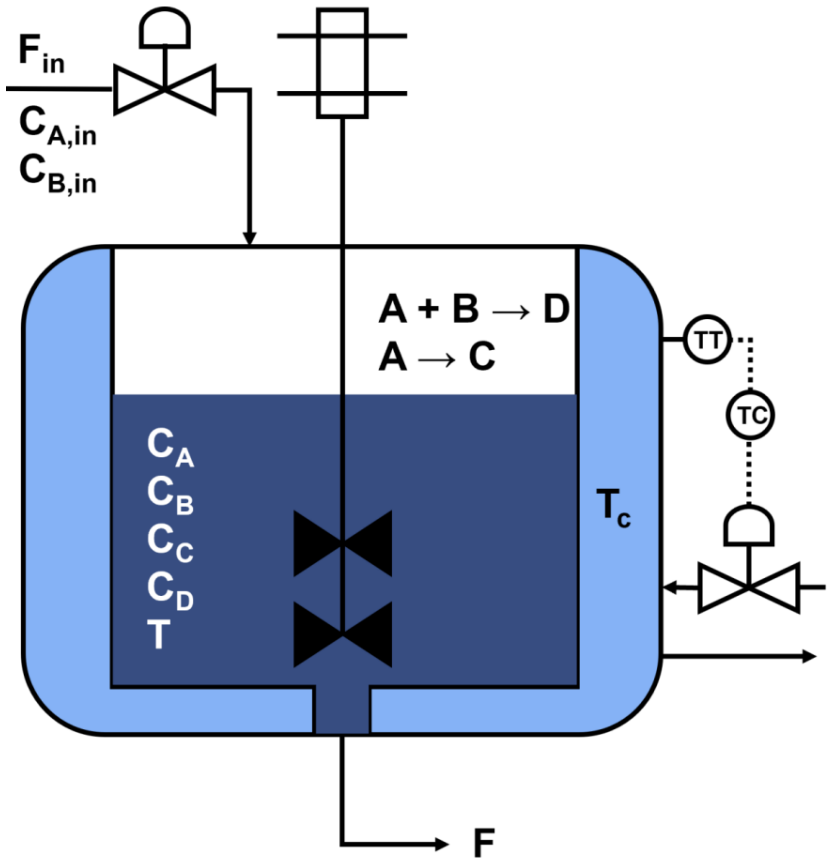


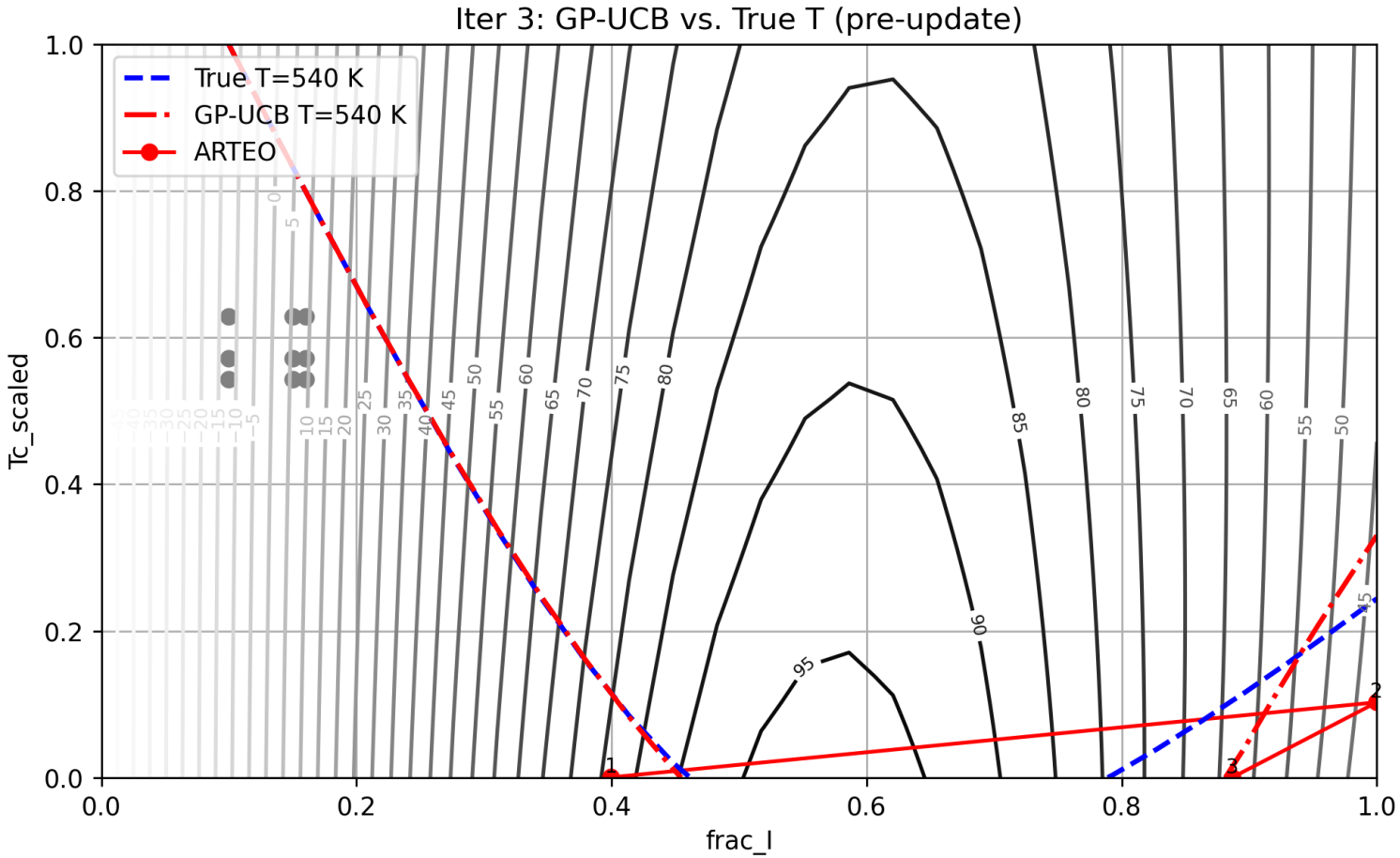
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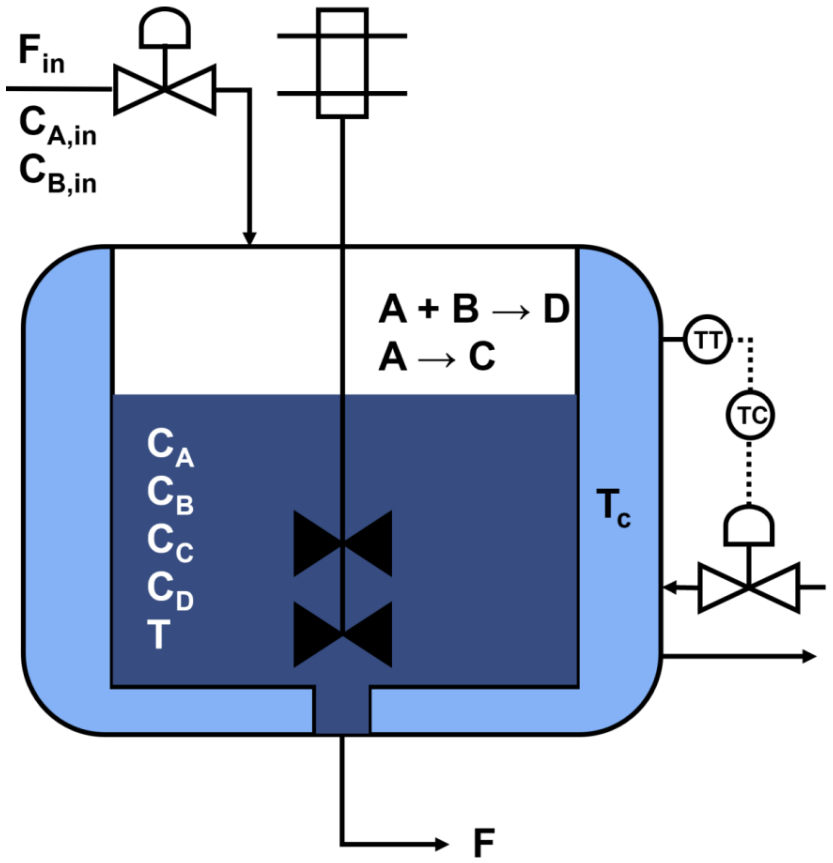


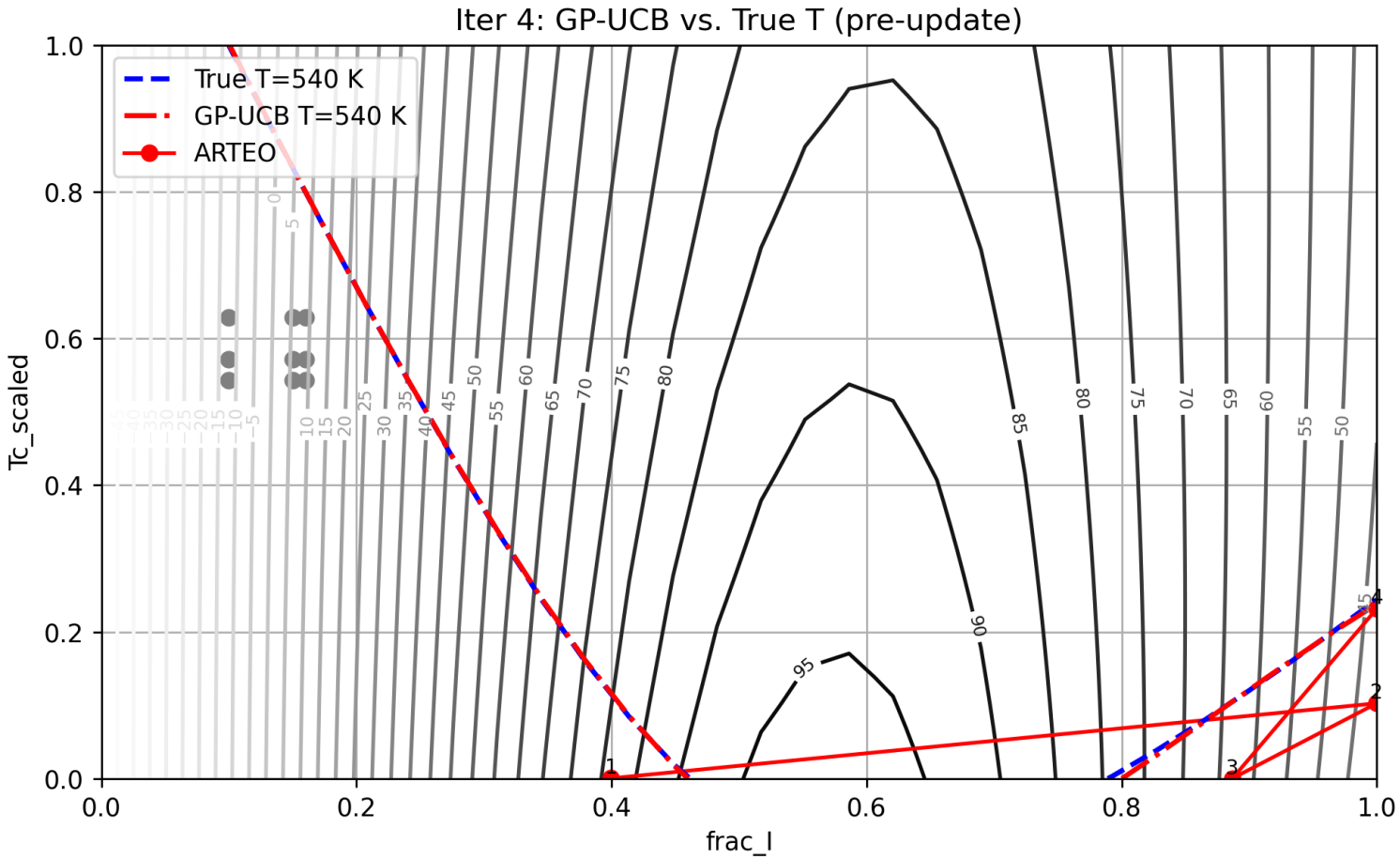
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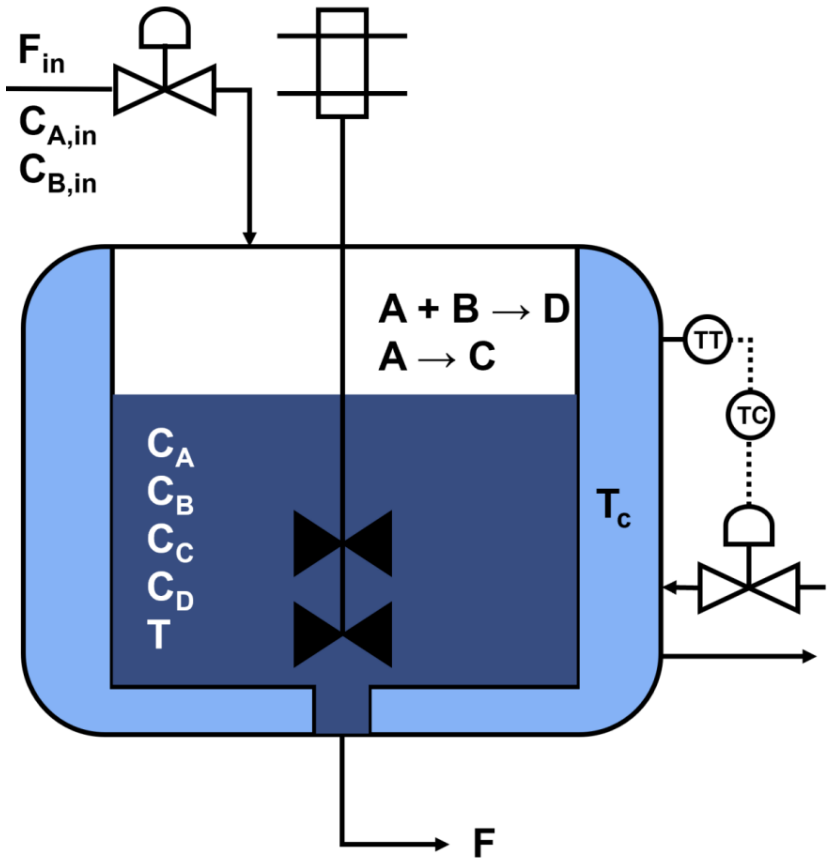


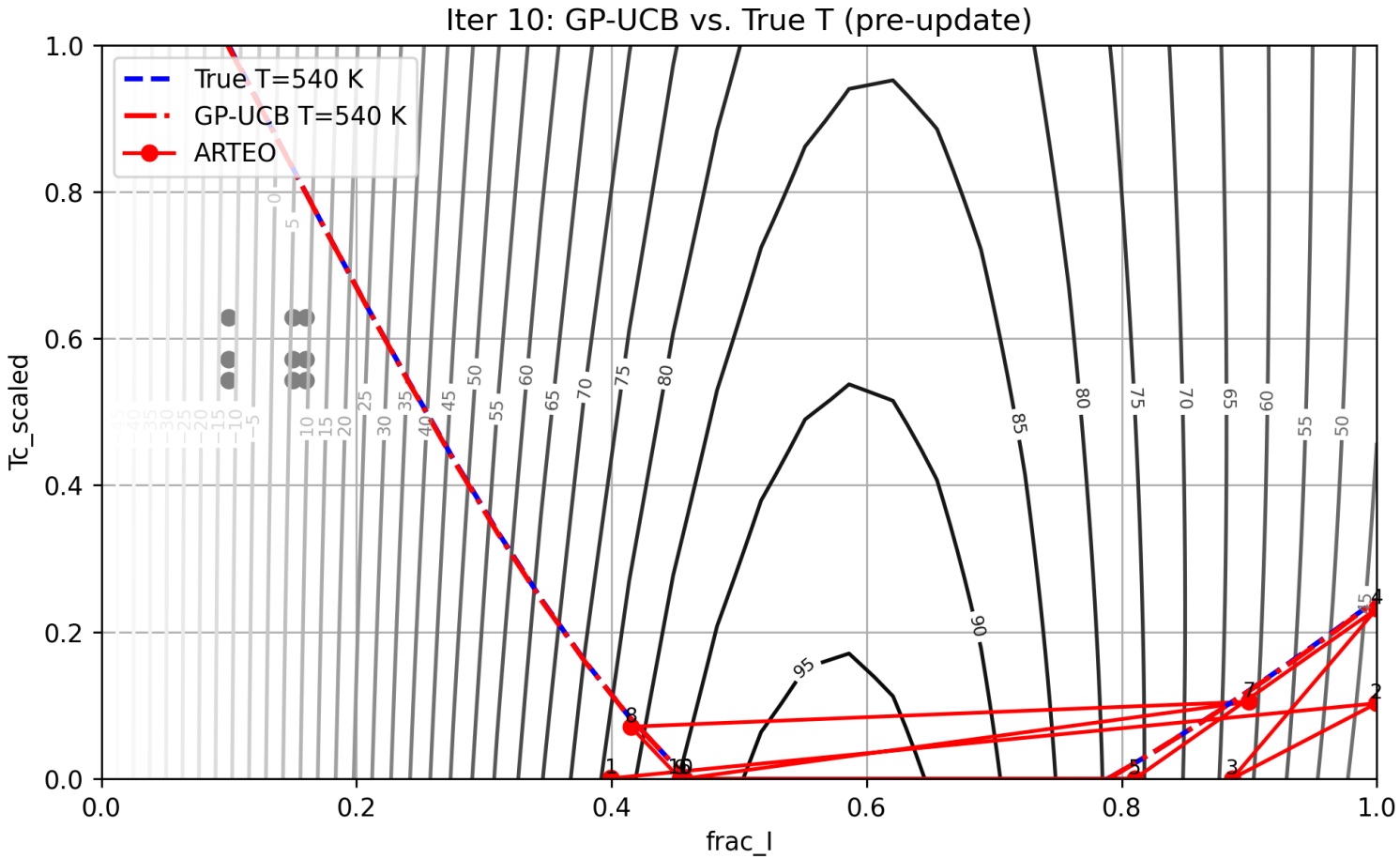
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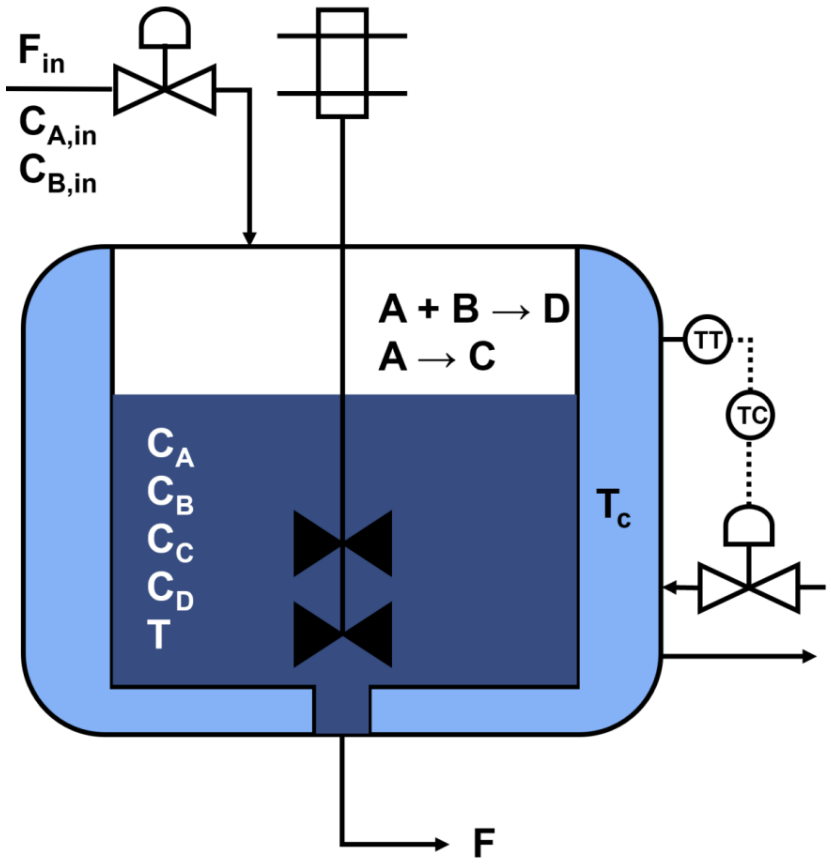


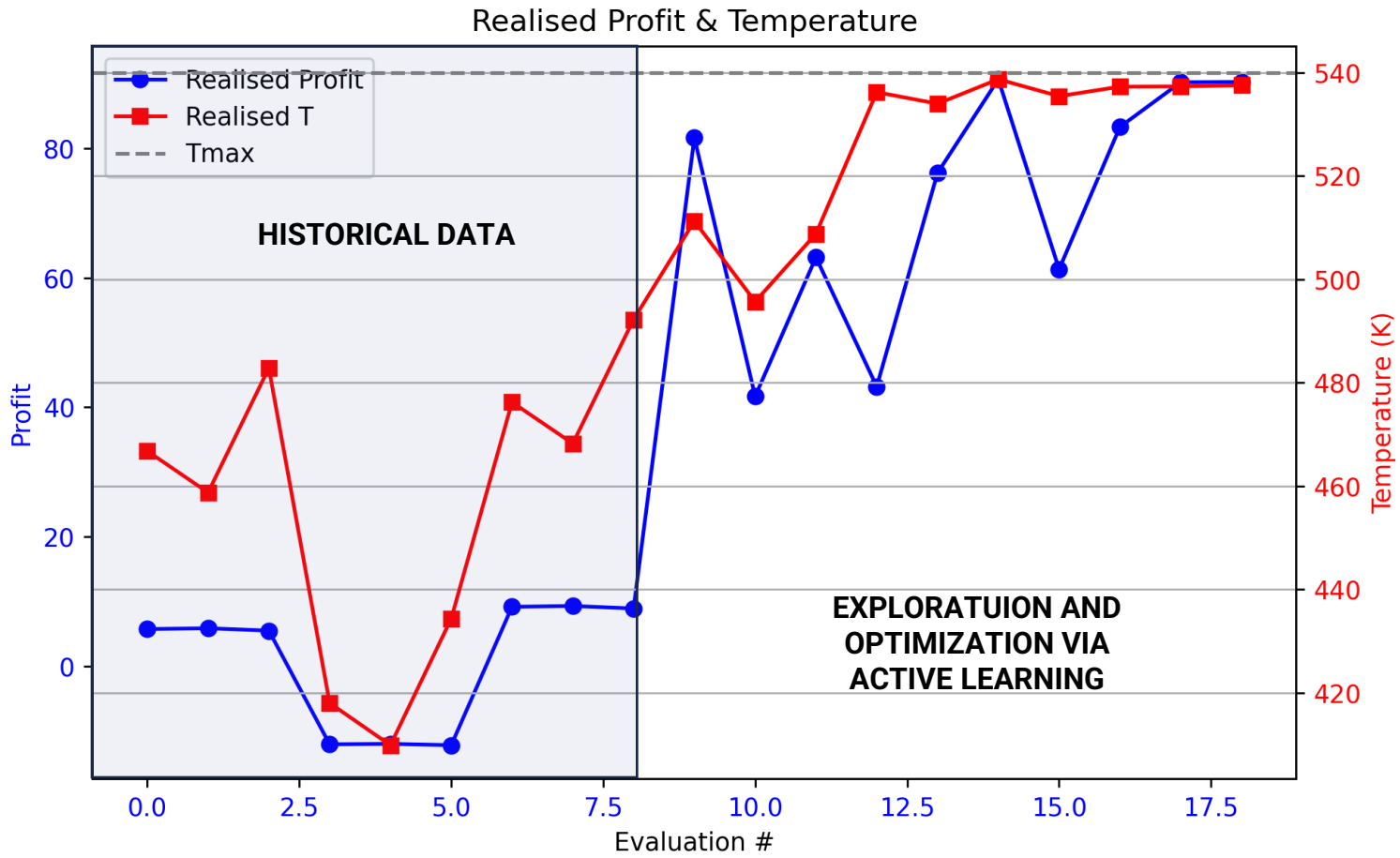
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Enhancing Process Simulation and Optimisation of Liquified Natural Gas Plants with Conditional Generative Models

Alex Durkin
Mehmet Mercangöz

Background

- Liquified natural gas (LNG) production is an energy intensive process involving multistream heat exchangers (MHEXs) which provide great potential for process optimisation, but which are notoriously difficult to model and simulate.
- MHEX process simulations require the specification of dependent process state variables such as tube exit temperatures.
- LNG plant data is complex and highly dimensional, often exhibiting one-to-many mappings due to unknown, unmeasured, or uncertain process characteristics.

Methodology

Conditional Generative Modelling
A conditional variational autoencoder (CVAE) was trained to generate dependent simulation inputs conditioned on the true degrees of freedom.

Generative Simulations
The CVAE was interfaced with a process simulator to provide a realistic simulation environment to represent the real plant.

Adaptive Optimisation and Exploration
Bayesian optimisation (BO) was used to determine optimal steady state operations of the LNG plant.

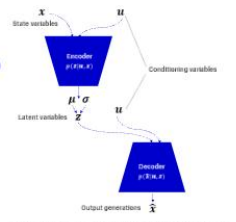
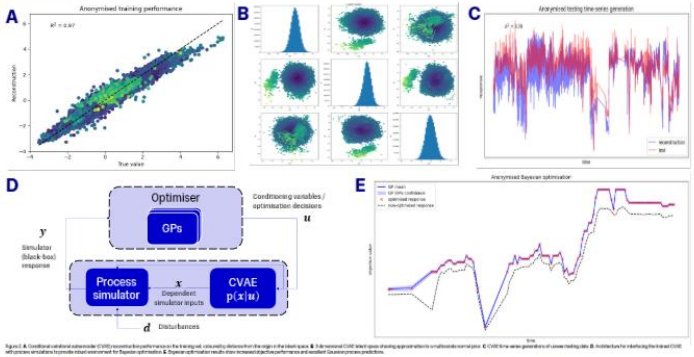


Figure 1: CVAE architecture. During training, the reconstruction loss and Kullback-Leibler (KL) divergence are minimised simultaneously. During conditional inference, outputs are generated from the latent space dependent on the conditioning variables.

Results



Conclusions

- CVAE provides data-driven framework to enhance process simulations and optimisation of complex MHEX processes.
- Enhanced simulation enabled feedback for BO to maximise LNG production efficiency & reduce power consumption.
- Significantly reduced simulation failures compared to optimising using the original simulator specification.

Future Work

- Incorporate more constraints to further improve simulation BO accuracy before deployment.
- Determine the effects of a fully data-driven CVAE simulator – requires physics constraints.
- Include process dynamics in the decision-making problem and compare BO to a reinforcement learning implementation.

Acknowledgements

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Safe Bayesian Optimization with Physics Residuals

Liqiu Dong,
Dr. Mehmet Mercangöz

Motivation

Optimizing chemical reactors requires finding the "sweet spot" (max profit) without violating safety constraints [1][2]

- **Data Scarcity:** Real plant experiments are expensive and time-consuming
- **The Risk:** Standard "Black-Box" Machine Learning (Bayesian Optimization) is blind to physics. It often explores dangerous regions just to learn where the boundaries are

We embed known physical laws (mass/energy balances) into optimization logic to prevent unsafe exploration

Methodology

We modified the standard Acquisition Function to include a **Physics Penalty**. This turns "Black Box" into "Gray Box"

$$\alpha_{PI}(\mathbf{x}) = \underbrace{J(\mathbf{x})}_{\text{Cost}} + \underbrace{\mathrm{E}(\mathbf{x})}_{\text{Exploration}} + \lambda \cdot \underbrace{\mathrm{P}_{\mathbf{safe}}(\mathbf{x})}_{\text{Safety (UCB)}} + \gamma \cdot \underbrace{\mathrm{P}_{\mathbf{phys}}(\mathbf{x})}_{\text{Physics Penalty}}$$

We substitute the GP mean predictions (μ) directly into the system's governing equations. If the GP's predictions are physically consistent, the residual is zero. We penalize deviations from this consistency:

$$\mathrm{P}_{\mathbf{phys}}(\mathbf{x}) = ||\mathcal{F}(\boldsymbol{\mu}_s(\mathbf{x}), \mathbf{x})||$$

- If the model predicts a state that violates physics, $\mathrm{P}_{\mathbf{phys}}$ spikes, discouraging the optimizer from sampling that region

Case Study: Non-isothermal CSTR with two parallel reactions

It is described by a set of 5 coupled Ordinary Differential Equations (ODEs) representing the dynamic behavior, e.g. the energy balance:

$$\frac{dT}{dt} = \frac{F}{V}(T_{in} - T) + \frac{1}{\rho c_p}\sum_{j=1}^{2}(-\Delta H_j)r_j + \frac{UA}{V\rho c_p}(T_c - T)$$

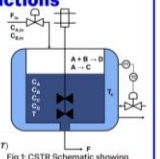


Fig 1: CSTR Schematic showing inputs (A, B) and outputs (C, D, Heat)

Target: Maximize the profit

Constraint: Temperature must stay below **670K**

Results & Discussion

Trajectory Analysis
- We compared **Baseline** vs. **SafeOpt** vs. **Our Algorithm**

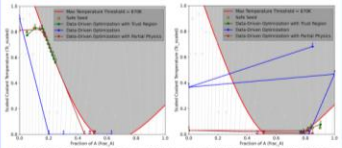


Fig 2: Optimization trajectories overlaid on ground-truth economic profit landscape

Robustness & Convergence
- Tested on 15 independent runs with random seed pairs

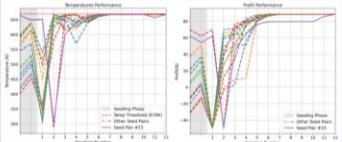


Fig 3: Temperature and Profit performance profiles

Residual Evolution
- Shown for a case with four seed points

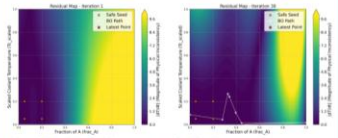


Fig 4: Global residual evolution over ground-truth landscape

Conclusion & Outlook

- This framework allows for **safe, real-time optimization** of chemical processes even with very limited initial data
- **Future Work:** Application to industrial systems with uncertain parameters or measurement noise.

References

[1] Sui, Y., Gotovos, A., Burdick, J., and Krause, A. (2015).Safe Exploration for Optimization with Gaussian Pro-cesses. In Proceedings of the 32nd International Confer-ence on Machine Learning, 997–1005. PMLR.
[2] Dong, L., Zagorowska, M., Liu, T., Durkin, A., and Mercangöz, M. (2024). PI-cof: A bilevel optimization framework for solving active learning problems using physics-information. In 2024 IEEE Conference on Control Technology and Applications (CCTA), pages 101–106. IEEE.

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Pareto Front Guided Sampling for Operator-in-the-Loop real time optimization of Chemical Processes

Samuel Stricker,
Dr. Antonio Del Rio Chanona,
Dr. Mehmet Mercangöz

Motivation

- Optimization in chemical systems can be expensive, gradient-free, and uncertainty-dominated [1].
- Classical BO performs well but offers limited interpretability.
- Expert priors (feasible regions, safety intuition) are rarely incorporated.

Pareto Front Guided Sampling

- Bayesian Optimization based DoE Algorithm for multi-objective optimization [2].
- Constructs Pareto fronts from surrogate model's predicted means and uncertainties.
- Explicit visualization of exploration–exploitation structure and constraint trade-offs:
  - **Objectives**: maximization of mean value
  - **Exploration**: maximization of uncertainty
  - **Constraints**: probabilities of satisfying the constraint
- Enables expert-driven selection and systematic exclusion of unsafe or low-value regions.

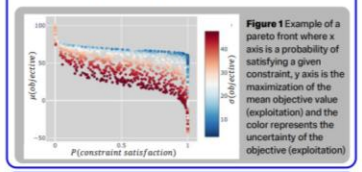


Figure 1 Example of a pareto front where x axis is a probability of satisfying a given constraint, y axis is the maximization of the mean objective value (exploitation) and the color represents the uncertainty of the objective (exploitation)

Case Study: Non-isothermal CSTR

- Nonlinear, exothermic parallel reactions with strict safety bound (Temperature < 540 K).
- Desired product B is produced using A and D, while A reacts with itself to undesired product C.

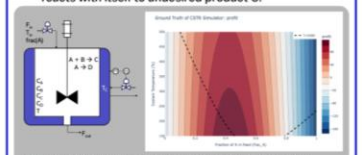


Figure 2 (left) scheme of CSTR reactor and reactions. (right) profit landscape as a function of fraction of A (frac(A)) in feed and cooling temperature (Tc).

Methods

- Training of 5 Gaussian Process regressors for concentrations of *A, B, C, D* and temperature *T*.
- Create Pareto Front using NSGA-II algorithm using the following optimization axis:
  - i. Probability of $T < 540\ K$ (maximization)
  - ii. Mean predicted Profit (maximization)
  - iii. Standard Deviation of predicted Profit (maximization)
- Exclusion of undesired experiments (automated):
  - i. $P(T < 540) > 0.8$
  - ii. $\mu(Profit) > \max(Profit(Experiments))$
  - iii. If possible: $\sigma(Profit) > 0.25 * \max(\sigma(Profit))$
- Sample experiment with min minimum L2 distance to Utopia Point (best of value of each objective)

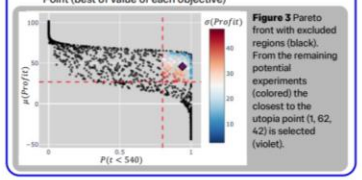


Figure 3 Pareto front with excluded regions (black). From the remaining potential experiments (colored) the closest to the utopia point (1, 62, 42) is selected (violet).

Results & Discussion

- Four start points on left/right site of constraint barrier
- Both converge to optima without any temperature violation
- Algorithm manages to jump the constraint barrier with low risk of constraint violation

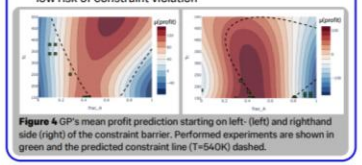


Figure 4 GP's mean profit prediction starting on left- (left) and righthand side (right) of the constraint barrier. Performed experiments are shown in green and the predicted constraint line (T=540K) dashed.

Outlook

- Human-in-the-loop: more sample efficient
- Multi-Fidelity optimization: to adhere to cost/time constraints

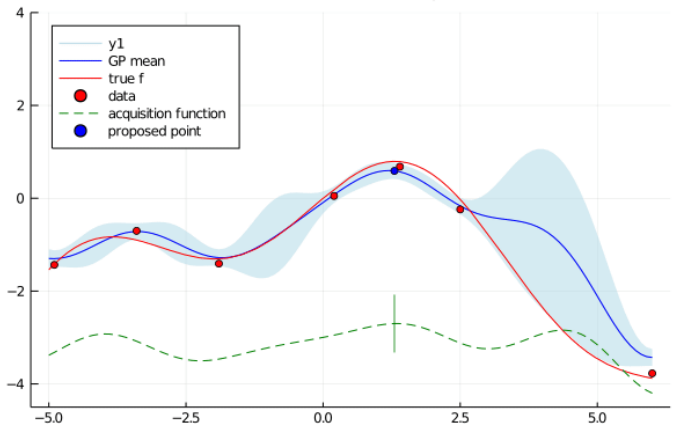
References

[1] T. Savage, E. A. d. R. Chanona, Human-Algorithm Collaborative Bayesian Optimization for Engineering Systems, 2024.
[2] S. Stricker, MSc. Thesis, ETH Zurich, Zurich, 2024.

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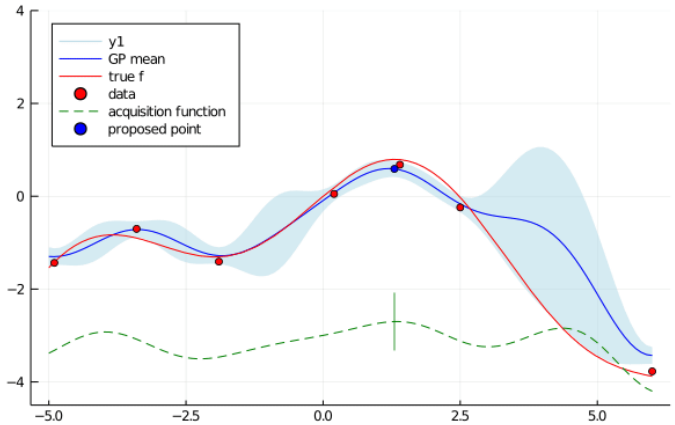
IMPERIAL UCL

AI for autonomy

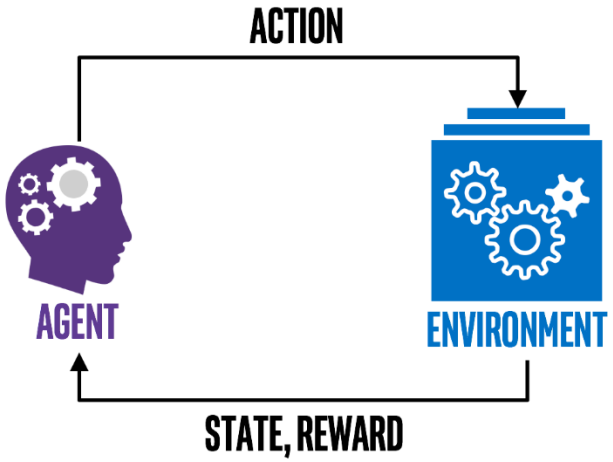


Learning a model
(+ model-based algorithms)

AI for autonomy



Learning a model
(+ model-based algorithms)

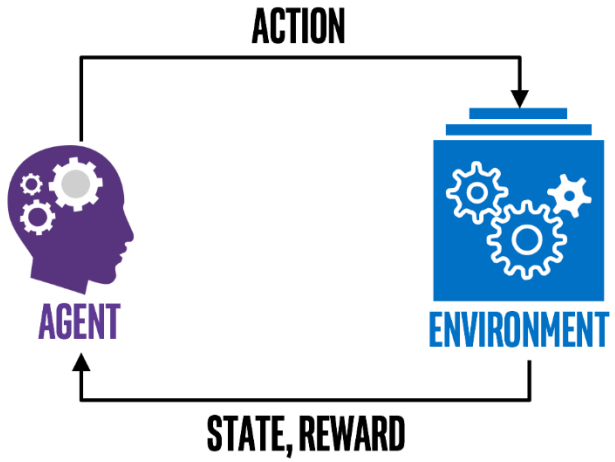


Learning a policy
(reinforcement learning)



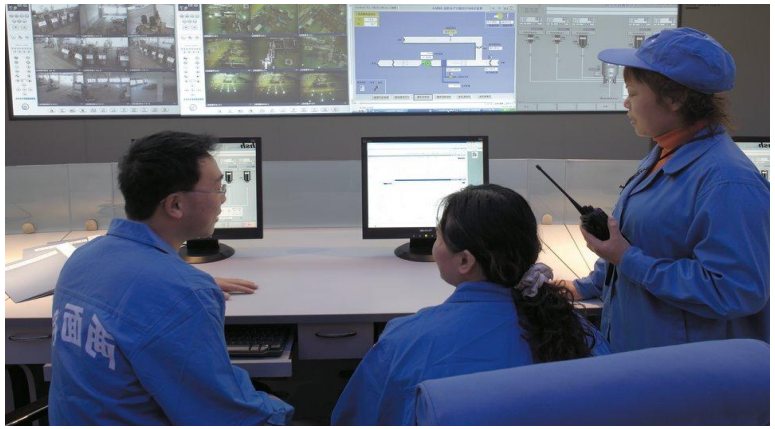
Learning how to mimic
human reasoning?

AI for autonomy



Learning a policy
(reinforcement learning)

Humans in the loop



PERFORMANCE IMPROVEMENT

Automating startup procedures can provide numerous benefits

To paraphrase a famous quote from the political arena, a dollar here, a dollar there, pretty soon you're talking about real money. And so it is with automating combined-cycle startup procedures. It doesn't take a genius to realize that the more times you start up the unit, the more opportunities you have to avoid losing money. And who isn't experiencing more unit starts these days?

If you still rely on manual startups with sign-off paper sheets, you are probably wasting fuel. Similarly, starts spread among different operators with different preferences and understandings of the boundaries around hot, warm, and cold starts can also lead to inconsistent performance. In other words, the more dynamic the operating modes are day-to-day, hour-to-hour, the more chances there are of losses from inconsistent and out-of-date procedures.

Jeff Williams, Emerson Automation Solutions, Power & Water, speaking at the 2017 7F Users Group conference in San Antonio, provided a laundry list of options offered for his company's Ovation™ platform to optimize combined-cycle operations by automating manual procedures, along with a few concrete examples of the monetary benefits to owner/operators.

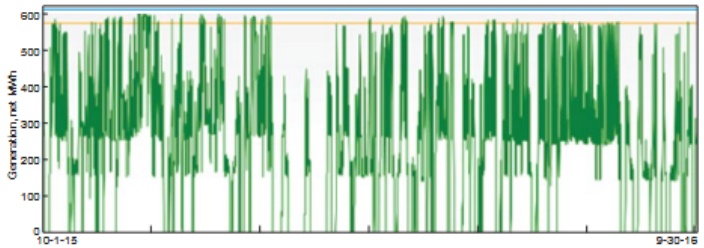
Regarding the latter:

- A southern US utility reduced fuel use by 54% over 55 starts and 290 load transitions, saving over $1-million in a year.
- A mid-Atlantic utility reduced fuel use by 26% over 50 starts, saving $250,000 annually.

Most recently installed facilities likely have a high degree of startup automation built into their controls. So the options up for consideration (table) are probably more relevant to older facilities originally designed for more stable operating regimes—such as baseload.

For example, according to Williams, what is "visible" at these plants are the number of starts from zero, the number of gas-turbine starts, capacity factor, heat rate, and minimum and maximum load points. What typically isn't so visible are the fuel use per start, time to start, starting reliability, dispatch time lost to starting, and AGC (automatic generation control) capability.

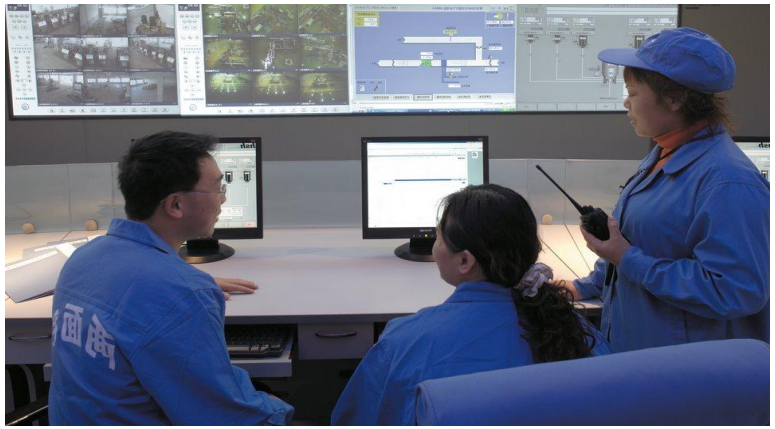
Extracting the added performance and revenue requires two things. First, the right metrics have to be installed. These include fuel used during hot, warm, or cold starts for your combined-cycle configuration; fuel used during load transitions and HRSG steam blending into the main header; instantaneous gas-turbine heat rate and overall plant heat rate; auxiliary (parasitic) megawatt-hour use; and fuel used during shutdown.



Annual load profile (Oct 1, 2015 to Sept 30, 2016) for a 2 × 1 supplementary-fired combined cycle in Colorado reveals dynamic operating requirements. Duct-burner firing can increase output by up to 75 MW

https://www.ccj-online.com/performance-improvement-automating-startup-procedures-can-provide-numerous-benefits/

Humans in the loop

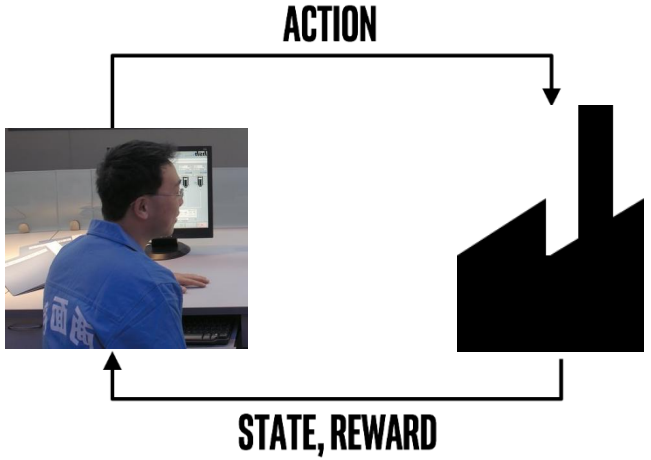


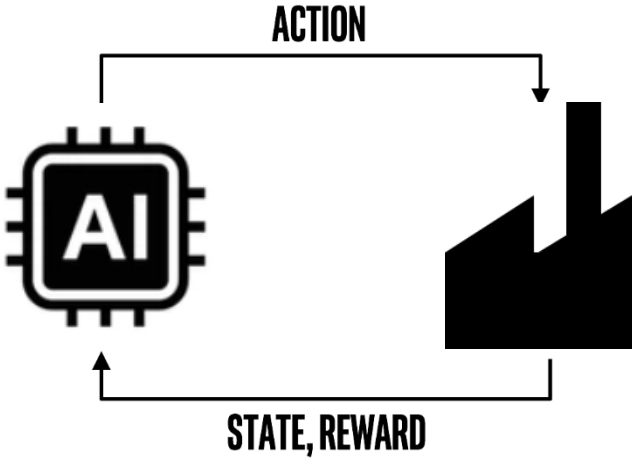
Before introducing the Exapilot operation efficiency improvement software package, this change was manually done by operators. The grade-change operation was manually intensive with 100 - 200 manual adjustments and relied on the use of trend displays because of the multi step construction of the polymerization tank and due to short discharge times. To improve this situation, they initiated a project that would reduce the operator's work-load.

Since the skill level of each operator was different, there was a large variation in transition product volume (+50% to -20%). As a result, there was also a need to quickly optimize operational procedures to avoid an impact to the production schedule. Upon a grade change, it was necessary to quickly match the properties of the product in transit to pre-defined targets, minimize overshoot control and to stabilize the process.

https://web-material3.yokogawa.com/suc-POLYCHEMen.pdf

From humans in the loop to AI agents in the loop





- Data collection / digitalization widespread
- Considerable volume of data from operations (more for grade changes – less for start-ups)

- **Data contains both good average and bad cases…**

- Training on actual operation is infeasible (slow and risky)
- Mechanistic models can be costly to build
- System ID experiments challenging in cases of interest
- **If you have a model, you can also build an MPC…**

Enter offline RL

Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems

Sergey Levine[1,2], **Aviral Kumar**[1], **George Tucker**[2], **Justin Fu**[1]
[1]UC Berkeley, [2]Google Research, Brain Team

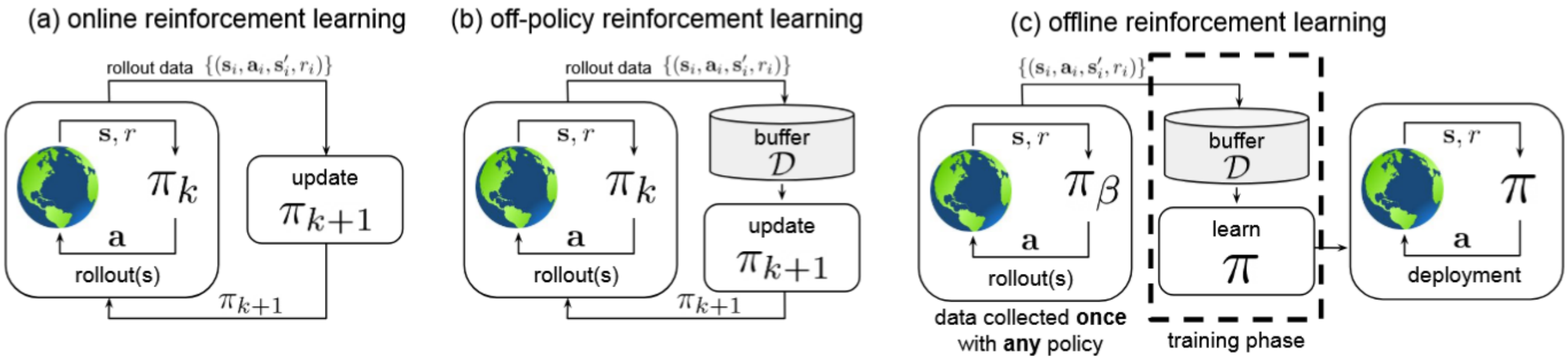
Abstract

In this tutorial article, we aim to provide the reader with the conceptual tools needed to get started on research on offline reinforcement learning algorithms: reinforcement learning algorithms that utilize previously collected data, without additional online data collection. Offline reinforcement learning algorithms hold tremendous promise for making it possible to turn large datasets into powerful decision making engines. Effective offline reinforcement learning methods would be able to extract policies with the maximum possible utility out of the available data, thereby allowing automation of a wide range of decision-making domains, from healthcare and education to robotics. However, the limitations of current algorithms make this difficult. We will aim to provide the reader with an understanding of these challenges, particularly in the context of modern deep reinforcement learning methods, and describe some potential solutions that have been explored in recent work to mitigate these challenges, along with recent applications, and a discussion of perspectives on open problems in the field.

Levine, S., Kumar, A., Tucker, G., & Fu, J. (2020). Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643*.

Enter offline RL

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Main challenge. Updates look for max value actions and if they are not in the dataset → could be that those are actually bad actions to take at that state

Stay close to the data. Constrain the policy to act like the behaviour policy (BC-style regularization, behaviour cloning with advantages, behaviour-regularized offline RL) so it doesn't wander into never-seen actions.

Be pessimistic about what you haven't seen. Push down Q-values for actions that aren't supported by the dataset, so the policy prefers in-dataset actions.

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Safe Deployment of Offline Reinforcement Learning via Input Convex Action Correction

Alex Durkin[1], Jasper Stolte[2], Matthew Jones[2], Raghuraman Pitchumani[3], Bei Li[2], Christian Michler[2], and Mehmet Mercangöz[1]

[1]Department of Chemical Engineering, Imperial College London, SW7 2AZ, UK
[2]Shell Information Technology International BV, 1031 HW Amsterdam, NL
[3]Shell Global Solutions International BV, 1031 HW Amsterdam, NL

Abstract

Offline reinforcement learning (offline RL) offers a promising framework for developing control strategies in chemical process systems using historical data, without the risks or costs of online experimentation. This work investigates the application of offline RL to the safe and efficient control of an exothermic polymerisation continuous stirred-tank reactor. We introduce a Gymnasium-compatible simulation environment that captures the reactor's nonlinear dynamics, including reaction kinetics, energy balances, and operational constraints. The environment supports three industrially relevant scenarios: startup, grade change down, and grade change up. It also includes reproducible offline datasets generated from proportional-integral controllers with randomised tunings, providing a benchmark for evaluating offline RL algorithms in realistic process control tasks.

We assess behaviour cloning and implicit Q-learning as baseline algorithms, offline agents face, including steady-state offsets and degraded performance near issues, we propose a novel deployment-time safety layer that performs gradient-b input convex neural networks (PICNNs) as learned cost models. The PICNN tiable correction of policy actions by descending a convex, state-conditioned cos retraining or environment interaction.

Experimental results show that offline RL, particularly when combined wit can outperform traditional control approaches and maintain stability across al demonstrate the feasibility of integrating offline RL with interpretable and safety stakes chemical process control, and lay the groundwork for more reliable data-dri systems.

HOFLON: Hybrid Offline Learning and Online Optimization for Process Start-Up and Grade-Transition Control

Alex Durkin[1], Jasper Stolte[2], and Mehmet Mercangöz[1]

[1]Department of Chemical Engineering, Imperial College London, SW7 2AZ, UK
[2]Shell Information Technology International BV, 1031 HW Amsterdam, NL

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Start-ups and product grade-changes are critical steps in continuous-process plant operation, because any misstep immediately affects product quality and drives operational losses. These transitions have long relied on manual operation by a handful of expert operators, but the progressive retirement of that workforce is leaving plant owners without the tacit know-how needed to execute them consistently. In the absence of a process model, offline reinforcement learning (RL) promises to capture—and even surpass—human expertise by mining historical start-up and grade-change logs, yet standard offline RL struggles with distribution-shift and value-overestimation whenever a learned policy ventures outside the data envelope. We introduce HOFLON (Hybrid Offline Learning + Online Optimization) to overcome those limitations. Offline, HOFLON learns (i) a latent data manifold that represents the feasible region spanned by past transitions and (ii) a long-horizon Q-critic that predicts the cumulative reward from state–action pairs. Online, it solves a one-step optimization problem that maximizes the Q-critic while penalizing deviations from the learned manifold and excessive rates of change in the manipulated variables. We test HO
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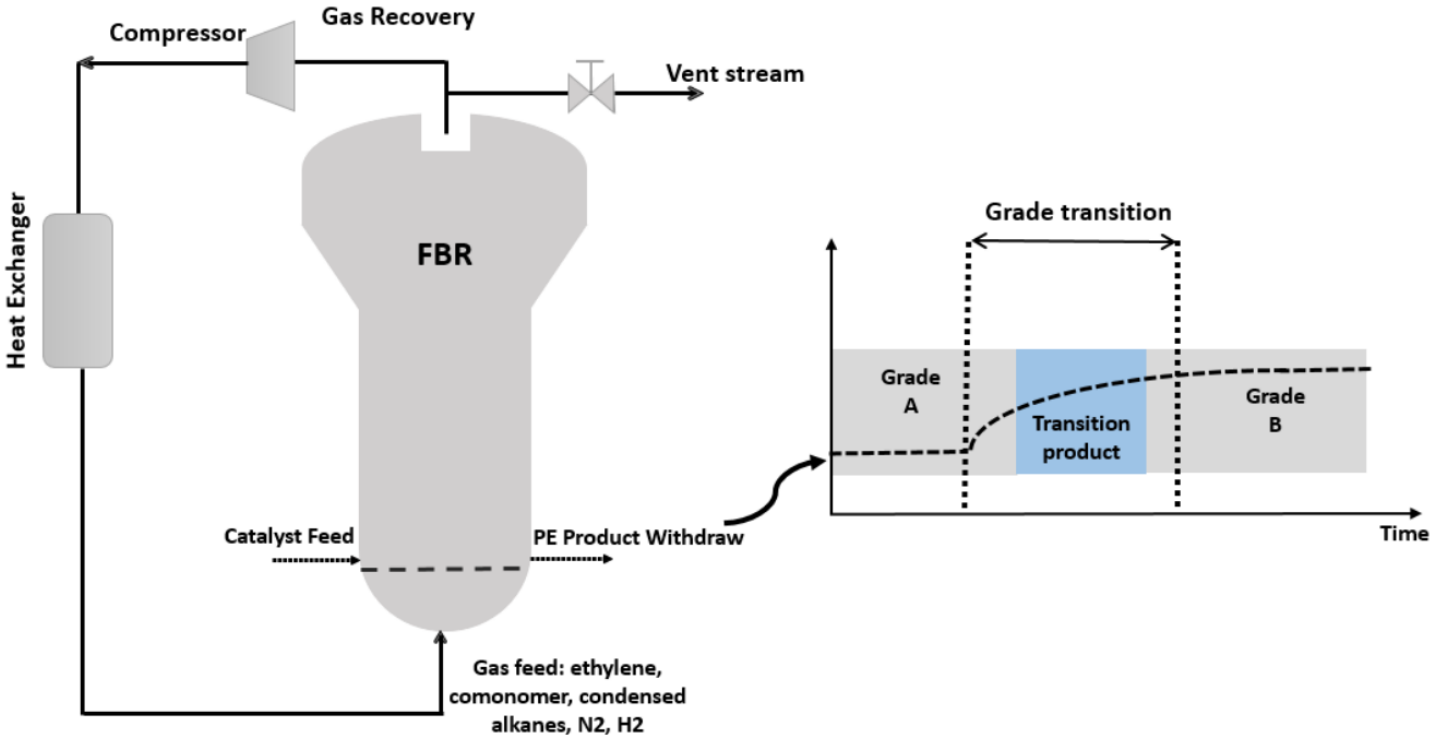
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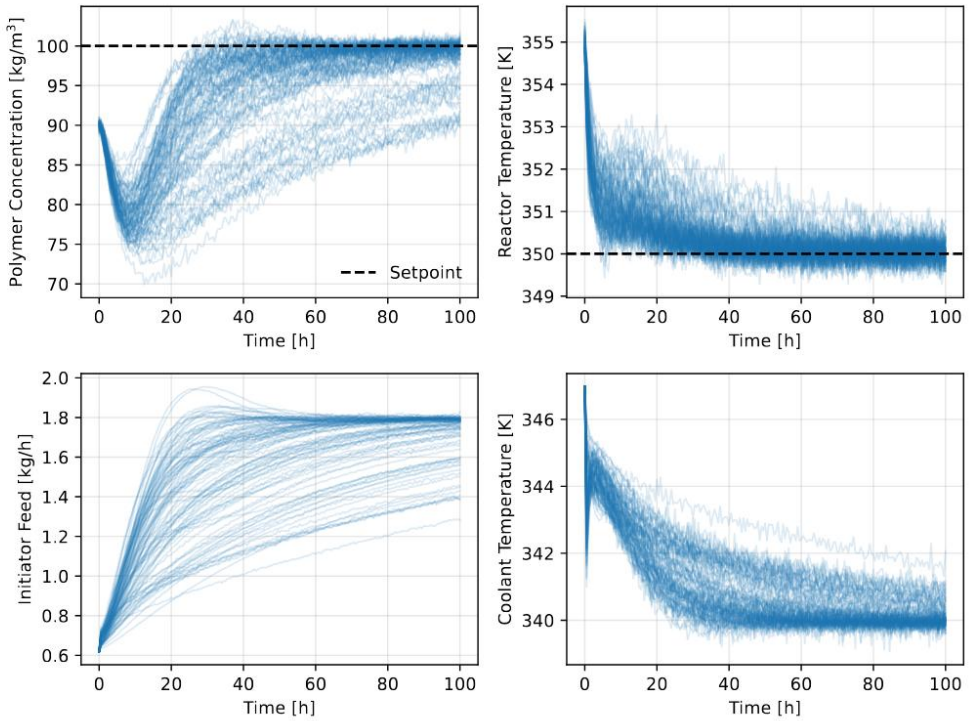
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Learning to do better than a human operator

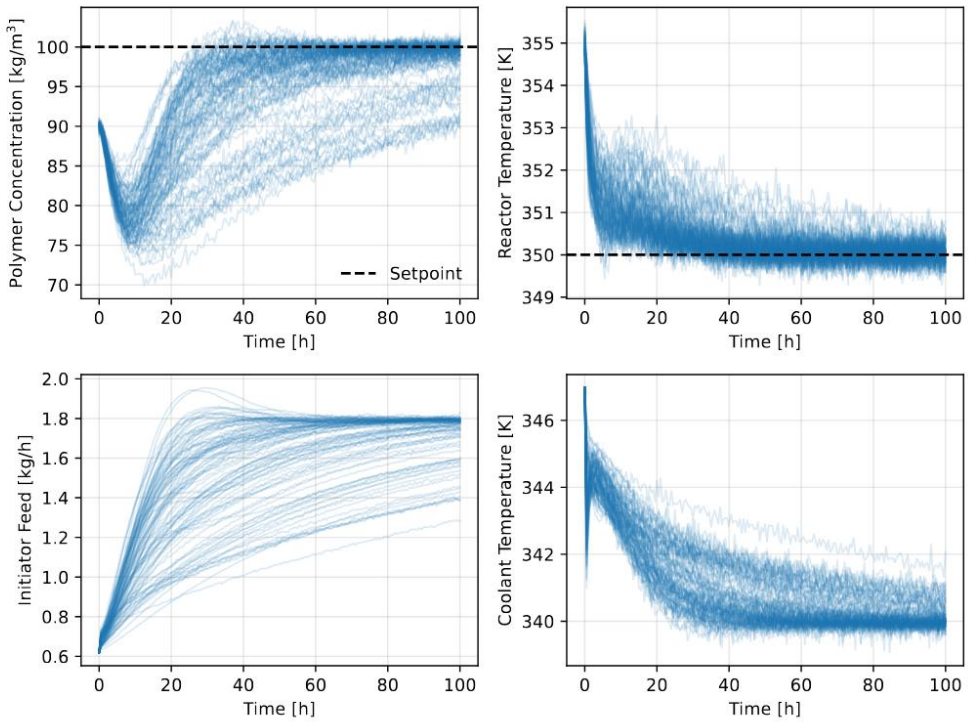




Kardous, S., McKenna, T. F., & Sheibat-Othman, N. (2020). Thermodynamic Effects on Grade Transition of Polyethylene Polymerization in Fluidized Bed Reactors. *Macromolecular Reaction Engineering*, *14*(4), 2000013.

Learning to do better than a human operator

- **Behavioural cloning.** Supervised learning from data. Regress a_t as a function of state s_t
- **Weighted regression.** You can think of assigning a score to each trajectory, use the data from high score trajectories with higher weight
- **Advantage weighted regression.** Define a more structured score definition (reward). Calculate an advantage to individual actions (not a complete trajectory). Do weighted regression



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Q learning

Objective: learn how good it is to take action *a* in state *s* (build a **state action value function**)

Update: take an action a_t at state s_t → observe a reward r_t and next state s_{t+1} and then update Q

$$Q_{\text{new}}(s_t, a_t) = Q_{\text{old}}(s_t, a_t) + \alpha\left[r_t + \gamma \max_{a_{t+1}\in\mathcal{A}} Q_{\text{old}}(s_{t+1}, a_{t+1}) - Q_{\text{old}}(s_t, a_t)\right]$$

- s_t: state at time t
- a_t: action taken at time t
- r_t: reward observed right after taking a_t
- s_{t+1}: next state
- α: learning rate
- γ: discount factor

Q learning using a Monte Carlo approach

- Assume we have full episodes:
  $(s_0, a_0, r_0), (s_1, a_1, r_1), \ldots, (s_T, a_T, r_T)$
- For each time step t, compute the return

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{T-t} r_T$$

- Treat each visited pair as a supervised sample:

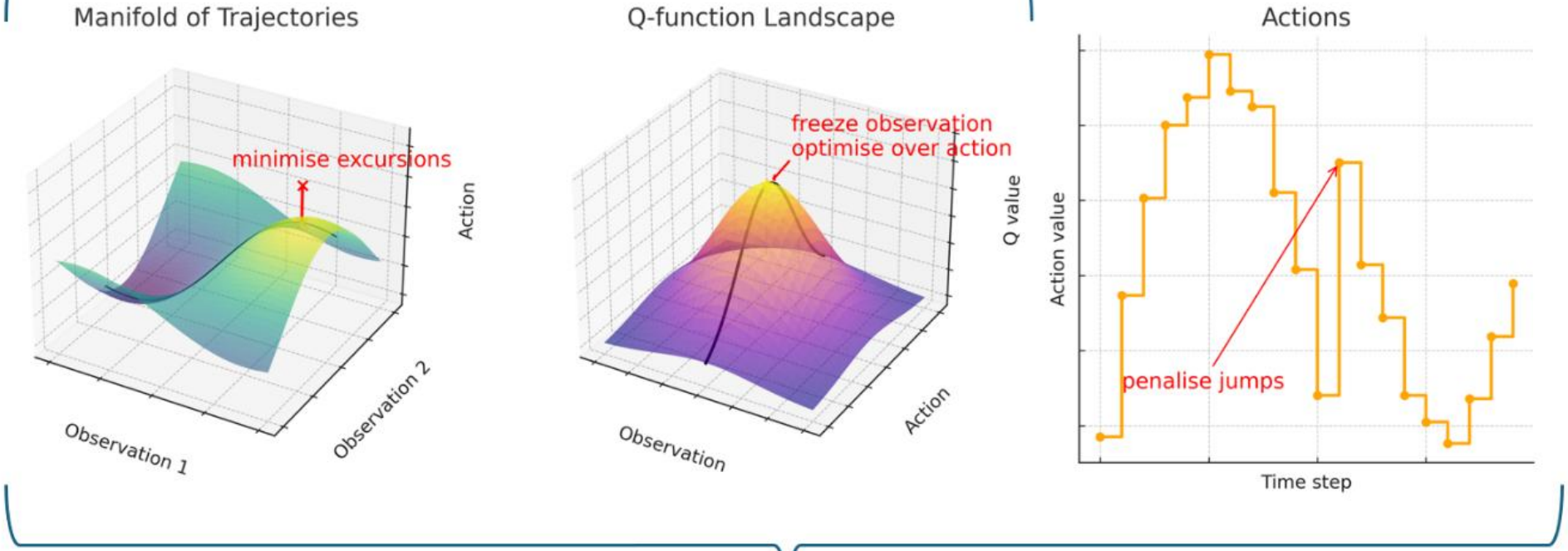
$$(s_t, a_t) \;\Rightarrow\; G_t$$

- Train a function approximator (network) to map

$$(s, a) \mapsto Q(s, a)$$

HOFLON

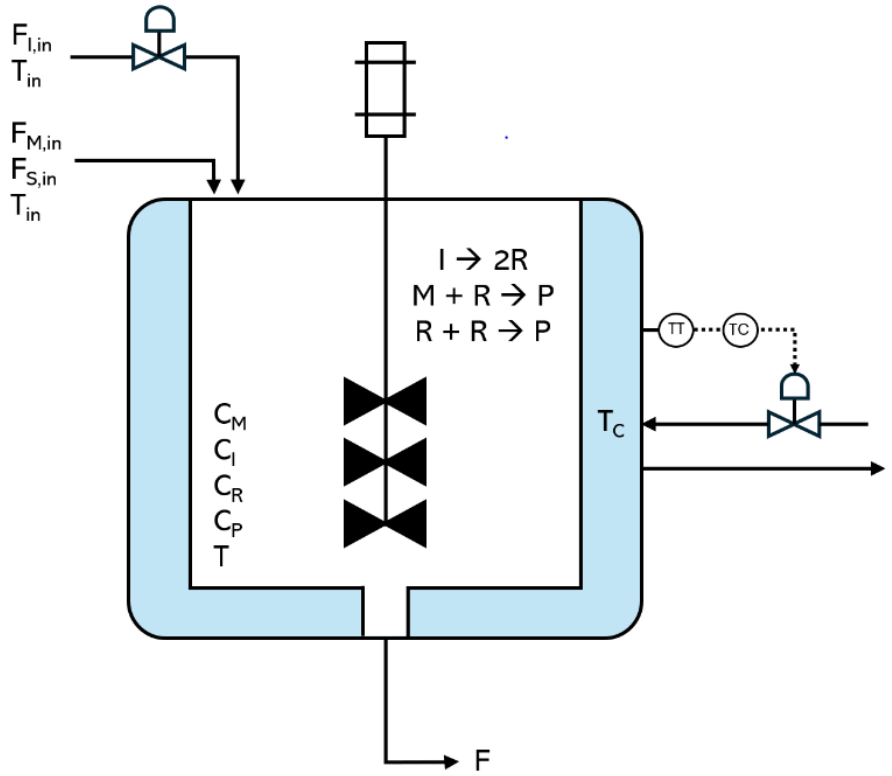
Learn offline



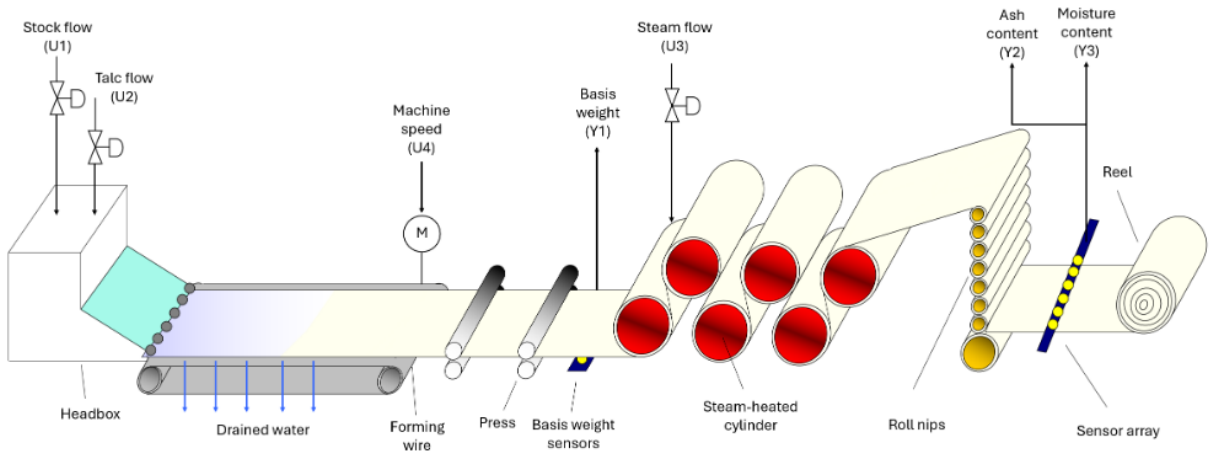
Optimize online

$$a_k^\star = \arg\max_{a\in[0,1]^m} \left[\hat{Q}_\theta(s_k, e_k, I_k, a) - \lambda_1 \phi_{\text{lat}}(s_k, a) - \lambda_2 \|a - a_{k-1}\|_2^2\right]$$

HOFLON - Case studies

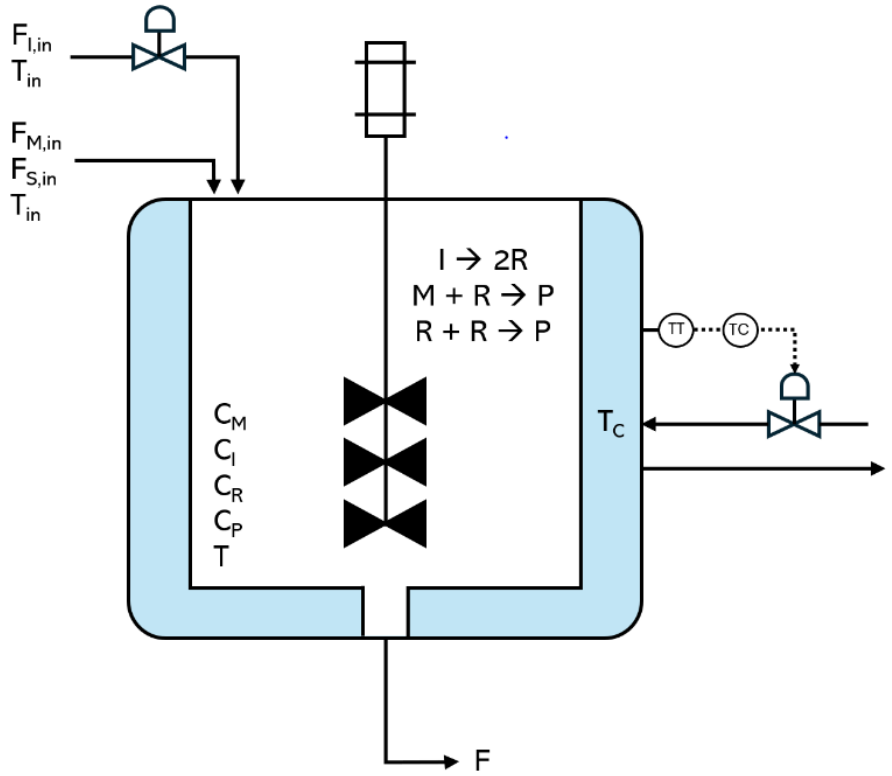


- **Polymerization reactor**

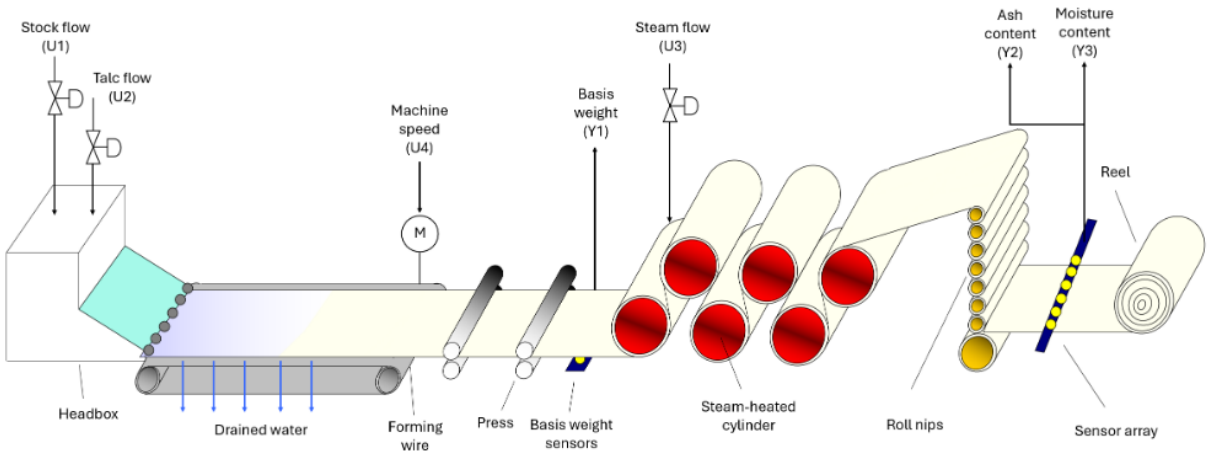


- **Paper machine**

HOFLON - Case studies

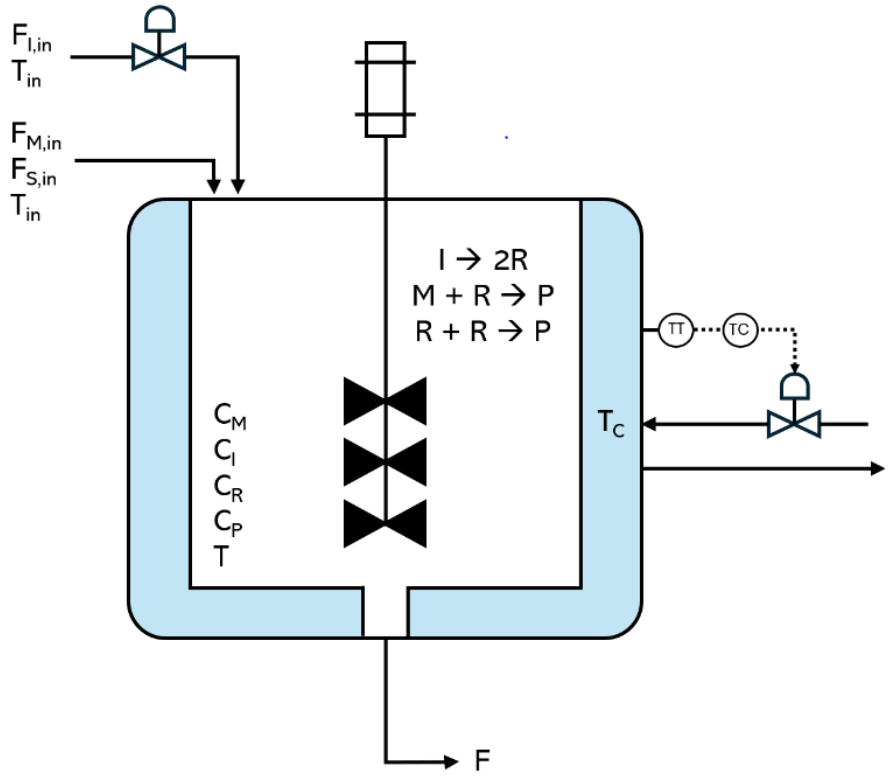


- **2 MVs and 2 CVs**

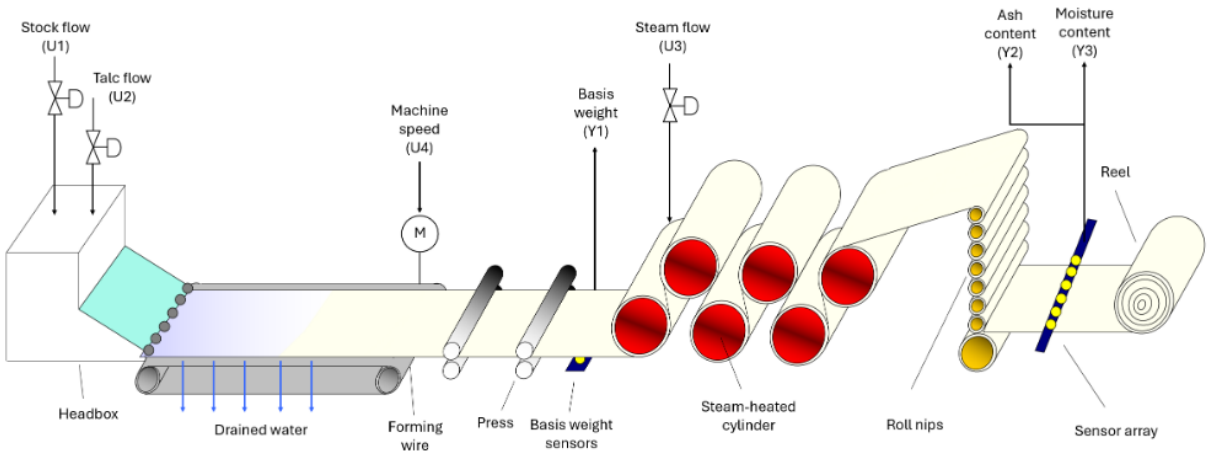


- **4 MVs and 3 CVs**

HOFLON - Case studies

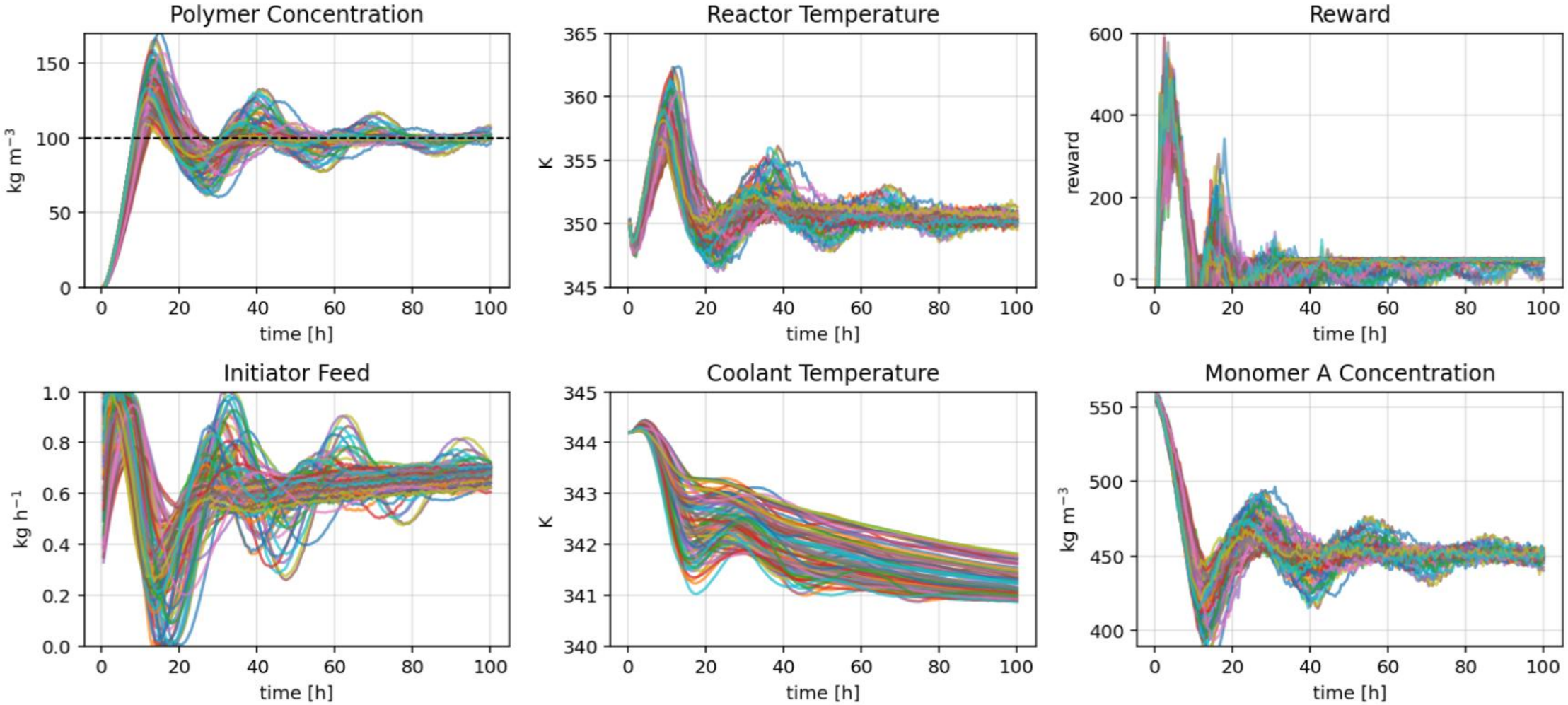


Start up



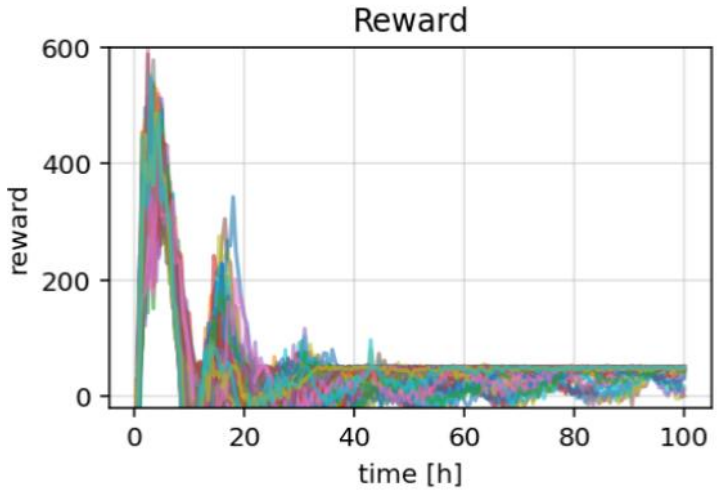
Paper grade	A	B	C	D	E	F	G
Basis weight (g/m^2)	64	49	78	102	63	100	78.5
Ash content (%)	7	7	17	17	9	17	14
Moisture content (%)	9	8	9	9	9	9	9

HOFLON – Polymerization reactor



Historical data

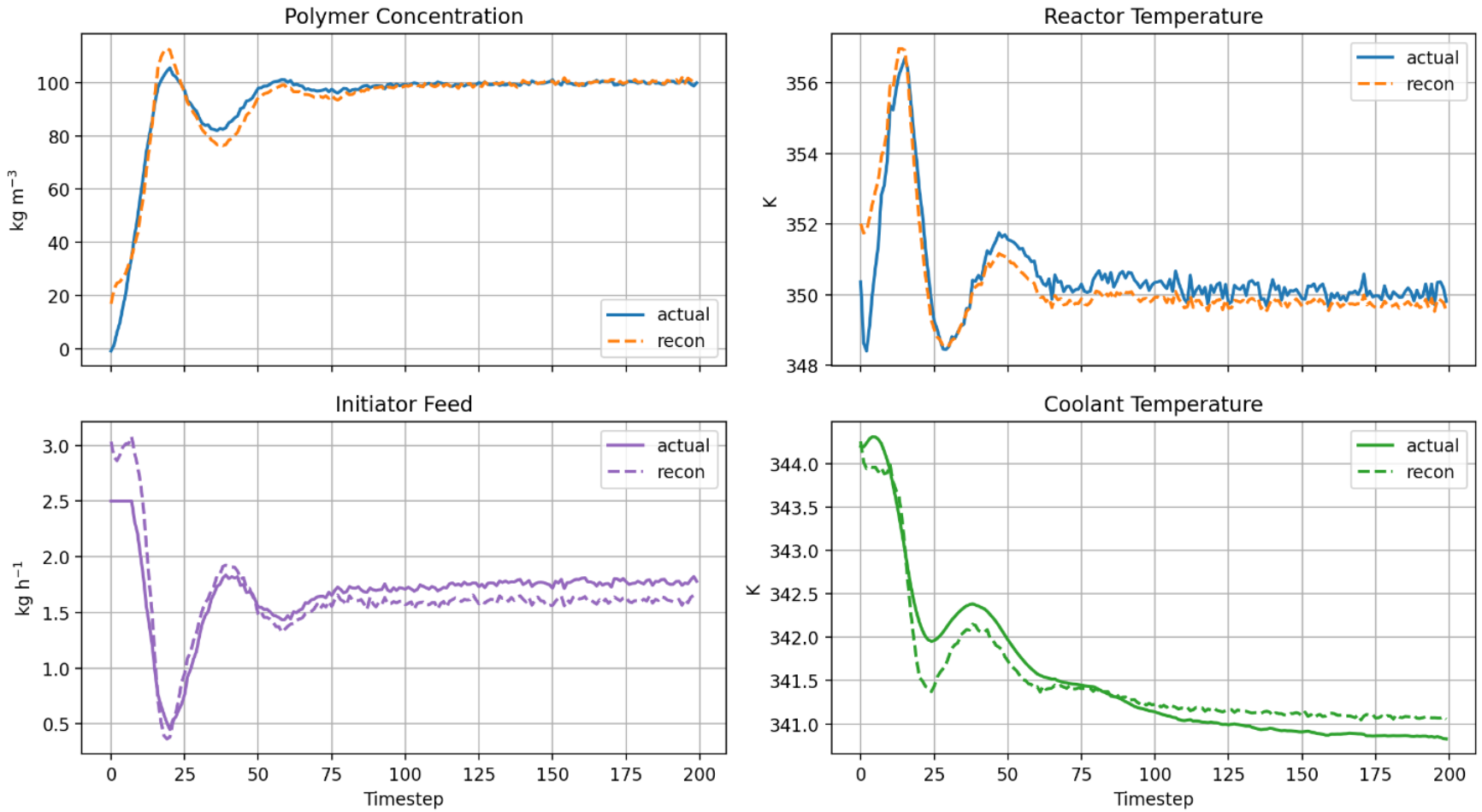
HOFLON – Polymerization reactor



- Reward with three components
- Encourage progress
- Encourage staying at the target
- Discourage large changes in MVs

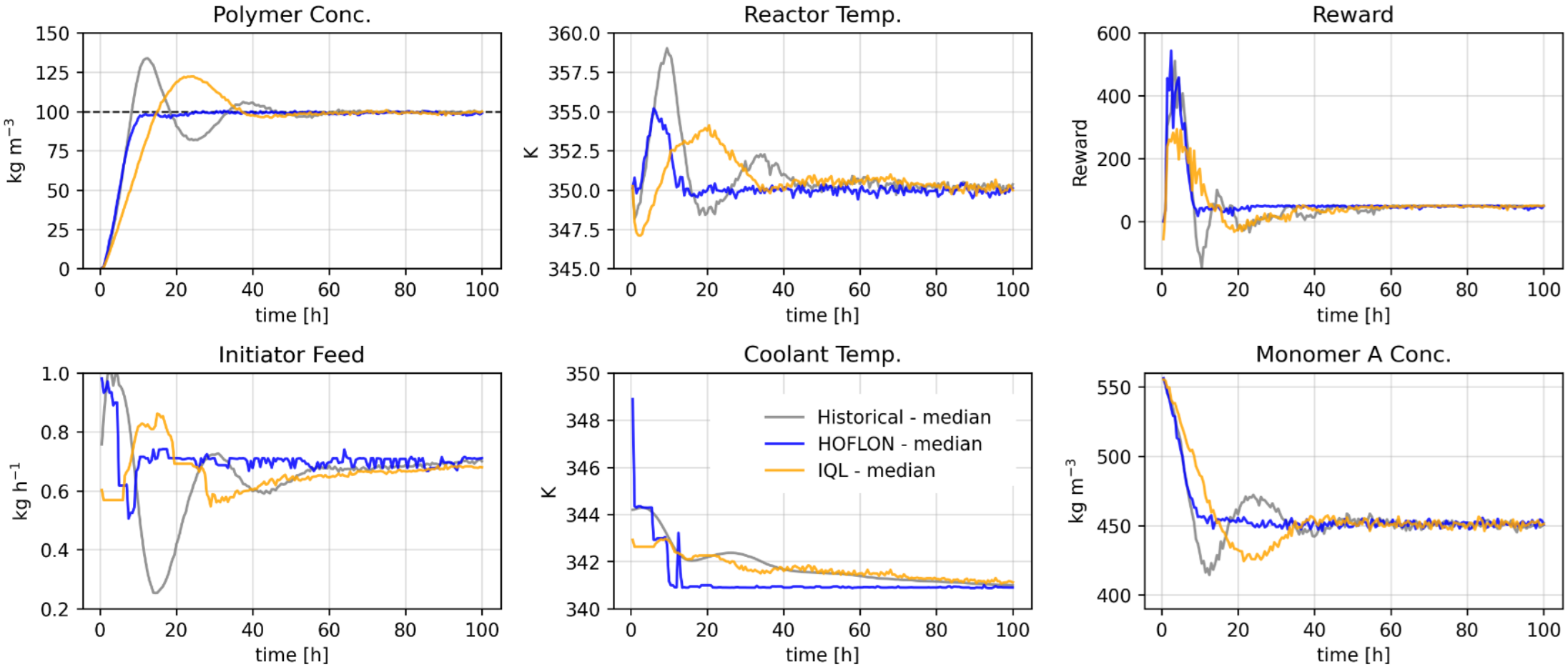
$$r(k) = \alpha\left[-e(k) + \exp\left(-\frac{e(k)}{\sigma^2}\right)\right] + \Delta e(k) - \lambda \left\|\Delta u(k)\right\|_2^2$$

HOFLON – Polymerization reactor



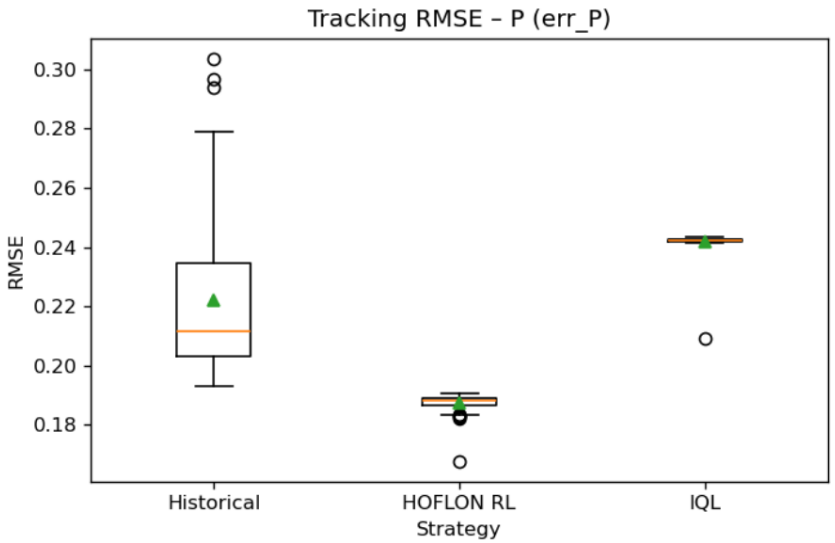
AAE reconstructions

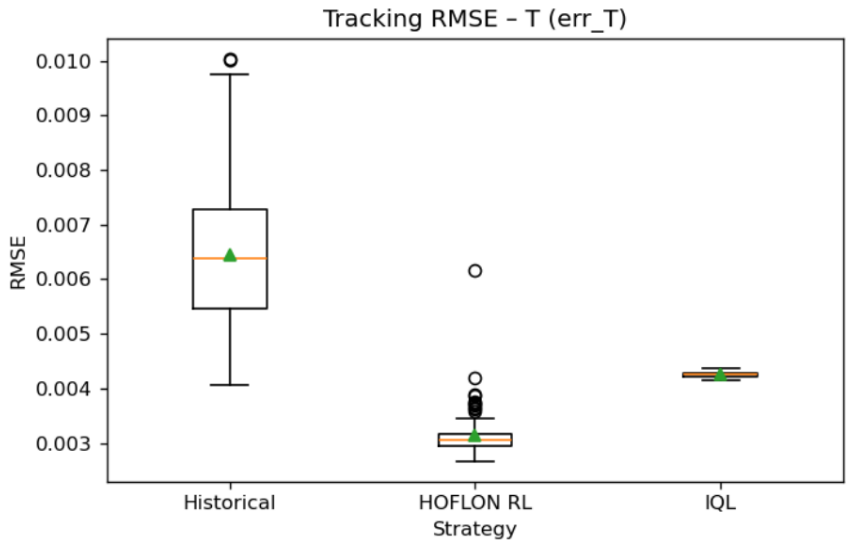
HOFLON – Polymerization reactor

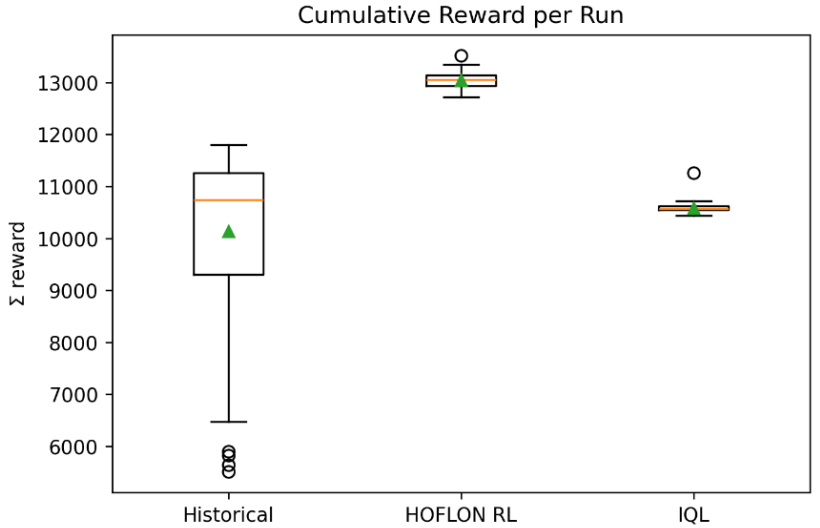


HOFLON performance

HOFLON – Polymerization reactor

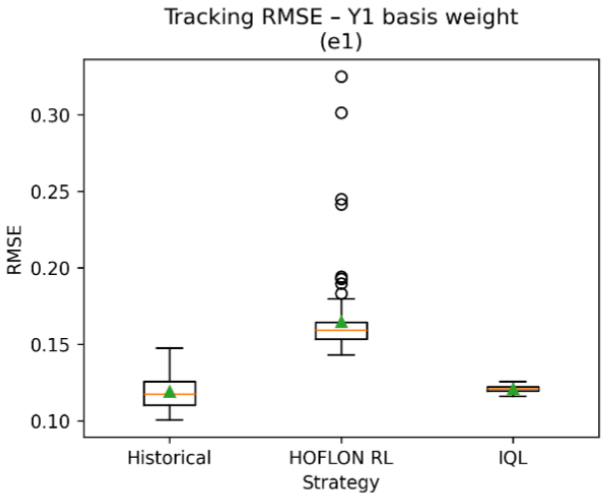


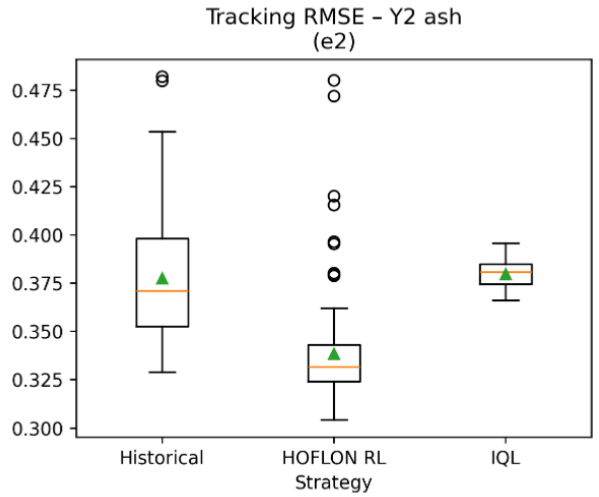


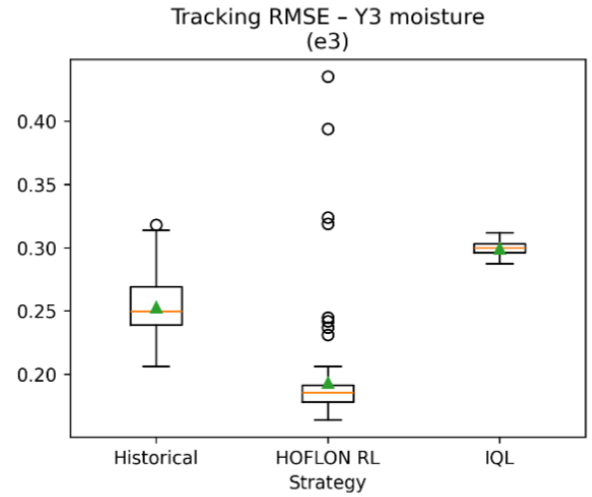


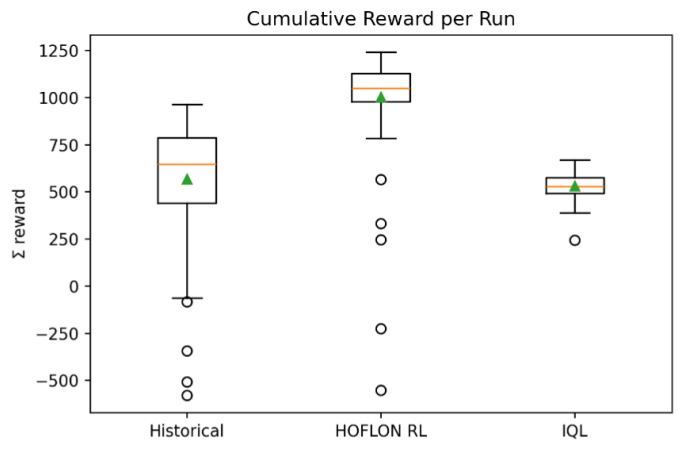
HOFLON performance

HOFLON – Paper machine



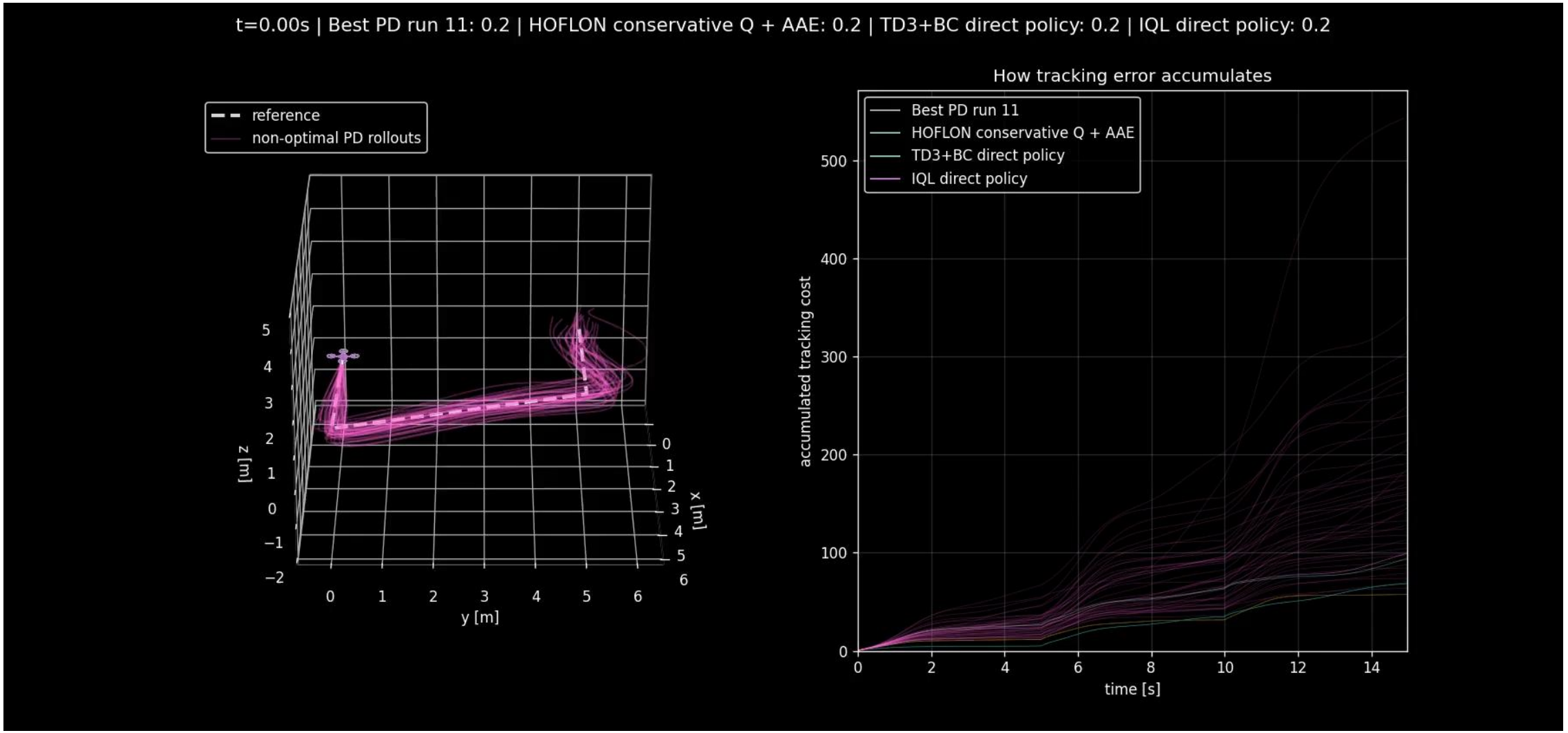




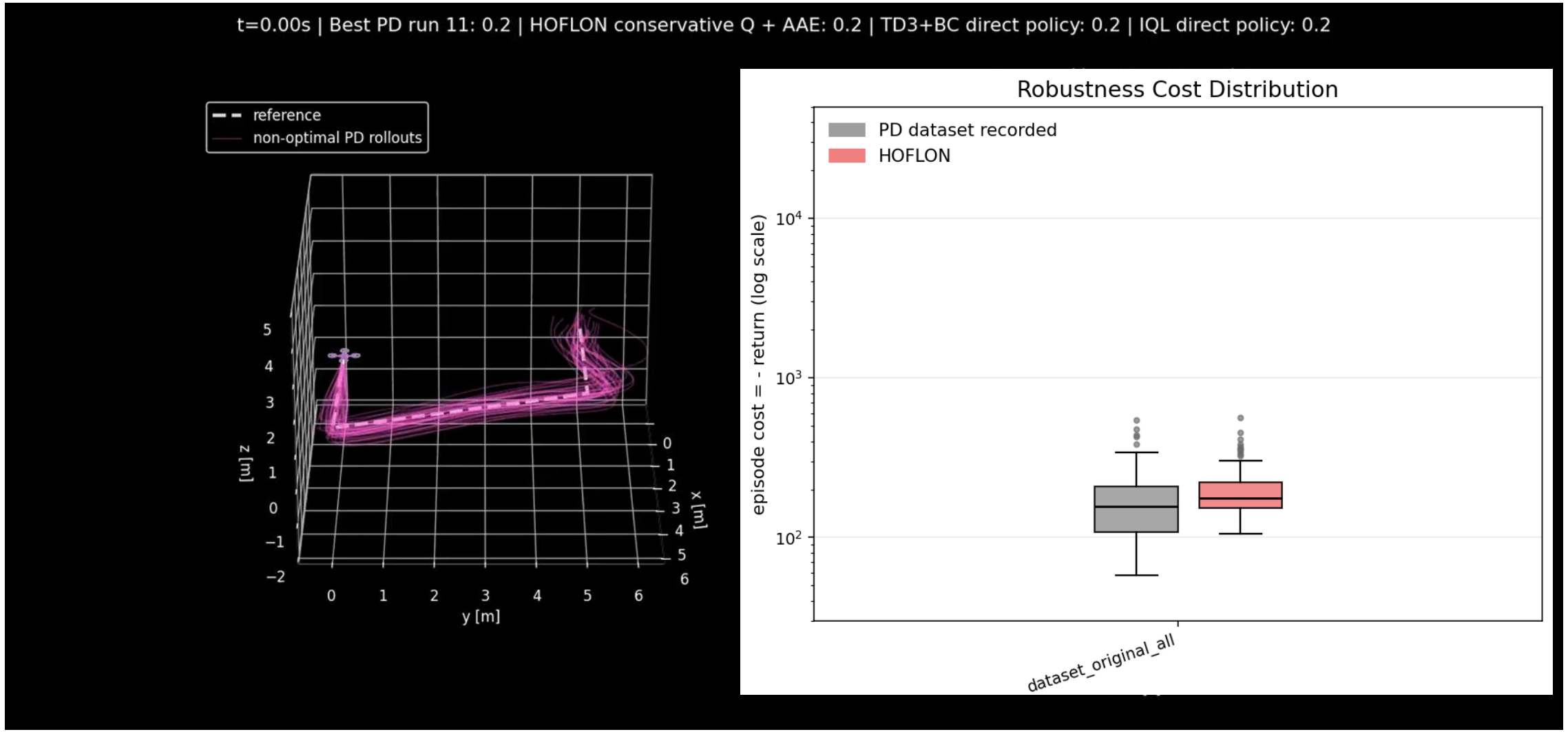


HOFLON performance

HOFLON – Drone racing

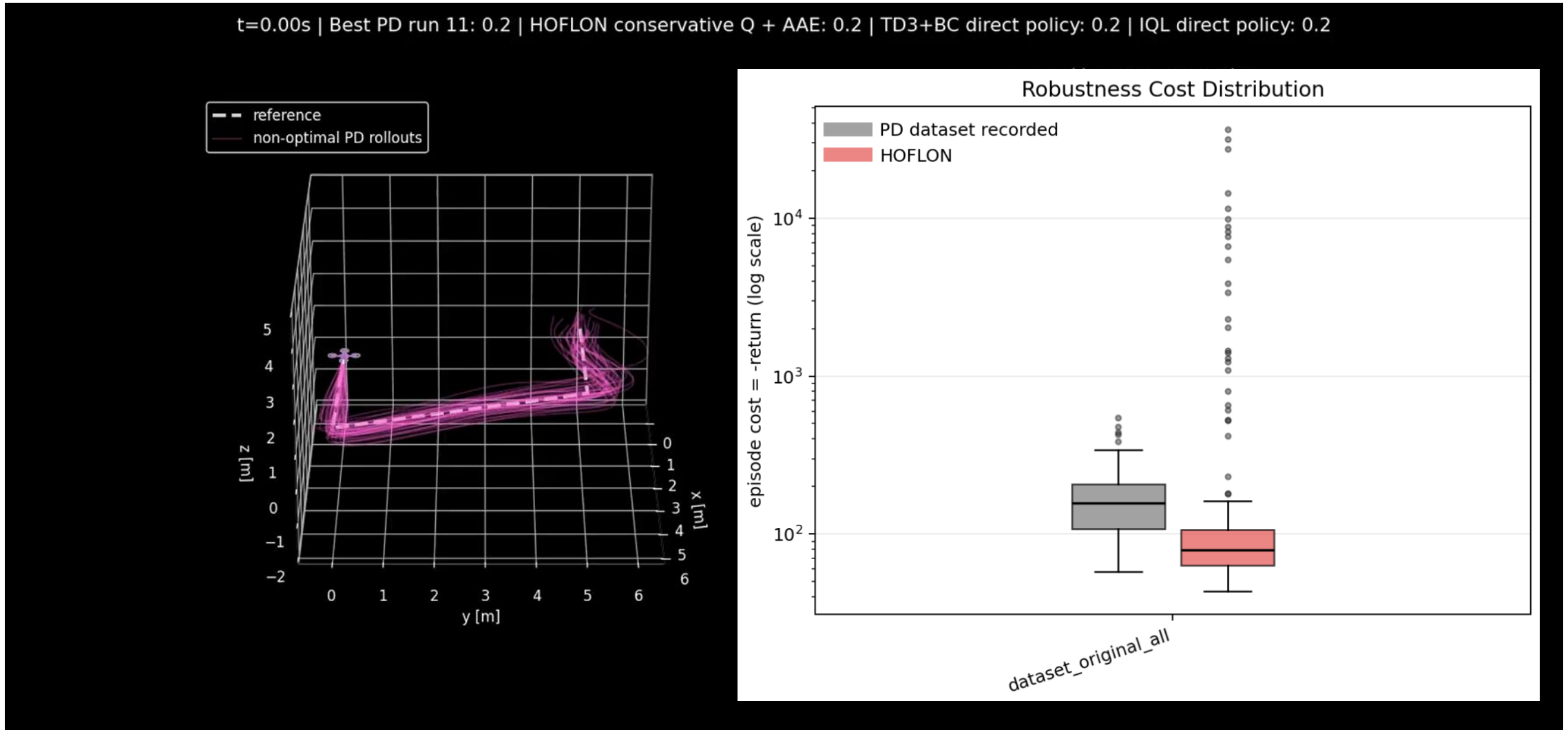


HOFLON – Drone racing



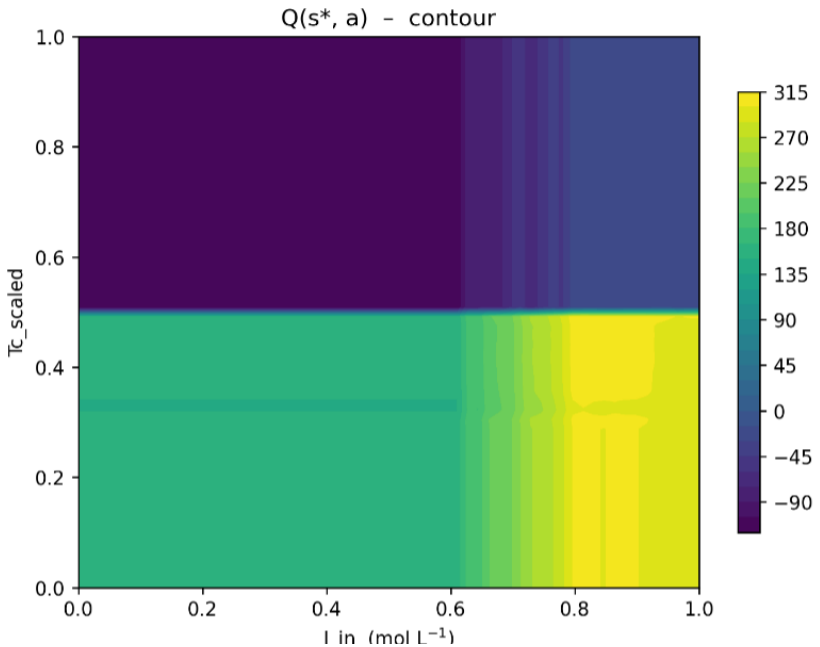
HOFLON performance

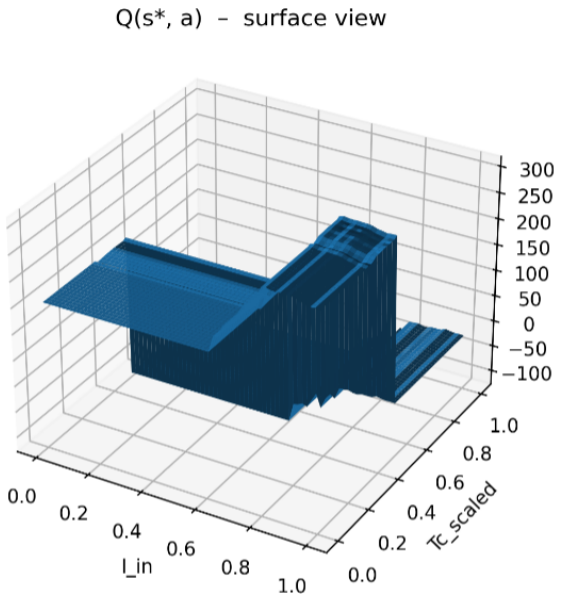
HOFLON – Drone racing



HOFLON performance

Discussion






```python
# ---------- 1. Nelder-Mead ----------------------------------------------
res_nm = minimize(obj, x0, method="Nelder-Mead",
                  options={"maxiter": 250, "disp": False})
u_nm_raw = res_nm.x
u_nm     = np.clip(u_nm_raw, 0.0, 1.0)
obj_nm   = obj(u_nm)

# ---------- 2. Powell (bounded) -----------------------------------------
res_pw = minimize(obj, x0, method="Powell",
                  bounds=bounds,
                  options={"maxiter": 250, "disp": False})
u_pw_raw = res_pw.x
u_pw     = np.clip(u_pw_raw, 0.0, 1.0)
obj_pw   = obj(u_pw)

# ---------- 3. DIRECT (pattern search) ----------------------------------
if HAVE_DIRECT:
    res_ps = direct(obj, bounds, maxiter=40, eps=1e-1)   # DIRECT options
    u_ps_raw = res_ps.x
    u_ps     = np.clip(u_ps_raw, 0.0, 1.0)
    obj_ps   = obj(u_ps)
else:
    # If DIRECT is unavailable, set cost huge so it never wins
    u_ps_raw, u_ps, obj_ps = (None, None, np.inf)

# ---------- choose best of the three ------------------------------------
objs = np.array([obj_nm, obj_pw, obj_ps])
idx  = np.argmin(objs)
```

- Devil is in the Q function and in solving the optimization problem
- AAE also needs to be good
- Iterative weight tuning is a pain (you should not tune in the first place remember we have no simulation model in reality)

Conclusion

Consistent gains on both tasks. Across the polymerization start-up and the paper-machine grade change, HOFLON gives the **highest cumulative reward** over 100 episodes → the decision-time optimization is producing reliable sequences

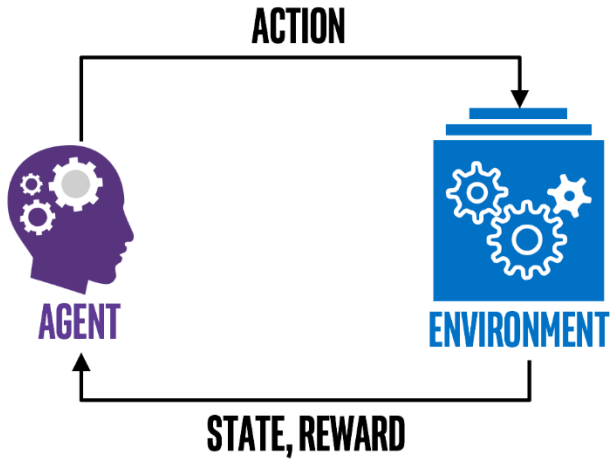
Some thoughts on next steps:

Single agent for all transitions
• Train one HOFLON-style agent that can handle start-up, grade-up, grade-down, and recovery by conditioning on the transition type

Try to build a convex Q critic
• Compare convex-in-action models (PICNN/ICNN) vs standard NNs for smoother, more reliable online solves

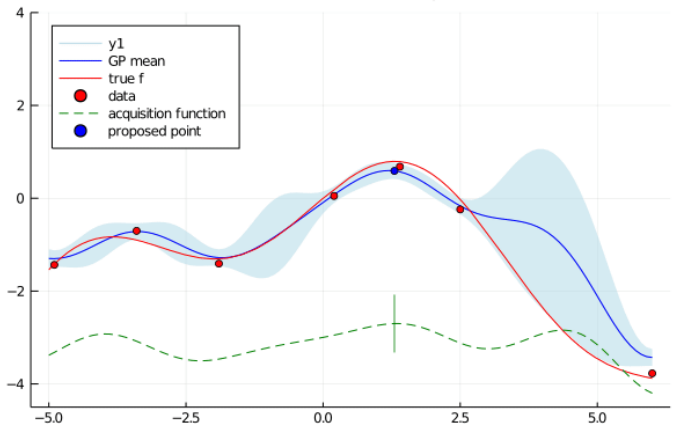
Diffusion models and capturing dynamics with the AEE
• Incorporate history to the unsupervised learning component

AI for autonomy

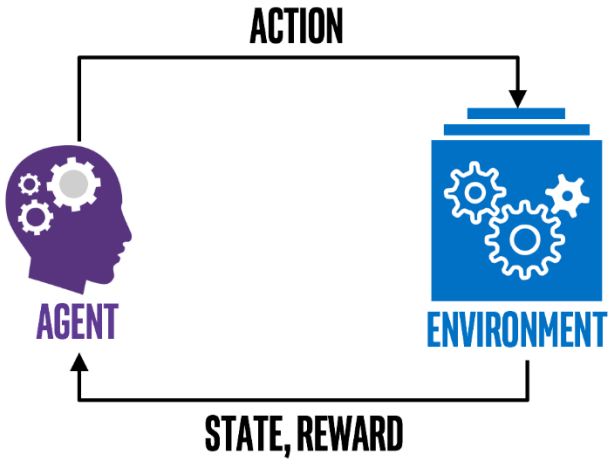


Learning a policy (reinforcement learning)

AI for autonomy



Learning a model
(+ model-based algorithms)



Learning a policy
(reinforcement learning)



Learning how to mimic
human reasoning?

AI for autonomy



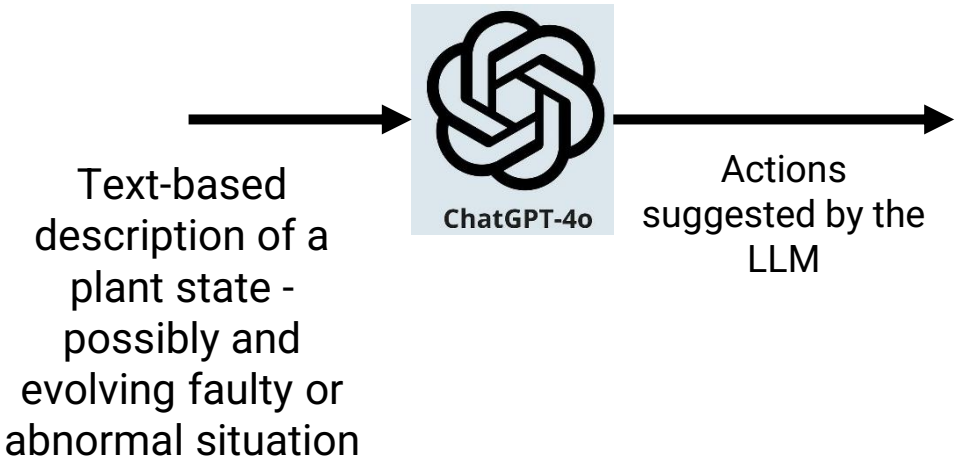
Learning how to mimic human reasoning?

The miracle

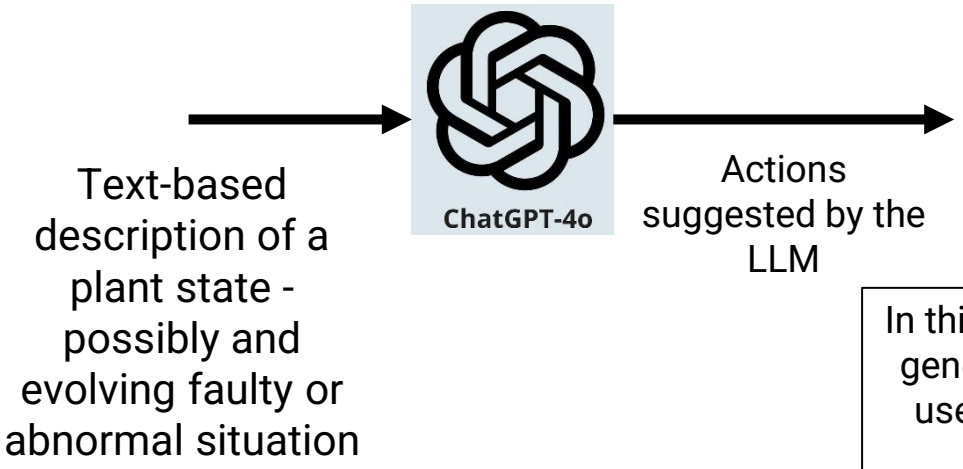


Learning how to mimic human reasoning?

Can LLMs play a role in autonomous operation?



Can LLMs play a role in autonomous operation?



In this work we force the LLMs to generate an output without tool use (e.g. generating a code to solve the problem)

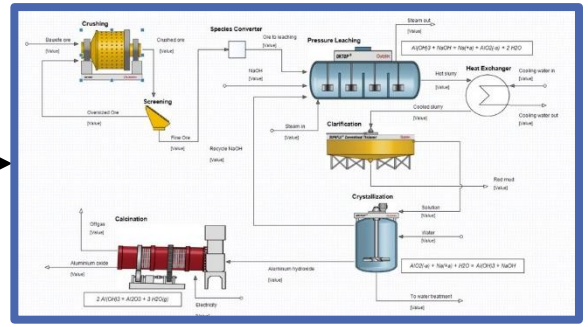
Tool use can also be a viable path, which we intend to explore in future work

Can LLMs play a role in autonomous operation?

Text-based description of a plant state - possibly and evolving faulty or abnormal situation

ChatGPT-4o

Actions suggested by the LLM



Use a validation tool: for example a digital-twin

Running a forward simulation on is easy – the inverse problem is difficult

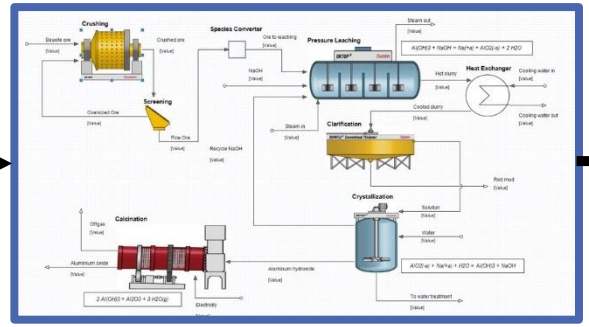
Having a digital twin to simulate a faulty or abnormal case is probably more difficult but this is our assumption at this point

Can LLMs play a role in autonomous operation?

Text-based description of a plant state - possibly and evolving faulty or abnormal situation

ChatGPT-4o

Actions suggested by the LLM



Collect sim outputs

Assess results

Good?

Implement recommended actions

Bad?

What the LLM suggest as the first solution might not be the right one

The analysis from the validation step could be complex result, which needs to be processed

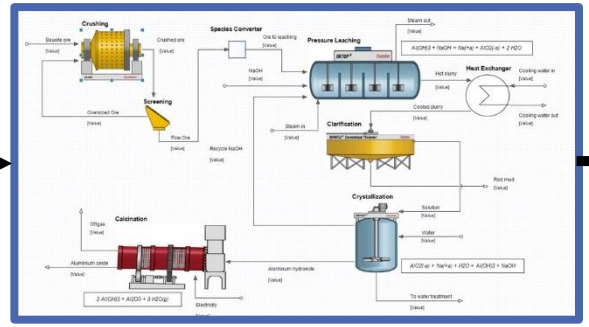
When working with LLMs the user typically provides feedback – we want the feedback to be autonomous

Can LLMs play a role in autonomous operation?

Text-based description of a plant state - possibly and evolving faulty or abnormal situation

ChatGPT-4o

Actions suggested by the LLM



Collect sim outputs

Assess results

Good?

Implement recommended actions

Bad?

ChatGPT-4o

Updated prompts

Use another LLM to process validation results

Generate updated prompts for the first LLM

Can LLMs play a role in autonomous operation?



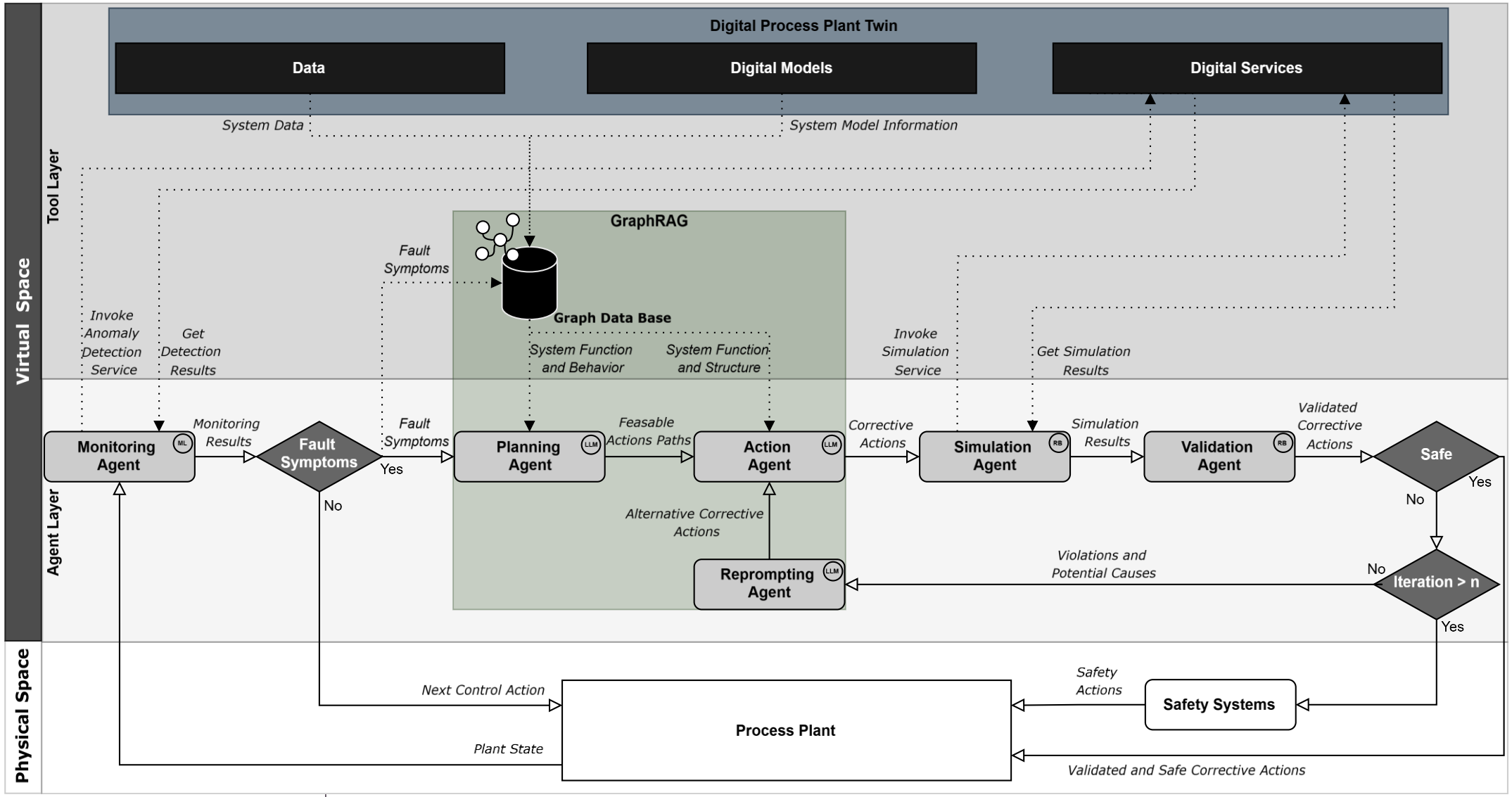
Because we have two (or more) LLMs involved we need a software layer to make them "**talk**" to each other

These configuration with multiple LLMs are referred to as **agentic** frameworks



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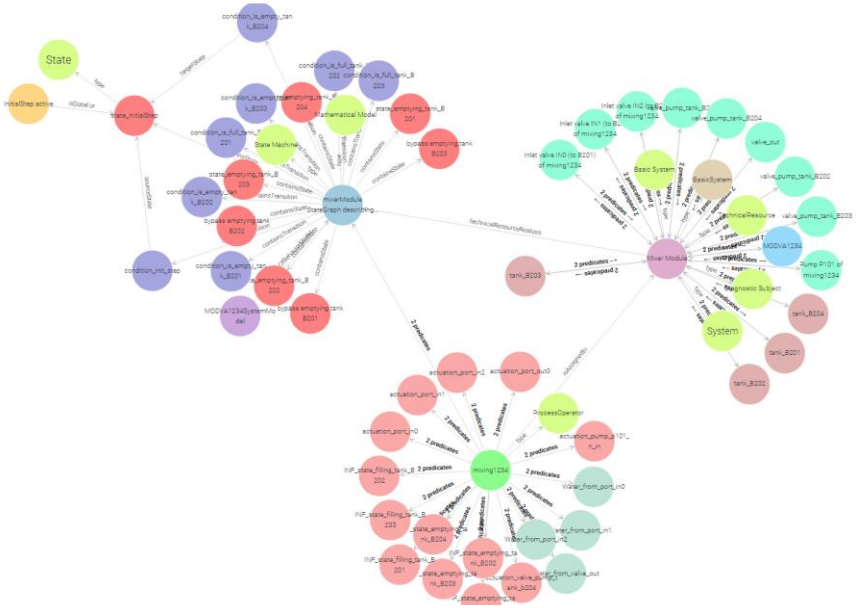
Active Fault-Tolerant-Control with LLM Agents



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Active Fault-Tolerant-Control with LLM Agents





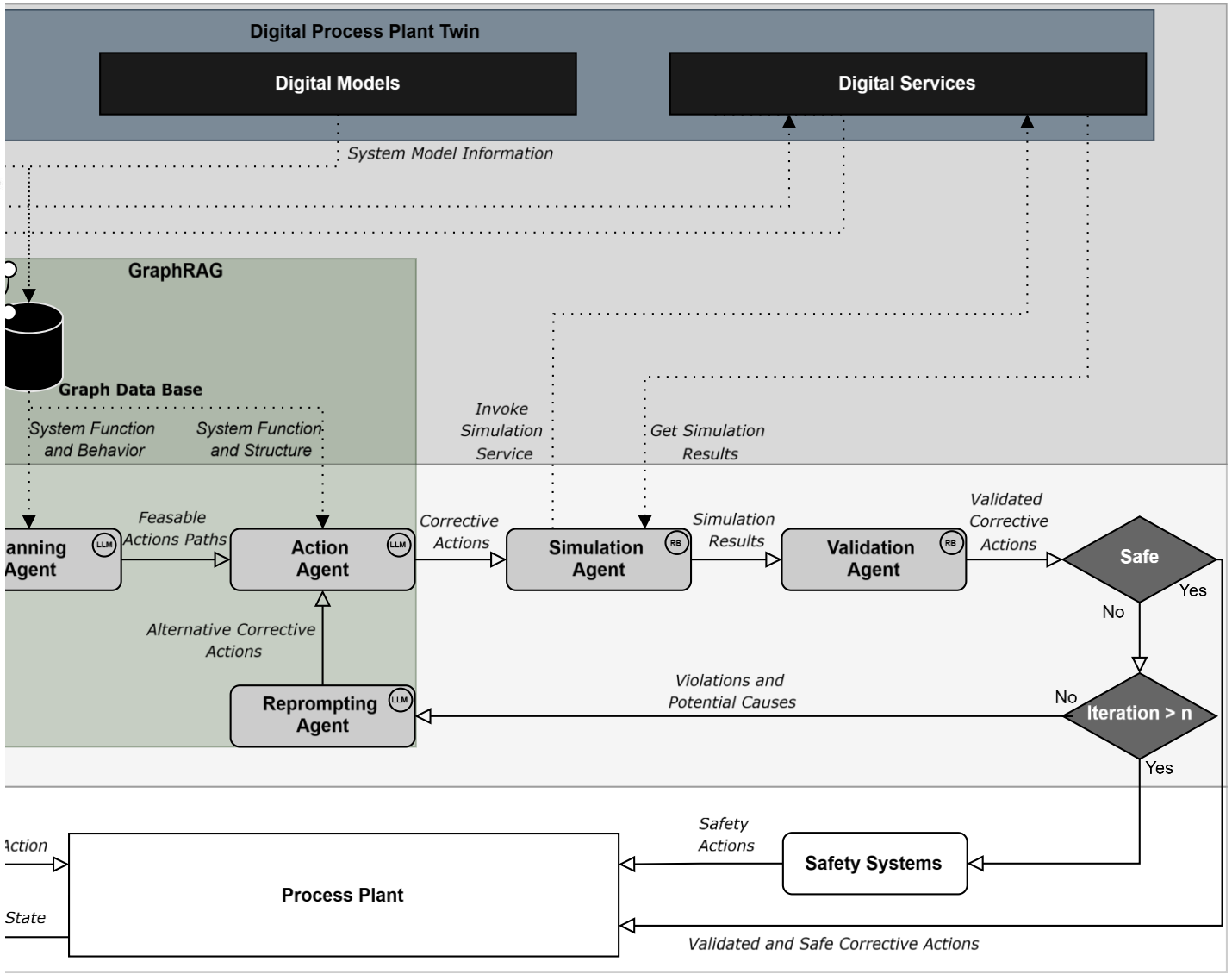
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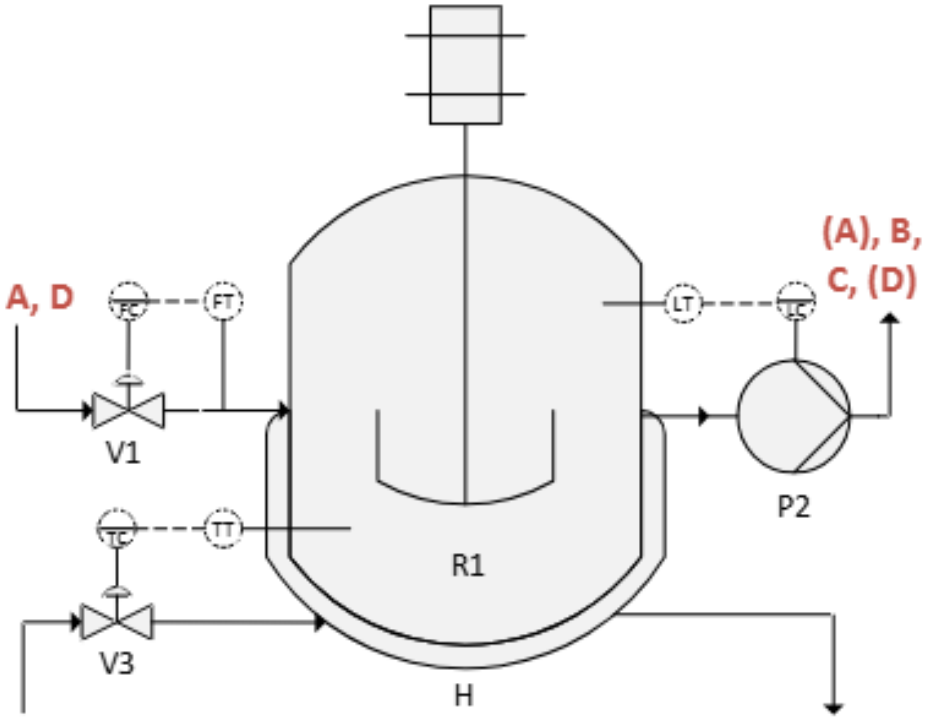
Active Fault-Tolerant-Control with LLM Agents

Plant Behaviour:

- Physics-driven reactor with coupled flow, level, and temperature dynamics
- PID loops stabilize operation; faults push the system out of nominal behavior

LLM agent as operator:

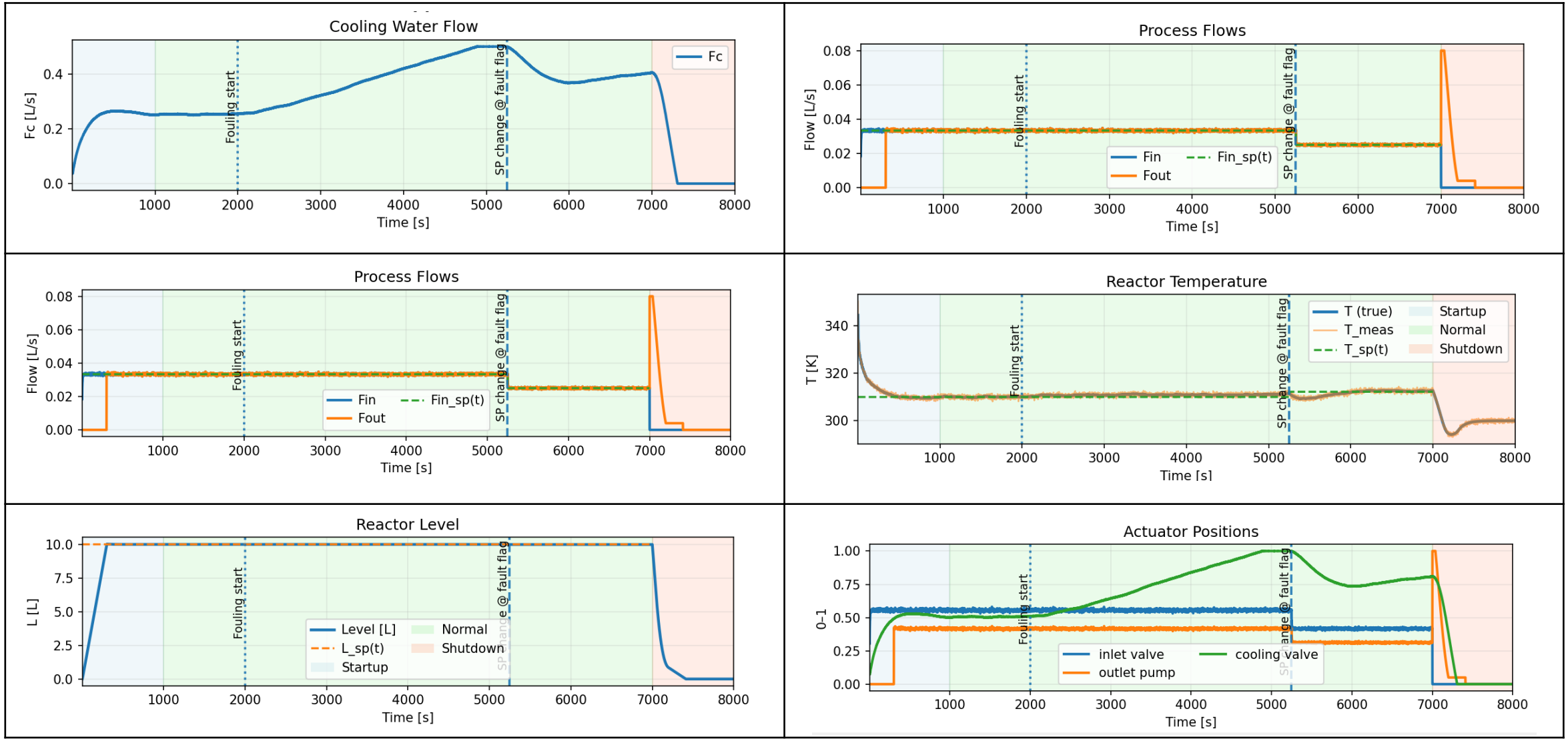
- Reads symptoms, understands the process state
- Decides *how the plant should operate under fault*
- Adjusts **setpoints**, not actuators — controllers do the rest



Plant: *Evolves by physics.*
PID: *Keeps it stable.*
LLM: *Decides what to do when things go wrong.*

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Active Fault-Tolerant-Control with LLM Agents



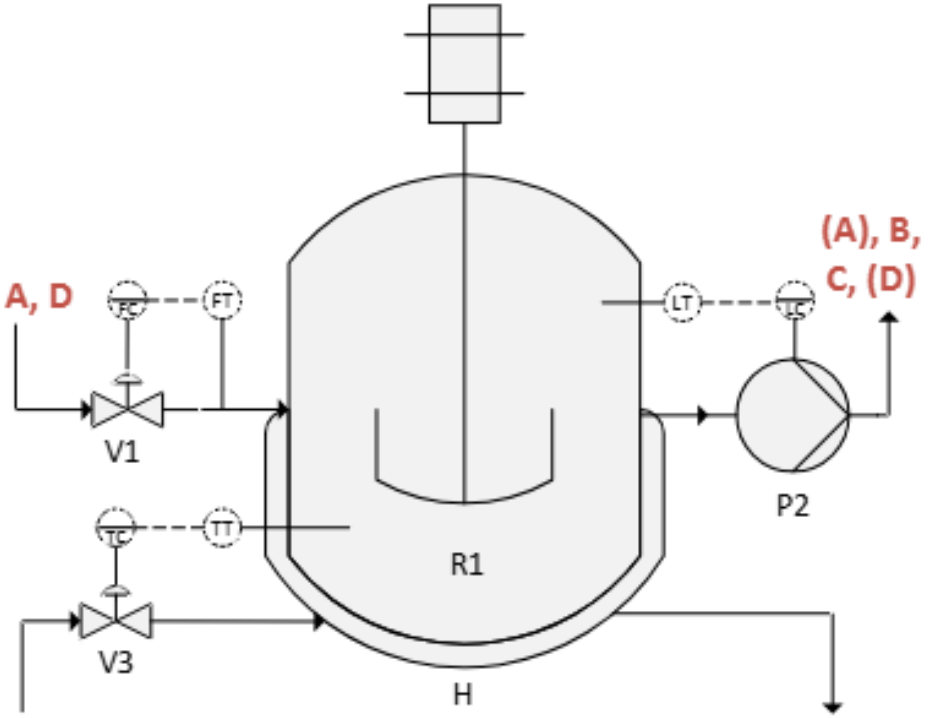
IMPERIAL UCL

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Active Fault-Tolerant-Control with LLM Agents

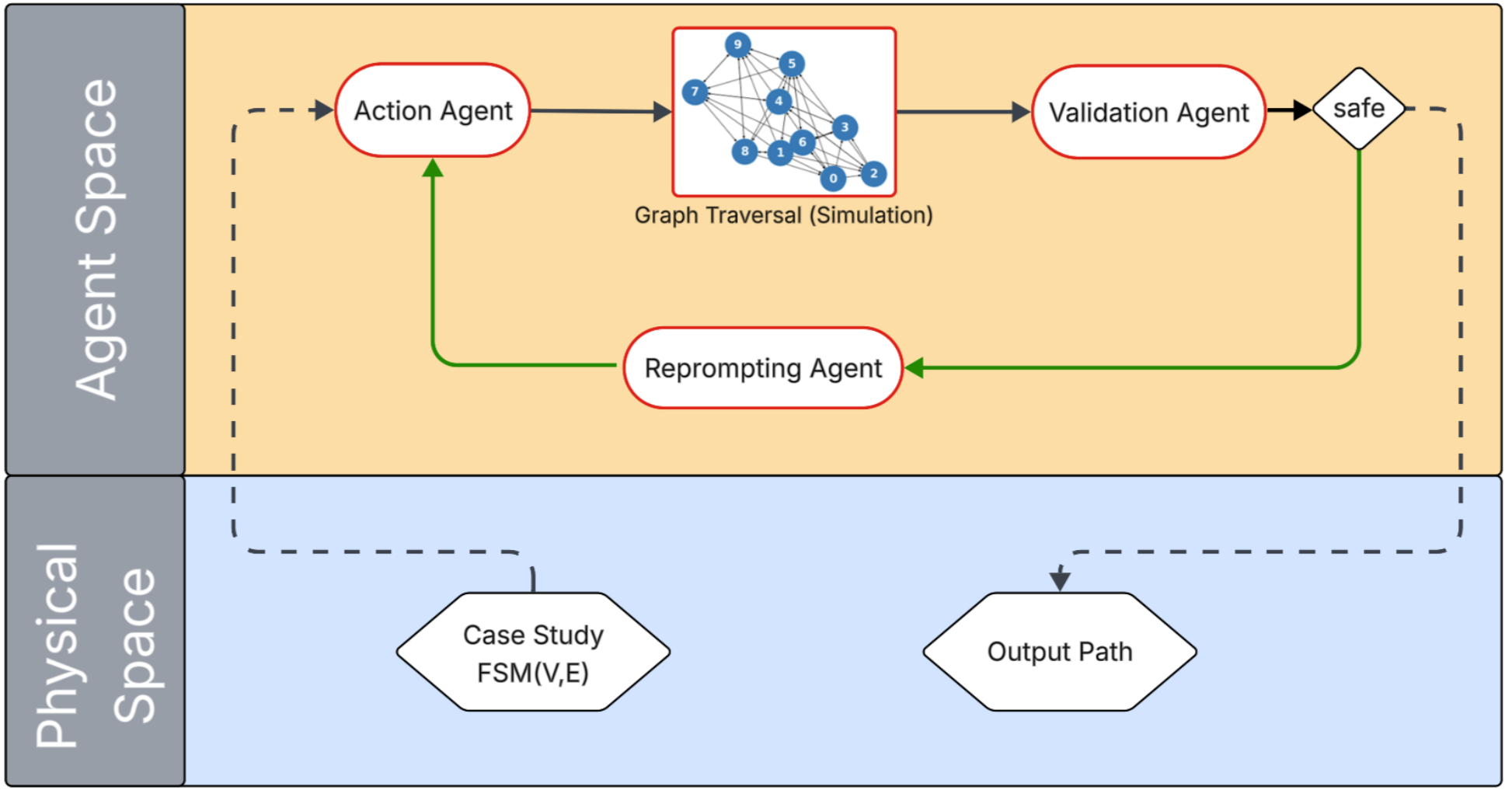
CSTR Module operational results (n = 10 per scenario)

Fault	Model	Success	Recovery	Time	Latency	Tokens in	Token out
Fouling	GPT-4.1-mini	8/10	43%	50.5s	1.9s	15.0k	0.2k
	GPT-4o-mini	9/10	50%	42.7s	2.3s	13.7k	0.2k
Pump degradation	GPT-4.1-mini	10/10	-	51.1s	1.9s	13.7k	0.2k
	GPT-4o-mini	10/10	-	36.8s	2.4s	11.6k	0.2k
Cool valve stuck	GPT-4.1-mini	10/10	100%	54.0s	2.4s	13.5k	0.3k
	GPT-4o-mini	4/10	0%	50.5s	3.5s	14.9k	0.3k
All	GPT-4.1-mini	28/30	75%	51.9s	2.1s	14.1k	0.2k
	GPT-4o-mini	23/30	50%	43.3s	2.7s	13.4k	0.2k



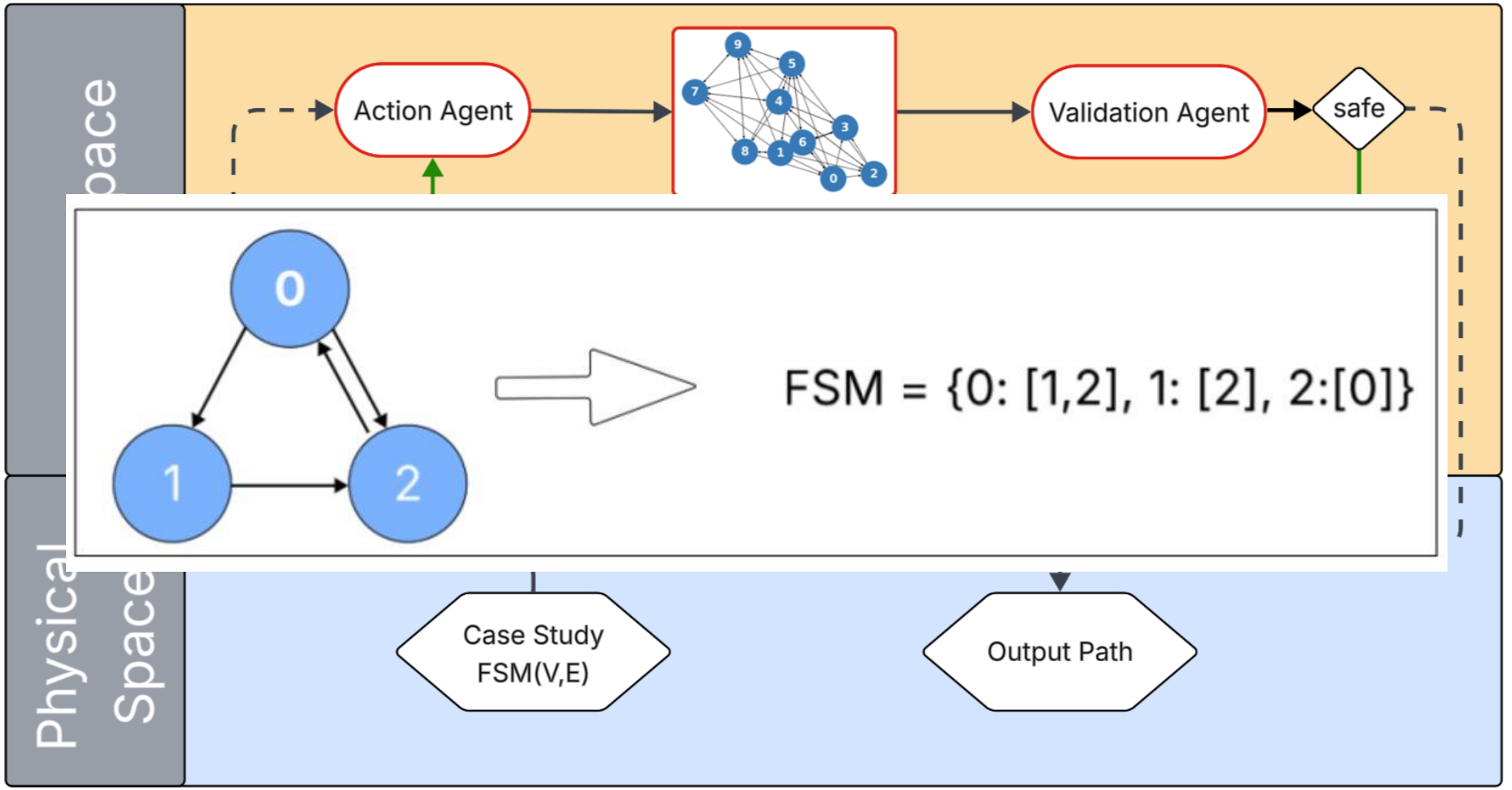
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Graph traversal with LLMs: Results



Vyas, J., & Mercangoz, M. (2025). Autonomous Control Leveraging LLMs: An Agentic Framework for Next-Generation Industrial Automation. *arXiv preprint arXiv:2507.07115*.

Graph traversal with LLMs



Vyas, J., & Mercangoz, M. (2025). Autonomous Control Leveraging LLMs: An Agentic Framework for Next-Generation Industrial Automation. *arXiv preprint arXiv:2507.07115*.

Conditionality: Reasoning under Constraints

Scenario:

- The agent must move from an initial position to a goal position within a grid-world.
- Actions: move left, right, forward, pickup, toggle, drop or done.

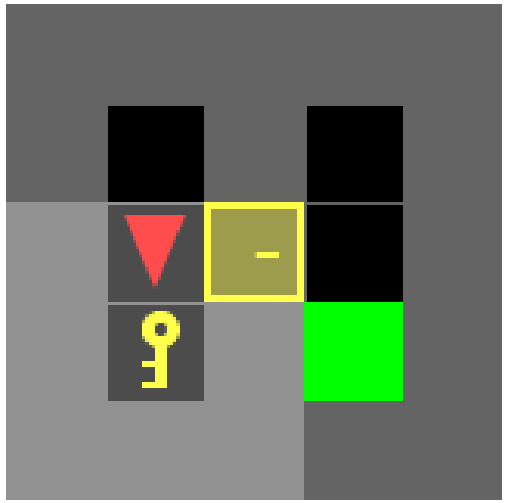
Constraints:

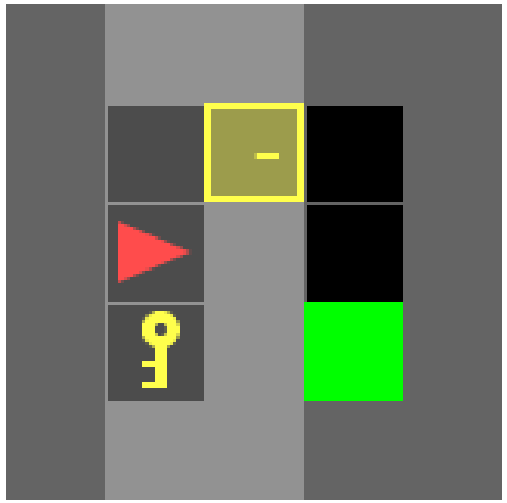
- Walls (gray cells) block movement — cannot pass through.
- The door (yellow square) can only be opened **after** collecting the key.
- The sequence of actions must respect these conditional dependencies.

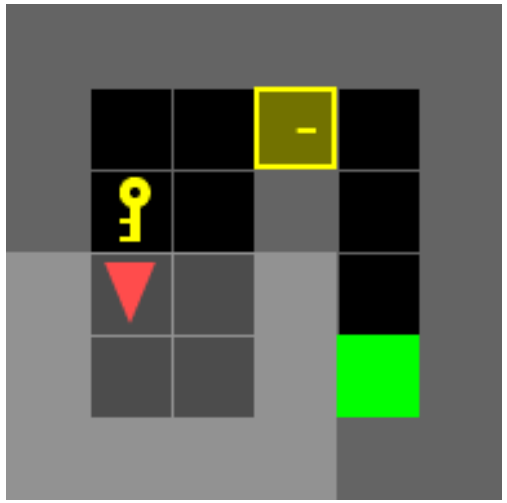


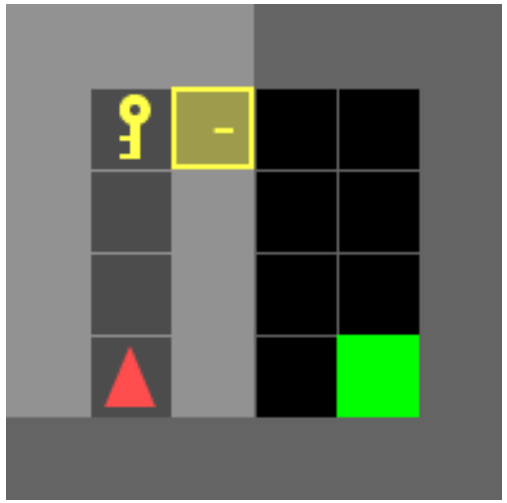
LLM Objective:
Generate an FSM-based reasoning path that satisfies all constraints to reach the goal.

Conditionality: Reasoning under Constraints

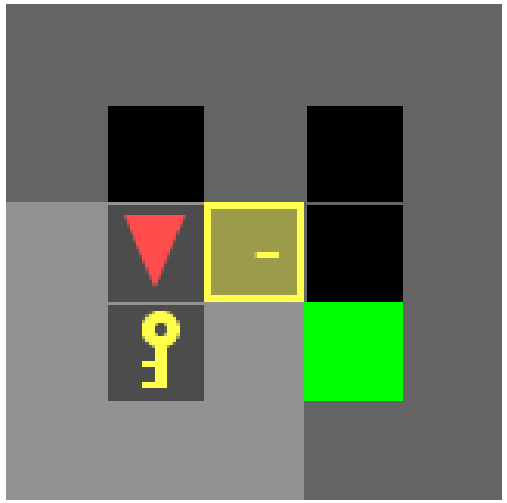


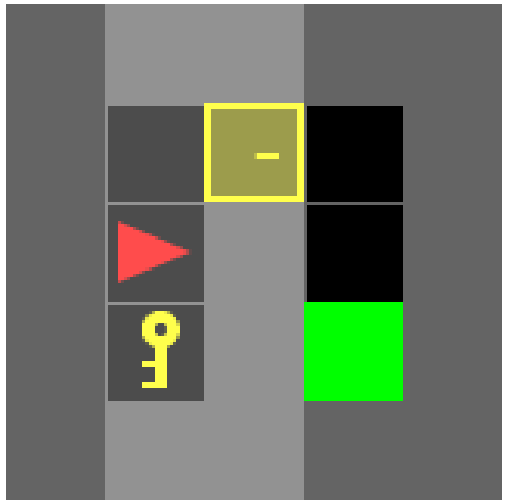


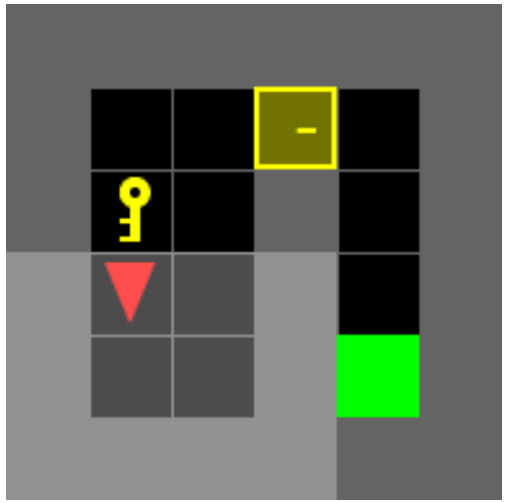


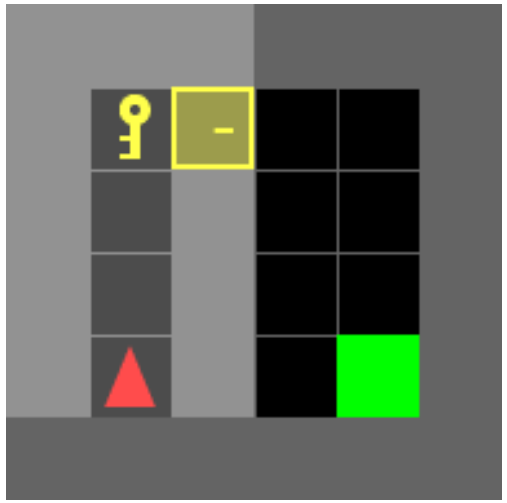


Conditionality: Reasoning under Constraints







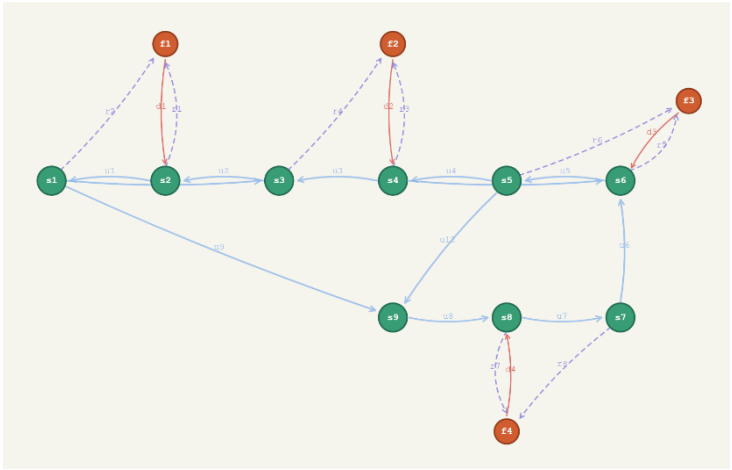


Fraction of Minigrid instances solved successfully with GPT-4o-mini

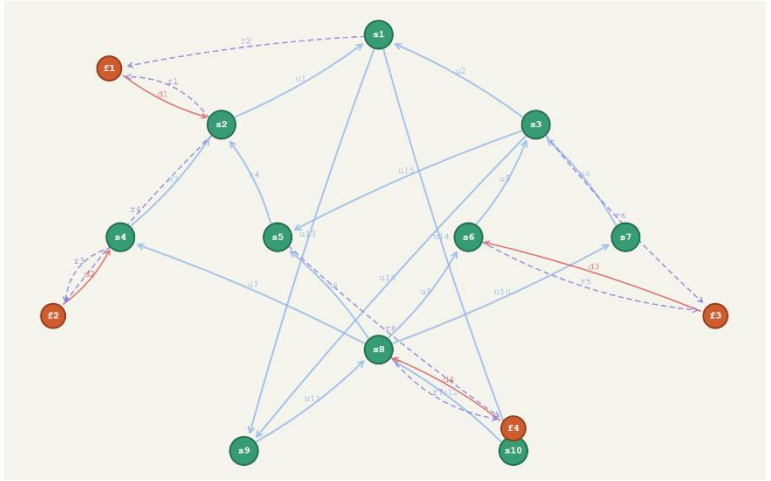
Grid	No	Naïve	Cut	Strong Cut
5 x 5	0.8	1.0	1.0	1.0
6 x 6	0.4	0.7	0.8	0.9
8 x 8	0.2	0.2	0.3	0.3
16 x 16	0	0	0	0.1

- Increasing grid size = increasing reasoning horizon
- Single-shot and naïve reprompting **fail catastrophically**
- Cut-based methods **delay failure**
- Strong cuts **preserve feasibility longest**

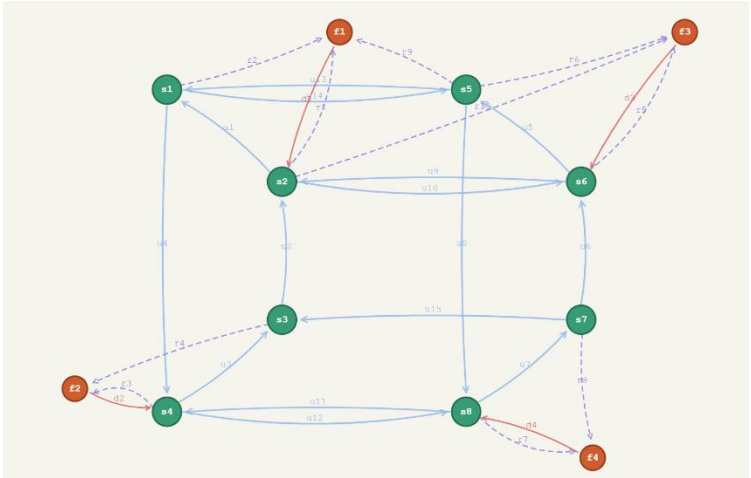
Different Case studies



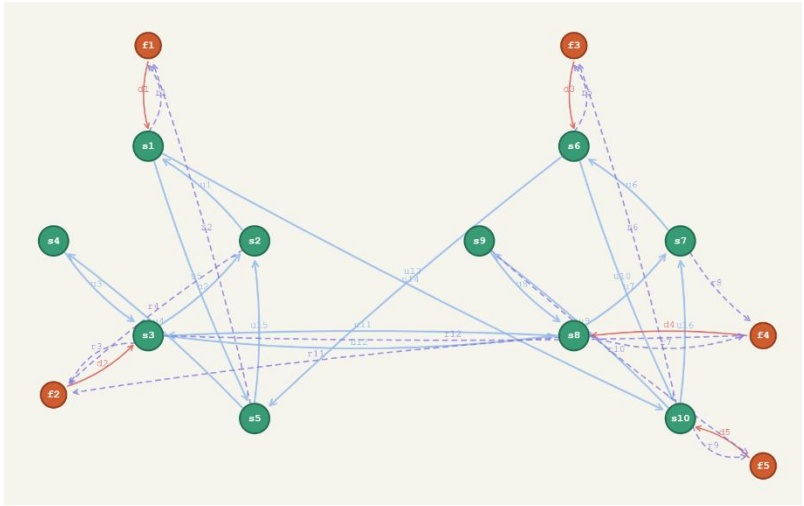
Pipeline



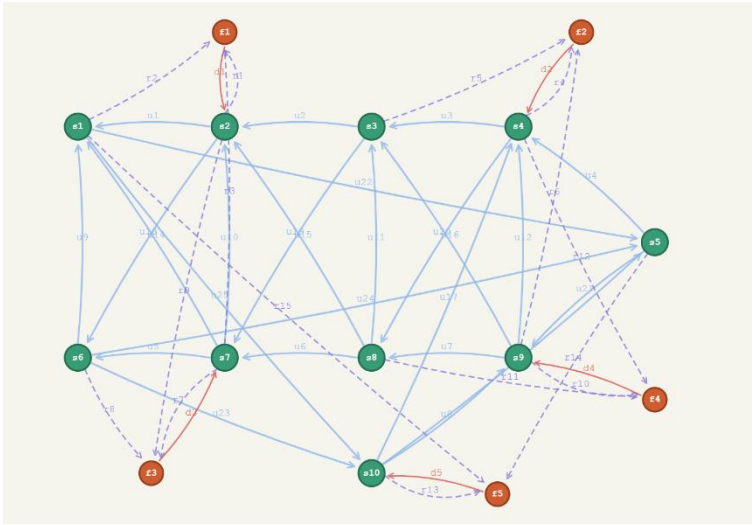
Branching



Cyclic



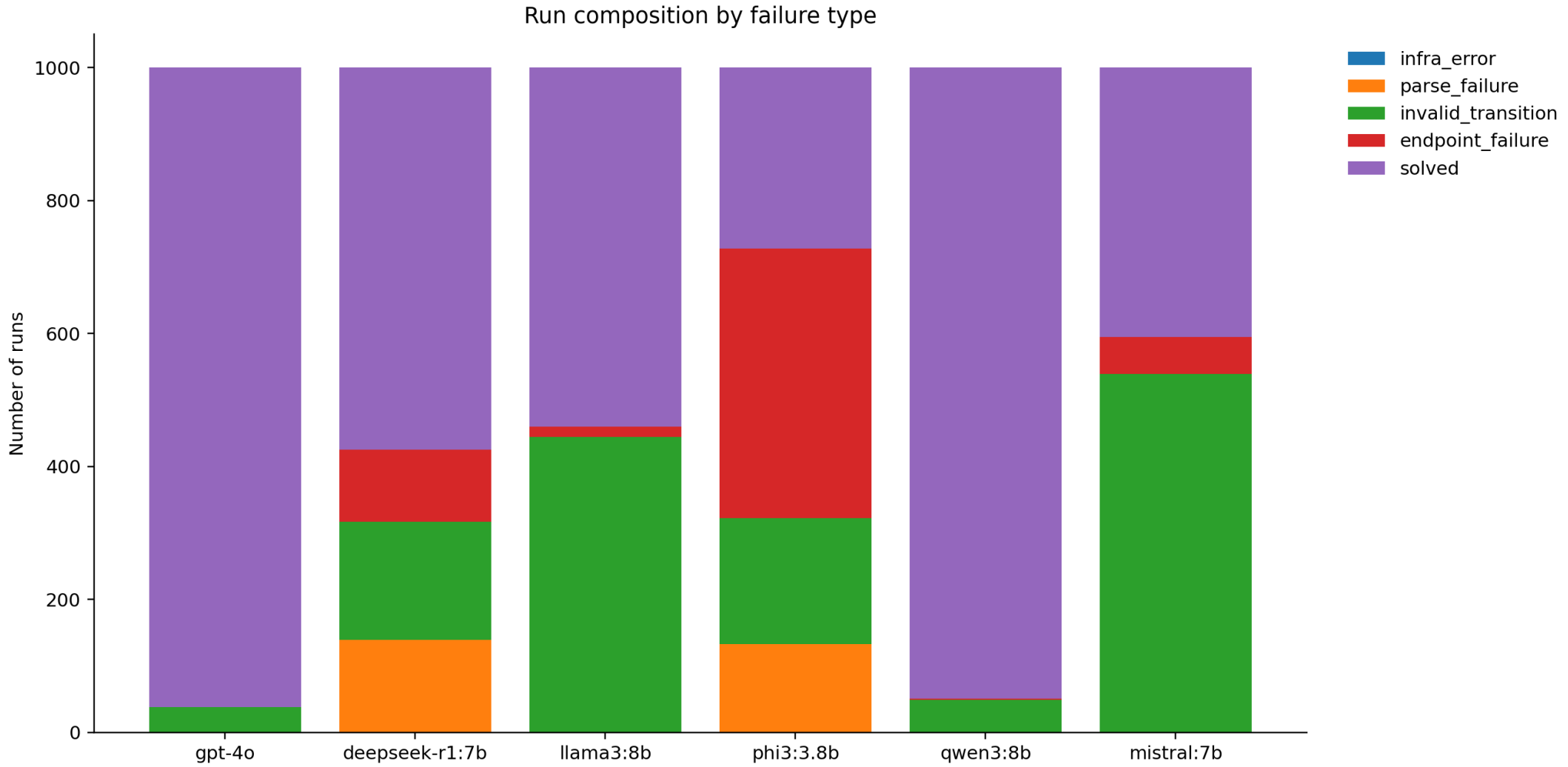
Hierarchical



Mesh

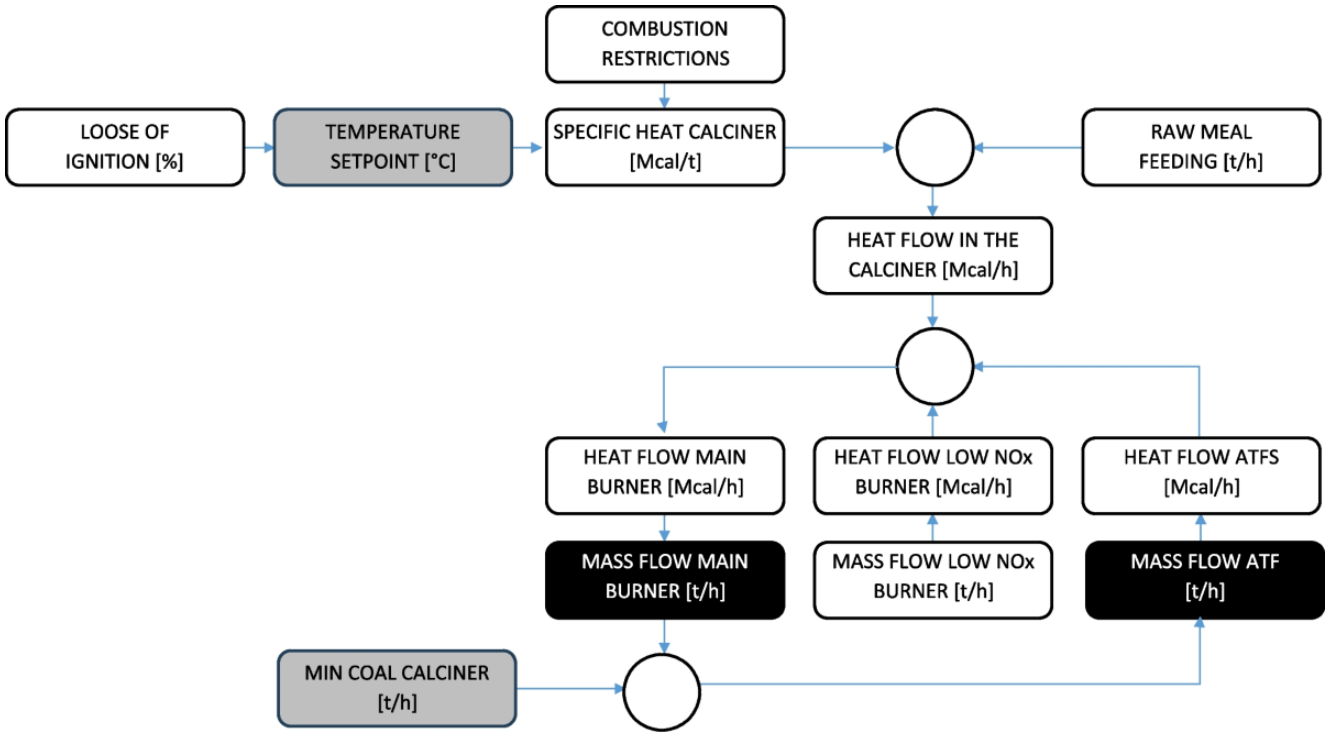
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Results



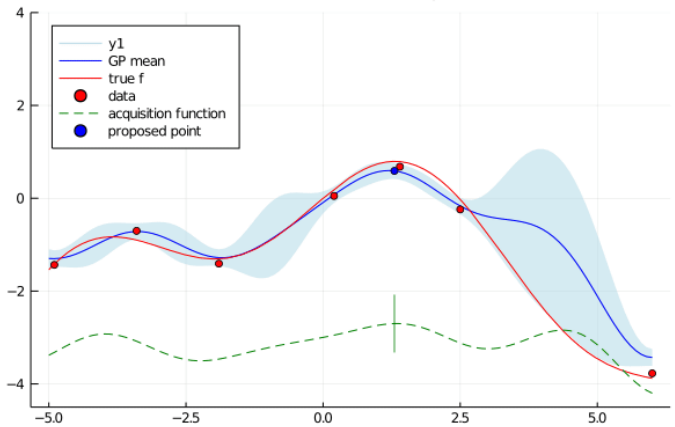
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Control with LLM Agents

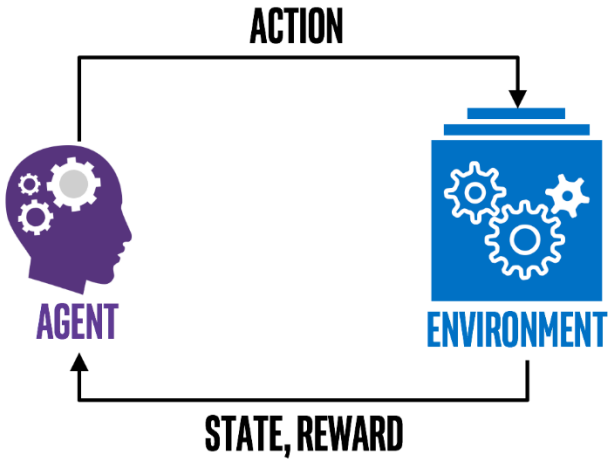




AI for autonomy



Learning a model
(+ model-based algorithms)



Learning a policy
(reinforcement learning)



Learning how to mimic
human reasoning?

AI personas in the engineering domain



Future systems will be collections of specialised agents that interact with humans, models, data, simulators and validation tools.

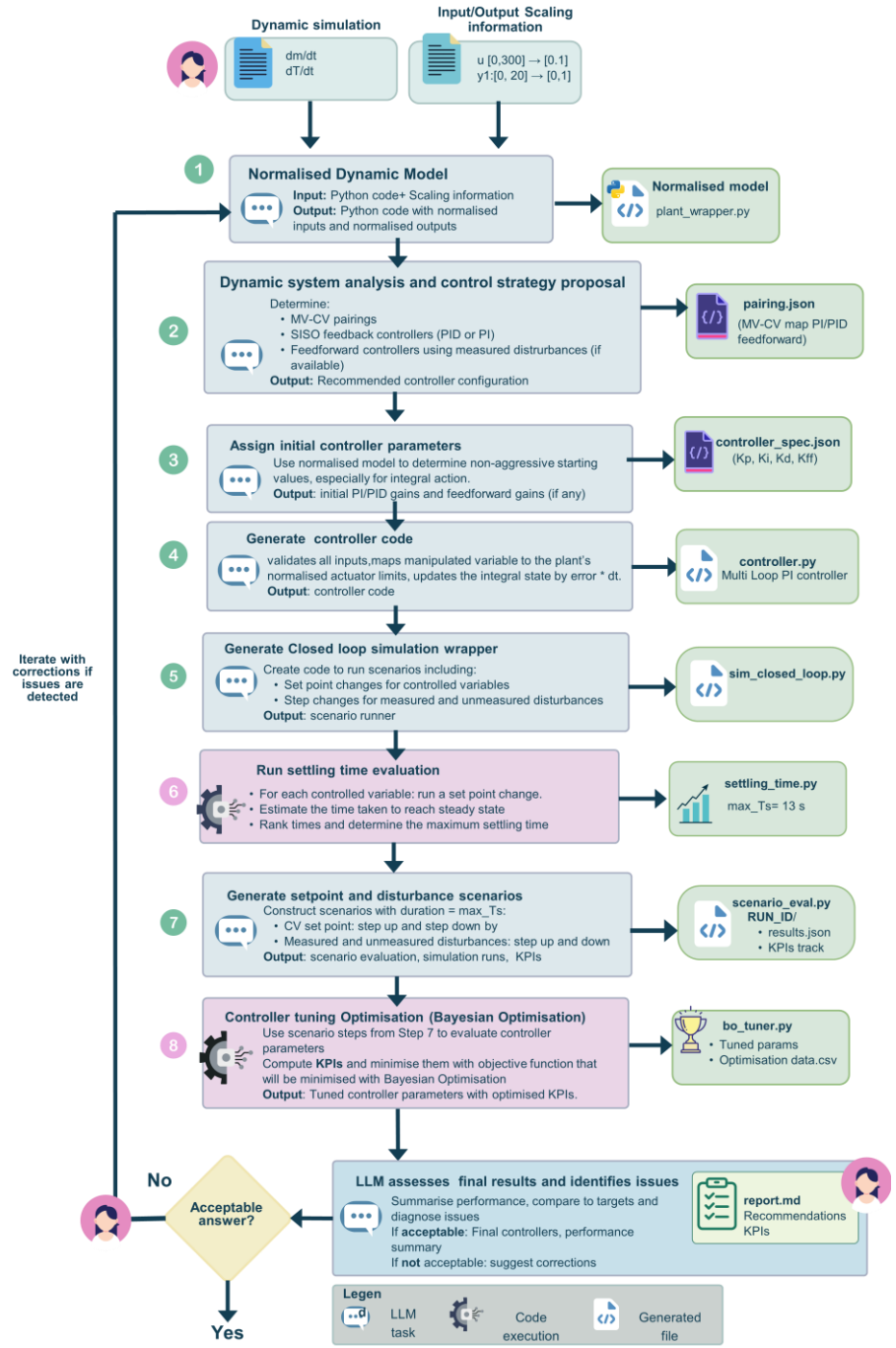
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AI personas: an example

Control system designer

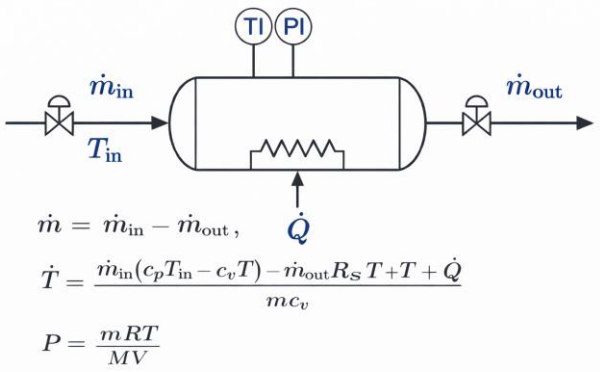
Loop pairing, tuning support, model structure and closed-loop analysis.

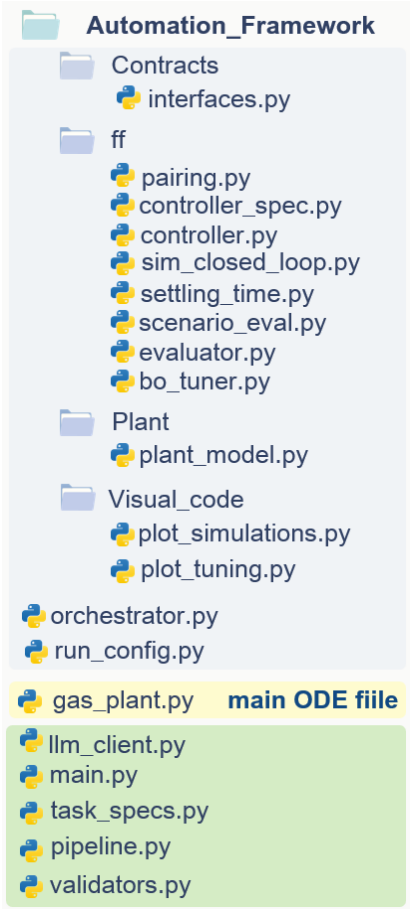
- The job/task description is decomposed into **domain-specific atomic units**.
- **LLM-generated code** is executed for simulation, numerical analysis, and algorithmic operations, while direct **LLM reasoning** is used for tasks requiring interpretation, judgement, and human-like decision-making.
- **Human users** provide **problem specifications and requirements** – can **optionally** assess outputs and provide feedback, otherwise the workflow is fully **autonomous**

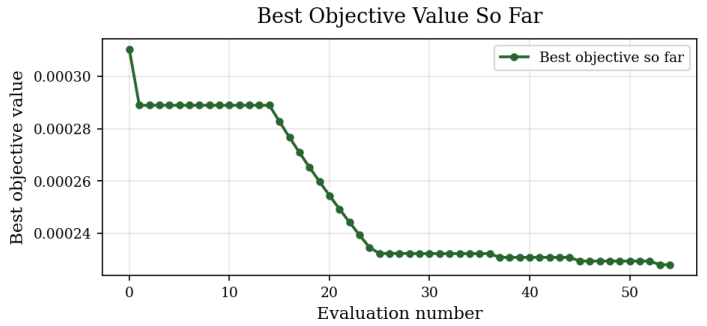


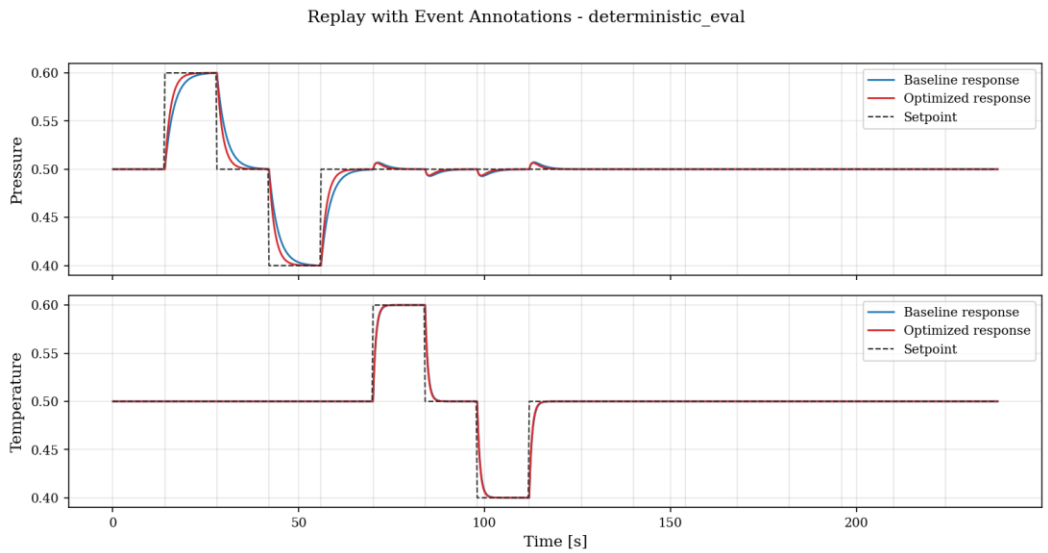
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AI personas: an example









In conclusion

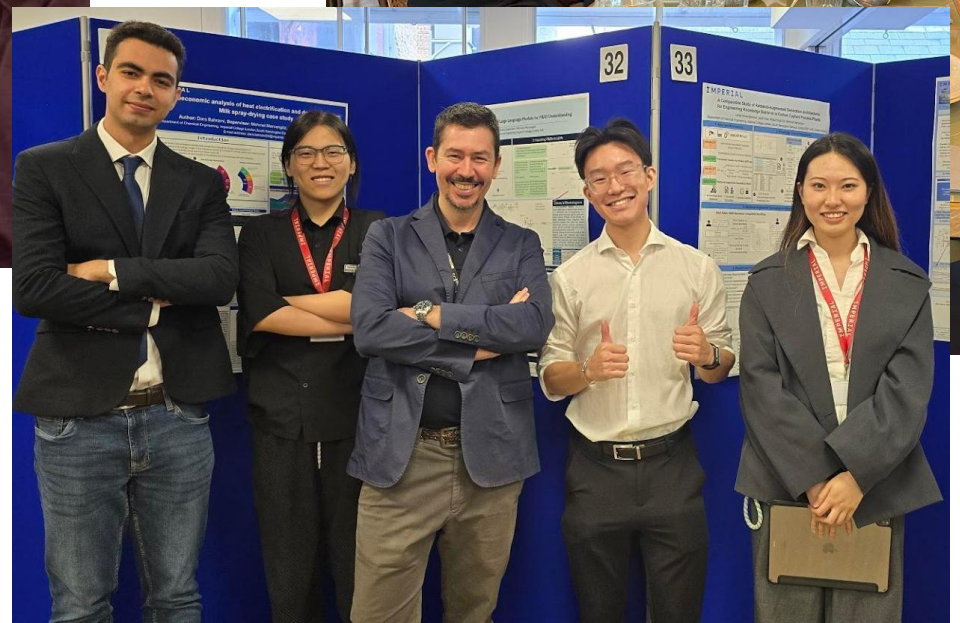
- **We are living in very interesting times**
- **With everything I showed there are still many things to do**
- **Autonomy is not a single technology (unless we get AGI – which could be a possibility but making AGI safe will also be a requirement)**

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People who do the actual work







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